

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

Well File No.
28636

MAR 18 2016

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas Division

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed September 4, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Chalmers 5300 21-19 8T					
Footages 2226 F N L 327 F W L		Qtr-Qtr LOT 2	Section 19	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 09/04/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 10215.75'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 10156.5'

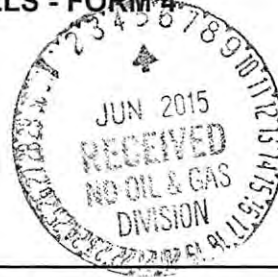
Company Oasis Petroleum North America LLC		Telephone Number 281 404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date March 17, 2016		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 4-7-2016	
By 	
Title JARED THUNE Engineering Technician	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

28633TH
28634TH
28635
28636TH
28648TH
28637TH
28649TH

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed March 14, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☒ Reclamation
☐ Other	Reserve pit reclamation

Well Name and Number See below					
Footages	Qtr-Qtr	Section	Township	Range	
F N L	LOT2	19	153 N	100 W	
Field	Pool	County			
Baker	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Neu Construction			
Address	City	State	Zip Code
602 W. 9th Street	Fairview	MT	59221

DETAILS OF WORK

Oasis Petroleum North America LLC plans to reclaim the reserve pit for the below referenced wells as follows:

Chalmers 5300 21-19 5T (28633)
Chalmers 5300 21-19 6B (28634)
Chalmers 5300 21-19 7T2 (28635)
Chalmers 5300 21-19 8T (28636)
Chalmers 5300 21-19 9B (28648)
Chalmers 5300 21-19 10T (28637)
Chalmers 5300 21-19 11T (28649)

The NDIC field inspector, Rick Dunn (NDIC) was notified on 03/06/2015

The surface owners, Wesley and Barbara Lindvig, 14075 41st Street NW, Alexander, ND 58831, were contacted on 03/06/2015

Spread material out in pit, cut top edge of liner and fold over cuttings, cover entire pit with line, back fill with clay
slope and contour well site to ensure proper drainage

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date June 4, 2015		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 9-23-15	
By 	
Title 	

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
28636

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Chalmers 5300 21-19 8T		Spacing Unit Description Sec. 19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		State TX	
Zip Code 77002		Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 2226 F N L	327 F WL	Qtr-Qtr LOT2	Section 19	Township 153 N	Range 100 W	County McKenzie
Spud Date October 5, 2014	Date TD Reached December 28, 2014	Drilling Contractor and Rig Number Nabors B22		KB Elevation (Ft) 2076	Graded Elevation (Ft) 2051	
Type of Electric and Other Logs Run (See Instructions) MWD/GR FROM KOP TO TD; CBL FROM INT. TD TO SURFACE						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	2175	17 1/2	54.5			1154	0
Vertical Hole	Intermediate	9 5/8	0	6158	13 1/2	36			905	4050
Vertical Hole	Intermediate	7	0	11165	8 3/4	32			789	3810
Lateral1	Liner	4 1/2	10255	20511	6	13.5			591	10255

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20524	Perforations	11165 20511			04/25/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11165' to 20511'						Name of Zone (If Different from Pool Name)				
Date Well Completed (SEE INSTRUCTIONS) July 21, 2015			Producing Method		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In)		
Date of Test 07/22/2015	Hours Tested 24	Choke Size 36 /64	Production for Test		Oil (Bbls) 1456	Gas (MCF) 1887	Water (Bbls) 2849	Oil Gravity-API (Corr.) 0	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI) 1300		Calculated 24-Hour Rate	Oil (Bbls) 1456	Gas (MCF) 1887	Water (Bbls) 2849	Gas-Oil Ratio 1296		

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 04/25/2015	Stimulated Formation Three Forks	Top (Ft) 11165	Bottom (Ft) 20511	Stimulation Stages 36	Volume 96178	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 4011690	Maximum Treatment Pressure (PSI) 9390		Maximum Treatment Rate (BBLS/Min) 40.0	
Details 100 Mesh White: 144690 40/70 White: 1554590 20/40 Ceramic: 2282410						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This is a supplemental completion report.		
I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com	Date 08/20/2015
Signature 	Printed Name Jennifer Swenson	Title Regulatory Specialist

**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE – Form 8**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

Well File No.

28636

NDIC CTB No.

To be assigned**228233**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number CHALMERS 5300 21-19 8T	Qtr-Qtr LOT2	Section 19	Township 153	Range 100	County Mckenzie
---	-----------------	---------------	-----------------	--------------	--------------------

Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective July 21, 2015
---	---	----------------------------	--

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Field Address	City	State	Zip Code
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Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective July 21, 2015
---	---	-----------------------------	--

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
---------------------------------	---------------------	--------------------	--------------------------

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Hofmann Trucking	25%	July 21, 2015
Other Transporters Transporting From This Lease	% Transported	Date Effective
		July 21, 2015
Comments		

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date August 13, 2015
Signature 	Printed Name Brianna Salinas	Title Marketing Assistant

Above Signature Witnessed By:		
Signature 	Printed Name Dina Barron	Title Mktg. Contracts Administrator



FOR STATE USE ONLY

Date Approved	AUG 28 2015
By 	
Title	Oil & Gas Production Analyst

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No. **28636**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
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Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Chalmers 5300 21-19 8T		Spacing Unit Description Sec. 19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		State TX	Zip Code 77002
		Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 2226 F N L	327 F WL	Qtr-Qtr LOT2	Section 19	Township 153 N	Range 100 W	County McKenzie
Spud Date October 5, 2014	Date TD Reached December 28, 2014	Drilling Contractor and Rig Number Nabors B22		KB Elevation (Ft) 2076	Graded Elevation (Ft) 2051	
Type of Electric and Other Logs Run (See Instructions) MWD/GR FROM KOP TO TD; CBL FROM INT. TD TO SURFACE						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
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Vertical Hole	Intermediate	9 5/8	0	6158	13 1/2	36			905	4050
Vertical Hole	Intermediate	7	0	11165	8 3/4	32			789	
Lateral1	Liner	4 1/2	10255	20511	6	13.5			591	10255

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1		Perforations							

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1-						Name of Zone (If Different from Pool Name)			
Date Well Completed (SEE INSTRUCTIONS)			Producing Method	Pumping-Size & Type of Pump			Well Status (Producing or Shut-In)		
Date of Test	Hours Tested	Choke Size /64	Production for Test	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Oil Gravity-API (Corr.)	Disposition of Gas	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Gas-Oil Ratio	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLs/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLs/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLs/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLs/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLs/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This is a preliminary completion report. A supplemental report will be filed upon first production of the well.

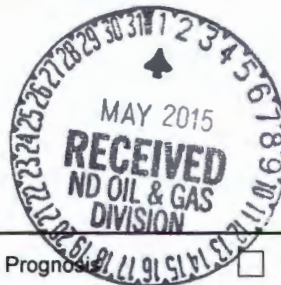
I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com	Date 07/21/2015
	Signature 	Printed Name Jennifer Swenson



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
28636



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date May 1, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Waiver from tubing/packer requirement

Well Name and Number Chalmers 5300 21-19 8T						
Footages			Qtr-Qtr	Section	Township	Range
2226 F N L	327 F W L	LOT2	19	153 N	100 W	
Field Baker	Pool Bakken	County McKenzie				

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date April 30, 2015		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date May 6, 2015	
By 	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28636

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
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☒ Notice of Intent

Approximate Start Date
May 1, 2015

☐ Report of Work Done

Date Work Completed

☐ Notice of Intent to Begin a Workover Project that may Qualify
for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis

☐ Redrilling or Repair

☐ Casing or Liner

☐ Plug Well

☐ Supplemental History

☐ Temporarily Abandon

☒ Other

☐ Spill Report

☐ Shooting

☐ Acidizing

☐ Fracture Treatment

☐ Change Production Method

☐ Reclamation

Change well status to CONFIDENTIAL

Well Name and Number

Chalmers 5300 21-19 8T

Footages	Qtr-Qtr	Section	Township	Range
2226 F N L	327	LOT2	153 N	100 W

Field	Pool
Baker	BAKKEN

County
McKenzie

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address	City	State	Zip Code
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DETAILS OF WORK

Effective immediately, we request **CONFIDENTIAL STATUS** for the above referenced well.

This well has not been completed.

OFF CONFIDENTIAL 11/01/15.

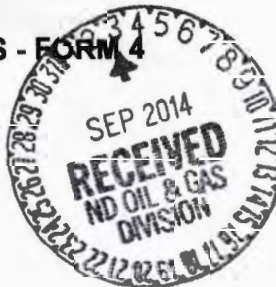
Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436
Address 1001 Fannin, Suite 1500		
City Houston	State TX	Zip Code 77002
Signature 	Printed Name Jennifer Swenson	
Title Regulatory Specialist	Date April 30, 2015	
Email Address jswenson@oasispetroleum.com		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 5/07/15	
By 	
Title Engineering Technician	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28636

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date September 2, 2014	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other Physical Address	

Well Name and Number Chalmers 5300 21-19 8T					
Footages	Qtr-Qtr	Section	Township	Range	
2226 F N L 327 F W L	LOT2	19	153 N	100 W	
Field BAKER	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully submits the physical address for the above referenced well:

**13762 45th Street NW
Alexander, ND 58831**

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Regulatory Assistant	Date September 2, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 3/12/2015	
By <i>Matthew Mason</i>	
Title ENGINEERING TECHNICIAN	



Oasis Petroleum North America, LLC

Chalmers 5300 21-19 8T

2,226' FNL & 327' FWL

Lot 2 Sec. 19, 153N, 100W

Baker / Three Forks

McKenzie County, North Dakota

BOTTOM HOLE LOCATION:

173.80' N & 9,924.83' E of surface location or approx.

2,052.20' FNL & 237.32' FEL SE NE Sec. 20, T153N, R100W

Prepared for:

Nathan Gabelman
Oasis Petroleum North America, LLC
1001 Fannin Suite 1500
Houston, TX 77002

Prepared by:

G. Wayne Peterson, Michelle Baker,
Zachary Moses, Molly Hagstrom
PO Box 80507; Billings, MT 59108
(406) 259-4124
geology@sunburstconsulting.com
www.sunburstconsulting.com

WELL EVALUATION



**Figure 1. Nabors B22 drilling the Oasis Petroleum North America, LLC - Chalmers 5300 21-19 8T during October-December, 2014 in Baker Field, McKenzie County, North Dakota.
(G. Wayne Peterson, Sunburst Consulting)**

INTRODUCTION

The **Oasis Petroleum North America, LLC Chalmers 5300 21-19 8T** [Lot 2 Section 19, T153N, R100W] is located approximately 7 miles south of the town of Williston in McKenzie County, North Dakota. The Chalmers 5300 21-19 8T is a horizontal Three Forks well within the Williston Basin consisting of one 9,358' lateral drilled toward the east. The vertical hole was planned to be drilled to approximately 10,341'. The curve would be built at 12 degrees per 100' to land within the Three Forks. This well is a two section lateral which originates in the northwest quarter of section 19, then drilled east to the northeast quarter of section 20. Directional drilling technologies and geo-steering techniques were used to land in the Three Forks reservoir and maintain exposure to the ideal target rock.

OFFSET WELLS

Offset well data used for depth correlation during curve operations are found in the 'Control Data' section appended to this report. Offset well control was essential in curve operations, to successfully land within the Three Forks. Formation thicknesses expressed by gamma ray signatures in these wells were compared to gamma data collected during drilling operations in order to successfully land the curve. The target landing true vertical depth (TVD) was periodically updated during drilling to ensure accurate landing of the curve.

GEOLOGY

The Charles Formation [Mississippian Madison Group] was logged 8,534' MD 8,533' TVD (-6,457' SS). Samples in the lower portion of the Charles Formation consisted of a limestone mudstone, which was light brown, light gray brown, off white in color. It was microcrystalline, friable, laminated, with an earthy texture. A trace of intercrystalline porosity, as was *rare spotty light brown oil stain*. Occasionally noted was a dolomite mudstone, which was light brown, light gray brown in color. It was microcrystalline, friable-firm, laminated, with an earthy texture. Also noted was a trace of intercrystalline porosity, and *occasional spotty light brown oil stain*. Rarely noted was anhydrite, which was off white, cream in color. It was soft, microcrystalline, and massive with an earthy to amorphous texture. Following connections or periods of non-circulation, gas peaks of 68 to 113 units were noted, as were drilling gas shows of 82 to 138 units.

The Mission Canyon Formation [Mississippian Madison Group] was logged 9,431' MD 9,430' TVD (-7,354' SS). The Mission Canyon Formation consisted of a lime mudstone that was described as light gray, light brown, gray brown, trace dark gray in color. The lime mudstone was predominately friable to firm, with an earthy to rarely crystalline in texture. Some intervals contained a trace of black-brown algal material, a trace of fossil fragments, and traces of disseminated pyrite. Also present was an argillaceous lime mudstone that was described as light gray, occasional medium gray, rare gray tan, rare off white, trace dark gray in color. The argillaceous lime mudstone was predominately firm to friable, crystalline to chalky texture. Some intervals contained a trace of disseminated pyrite. Following connections or periods of non-circulation, gas peaks of 101 to 114 units were noted, as were drilling gas shows of 143 to 249 units. Rare intercrystalline porosity was noted as well as traces to occasional *spotty light brown oil stain* was occasionally observed while logging the Mission Canyon Formation.



Figure 2. Limestone with spotty light to medium brown staining from the Mission Canyon Formation.

The Upper Bakken Shale [Mississippian-Bakken Formation] was drilled at 10,782' MD 10,723' TVD (-8,647' SS). Entry into this member was characterized by high gamma, elevated background gas and increased rates of penetration. The black to black gray carbonaceous and *petroliferous* shale was hard with a splintery to smooth texture. Fracture porosity was noted, and trace minerals were observed to include disseminated pyrite and calcite fracture fill. Hydrocarbons evaluated in this interval reached a maximum of 334 units.

The Middle Bakken Member [Mississippian-Devonian Bakken Formation] was drilled at 10,808' MD 10,739' TVD (-8,663' SS). Samples in the Middle Bakken Member were predominantly silty sandstone which was described as light gray brown, light brown, trace light gray in color. It was very fine grained, friable, subround, smooth, moderately sorted, with calcite cement, moderately cemented. A trace of disseminated and nodular pyrite was noted as was fair intergranular porosity. Also noted was *common light to medium brown spotty to even oil stain*. Hydrocarbons evaluated in this interval reached a maximum of 161 units drilling gas, with a connection gas of 206 units.

The Lower Bakken Shale [Devonian-Bakken Formation] was drilled at 10,892' MD 10,784' TVD (-8,708' SS). Entry into this interval was characterized by high gamma, elevated background gas and increased rates of penetration. The carbonaceous black, black gray shale is *petroliferous*, hard, splintery, smooth and exhibits possible fracture porosity. Trace minerals included disseminated pyrite. Drilling gas in this interval reached a maximum of 159 units.

The Pronghorn Member [Devonian-Bakken Formation] was reached at 10,918' MD 10,795' TVD (-8,719' SS). Entry into this interval was characterized by lower gamma than the Upper Bakken Shale, and slightly slower penetration rates. Samples from the Pronghorn were described as siltstone which was dark gray trace gray black, friable to firm, subblocky to subsplit. This siltstone was moderately dolomite cemented and included disseminated and nodular pyrite. Also noted was a trace of *spotty light brown oil stain*. Drilling gas in this interval reached a maximum of 215 units with a survey gas of 226 units.



Figure 3. Black carbonaceous and petroliferous shale from the Lower Bakken Shale of the Bakken Formation and gray siltstone from the underlying Pronghorn Member.

The Three Forks Formation [Devonian] was reached at 10,973' MD 10,813' TVD (-8,737' SS) which was 17' low to the Oasis Petroleum NA LLC Chalmers 5300 31-19H. The target zone of

the Three Forks was to be drilled in a predominately 10 foot zone beginning 17 feet into the Three Forks.

Samples in the Three Forks were predominantly dolomite which was described as light brown-tan, occasional cream, trace light gray brown in color. It was very fine crystalline, firm, laminated, with a microsucrosic texture. Occasional disseminated pyrite was noted as was occasional intercrystalline porosity. Also noted was *occasional spotty to trace even light brown oil stain*. Also observed was light green-light gray green, light gray blue shale that was firm, subblocky, with an earthy texture. Occasional disseminated pyrite was noted as was possible intergranular porosity.



Figures 4, 5, & 6. A predominately dolomitic sample in the middle of the preferred drilling zone of the Three Forks (left); a predominately dolomitic sample high in the preferred drilling zone of the Three Forks (middle); and sample of the underlying claystone (right).

Gas Show

Gas monitoring and fluid gains provided evidence of a hydrocarbon saturated reservoir during the drilling of the Chalmers 5300 21-19 8T. Oil and gas shows at the shakers and in samples were continuously monitored. In the closed mud system, hydrostatic conditions were maintained near balance, this allowed for gas and fluid gains from the well to be evaluated. Gas on the Chalmers 5300 21-19 8T varied according to stratigraphic position and penetration rates which may have reflected increased porosity. During the vertical, connection gas peaks of 68 to 114 units were noted, as were drilling gas shows of 82 to 249 units, against a 9.75-10.7 lb/gal diesel-invert mud weight. Background concentrations in the lateral ranged from 161 to 466 units, against a 9.5 lb/gal saltwater gel drilling fluid. Connection peaks of 1,119 to 1,207 units were observed, coinciding with the best shows. Drilling out of casing at 11,185 MD' yielded a trip gas of 2,232 units. The lateral was completed with one BHA, consequently no other trip gasses were noted. Chromatography of gas revealed typical concentrations of methane, characteristic of Three Forks gas.

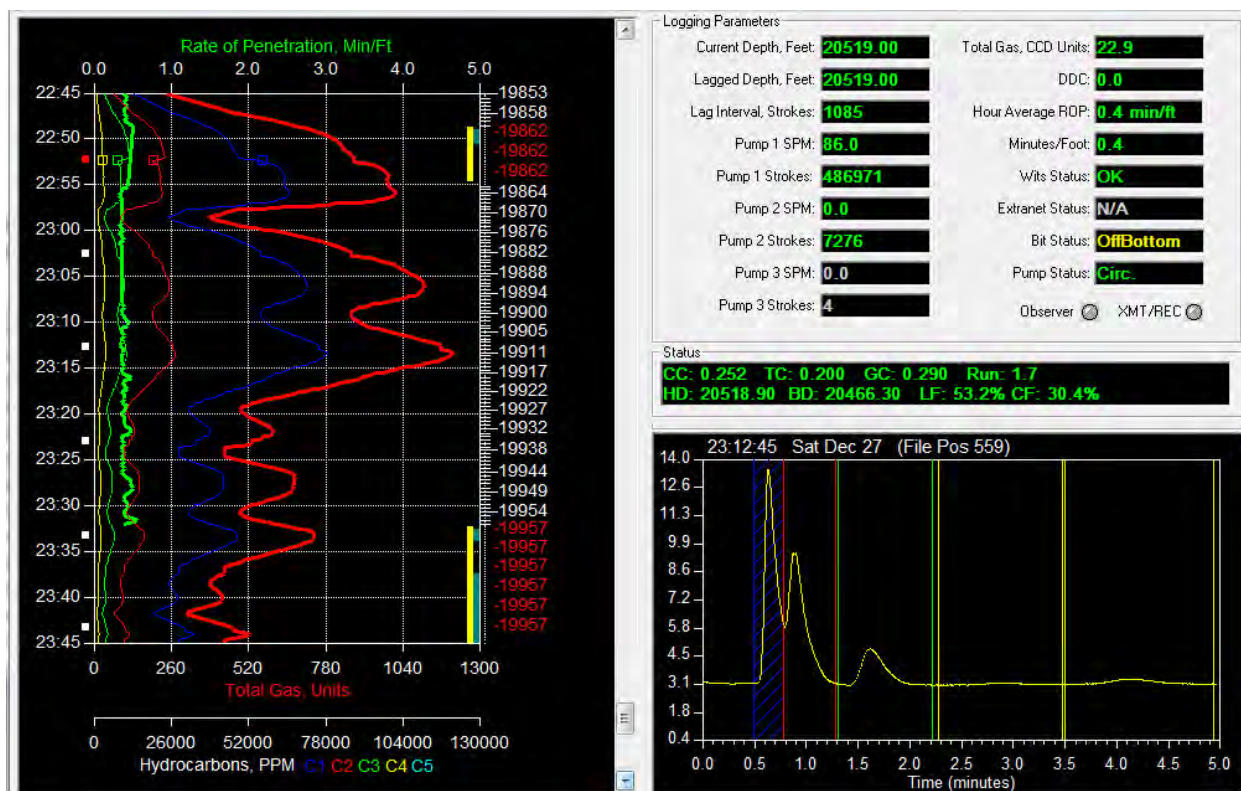


Figure 7. Gas chromatography of a 1,200 unit gas show.

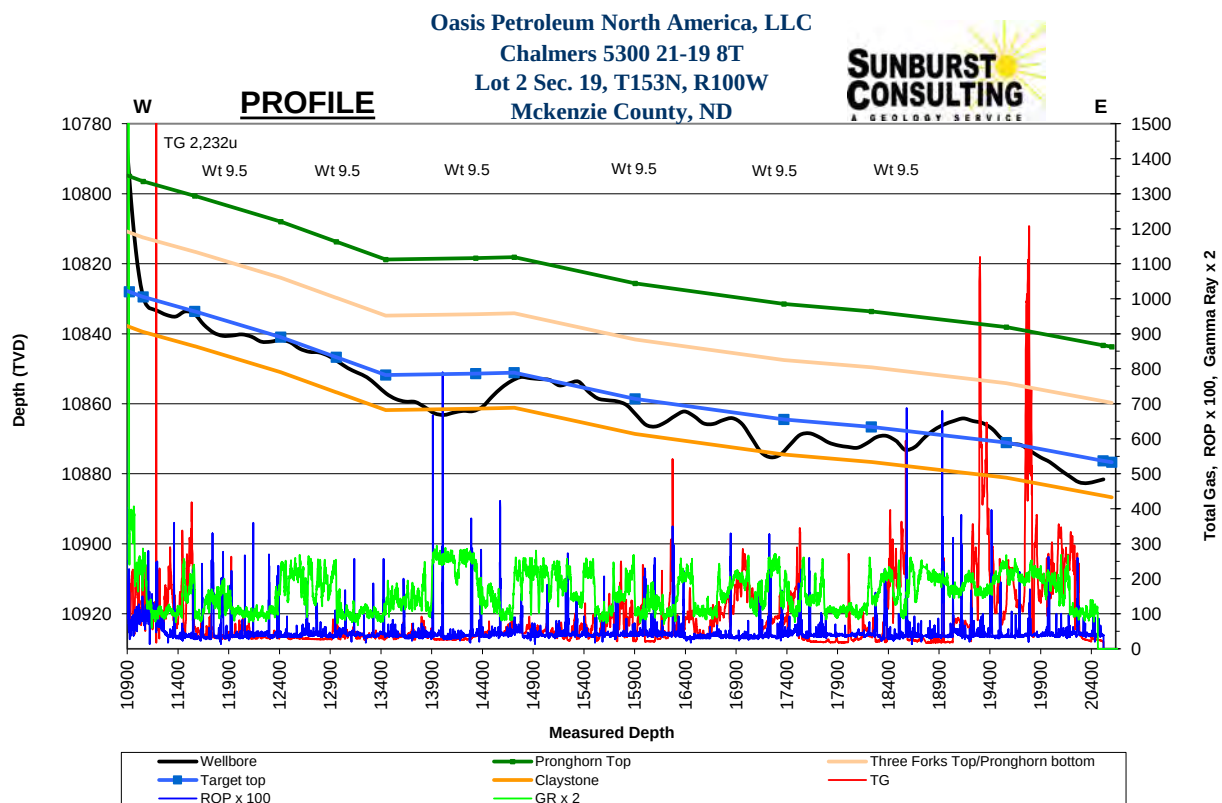


Figure 8. Profile displaying total gas, gamma ray and rate of penetration.

Geosteering

Ryan Energy Technologies provided personnel and equipment for measurement-while-drilling (MWD) services. The RPM directional drillers and MWD, and Sunburst Consulting personnel worked closely together throughout the project to evaluate data and make steering decisions to maximize the amount of borehole in the targeted zones and increase rate of penetration (ROP) of the formation.

The 875' curve was drilled in 23.5 hours with a bottom hole assembly (BHA) consisting of bit #4, a Smith MDI516 PDC bit, attached to a 2.38 degree fixed NOV 7/8 5.0 motor and MWD tools. The curve was successfully landed at 11,185' MD and 10,833' TVD, approximately 20' into the Three Forks Formation. Seven inch diameter 32# HCP-110 casing was set to 11,161' MD.

Geologic structure maps of the Chalmers 5300 21-19 8T and surrounding control wells had estimated formation dip to be a down dip at approximately -0.5° down for the length of the lateral. The preferred drilling interval of the Chalmers 5300 21-19 8T consisted of a ten foot zone located approximately seventeen feet into the Three Forks Formation. Penetration rates, gas shows, gamma ray data, and sample observations were utilized to keep the wellbore in the preferred stratigraphic position in the target zone. Using offset well data provided by Oasis representatives, projected porosity zones were identified in the preferred drilling areas.

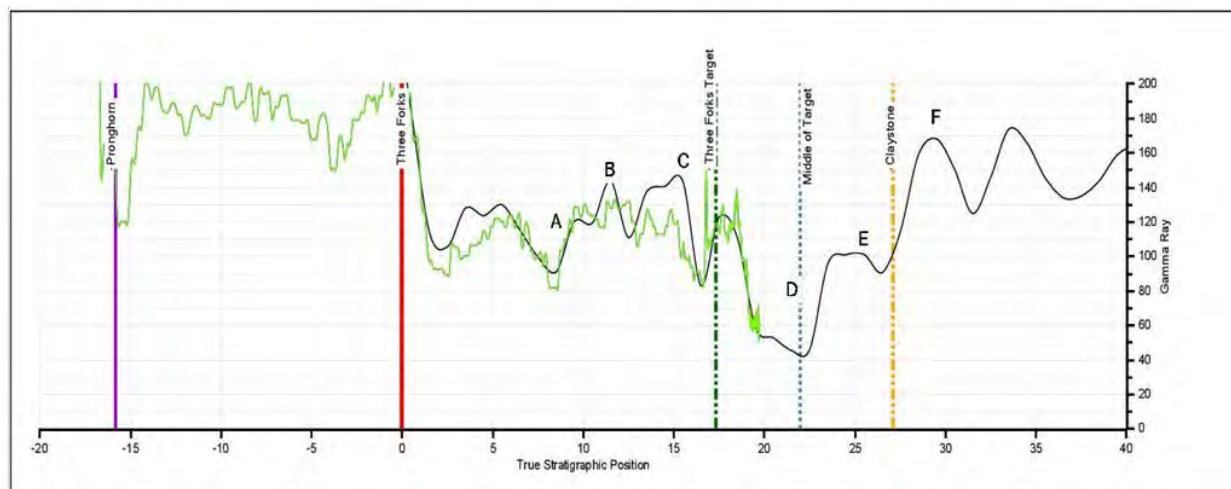


Figure 9. Offset well target definition, Indian Hills Prospect (Oasis).

Steering decisions were made by using target points and letter markers provided by Oasis personnel. The low gamma (D) in the middle of the target zone was often utilized to establish the well-bore's position in the target zone. If the well-bore moved toward the bottom of the target zone, the moderate gamma of marker (E) was observed. As the well-bore moved lower in formation, the higher gamma of the underlying claystone (F) was observed, as was the presence of claystone in collected samples. Slides were then utilized to move the well-bore back up into the target zone. In accordance with the drilling plan, the well-bore was steered down into the underlying claystone at prescribed intervals in order to definitively establish the well-bore's position in formation. As the well-bore moved higher, approaching the top of the target zone the

high to medium fluctuating gamma between markers (C & D) was noted. Later in the lateral the well-bore moved up in formation, and the marker (B) was observed, before the directional staff was able to steer the well-bore back down into the target zone. The TD of 20,519' MD was achieved at 04:50 hours CDT December 28, 2014. The well site team worked together to maintain the well bore in the desired target interval for 87% of the lateral, opening 9,358' of potentially productive reservoir rock. The hole was then circulated and reamed for completion.

SUMMARY

The Chalmers 5300 21-19 8T is a successful well in Oasis Petroleum's horizontal Three Forks development program in Baker Field. The project was drilled from surface casing to TD in 20 days. The TD of 20,519' MD was achieved at 04:50 hours CDT December 28, 2014. The well site team worked together to maintain the well bore in the desired target interval for 87% of the lateral, opening 9,358' of potentially productive reservoir rock.

Samples in the Three Forks were predominantly dolomite which was described as light brown-tan, occasional cream, trace light gray brown in color. It was very fine crystalline, firm, laminated, with a microsucrosic texture. Occasional disseminated pyrite was noted as was occasional intercrystalline porosity. Also noted was *occasional spotty to trace even light brown oil stain*. Also observed was light green-light gray green, light gray blue shale that was firm, subblocky, with an earthy texture. Occasional disseminated pyrite was noted as was possible intergranular porosity.

Gas on the Chalmers 5300 21-19 8T varied according to stratigraphic position and penetration rates which may have reflected increased porosity. The overall gas and hydrocarbon shows were encouraging and indicate a hydrocarbon rich system in the Three Forks.

The Oasis Petroleum North America, LLC. Chalmers 5300 21-19 8T awaits completion operations to determine its ultimate production potential.

Respectfully submitted,

G. Wayne Peterson

Sunburst Consulting, Inc.

29 December, 2014

WELL DATA SUMMARY

<u>OPERATOR:</u>	Oasis Petroleum North America, LLC
<u>ADDRESS:</u>	1001 Fannin Suite 1500 Houston, TX 77002
<u>WELL NAME:</u>	Chalmers 5300 21-19 8T
<u>API #:</u>	33-053-06021
<u>WELL FILE #:</u>	28636
<u>SURFACE LOCATION:</u>	2,226' FNL & 327' FWL Lot 2 Sec. 19, 153N, 100W
<u>FIELD/ PROSPECT:</u>	Baker / Three Forks
<u>COUNTY, STATE</u>	McKenzie County, North Dakota
<u>BASIN:</u>	Williston
<u>WELL TYPE:</u>	Three Forks Member Horizontal Lateral
<u>ELEVATION:</u>	GL: 2,051' KB: 2,076'
<u>SPUD/ RE-ENTRY DATE:</u>	October 5, 2014
<u>BOTTOM HOLE LOCATION</u>	173.80' N & 9,924.83' E of surface location or approx. 2,052.20' FNL & 237.32' FEL SE NE Sec. 20, T153N, R100W
<u>CLOSURE COORDINATE</u>	Closure Direction: 89.00° Closure Distance: 9,926.35'
<u>TOTAL DEPTH / DATE:</u>	20,519' on December 28, 2014 87% within target interval
<u>TOTAL DRILLING DAYS:</u>	20 days
<u>CONTRACTOR:</u>	Nabors #B22
<u>PUMPS:</u>	H&H Triplex (stroke length - 12")

<u>TOOLPUSHERS:</u>	Jessie Tibbets, Mark Rollins
<u>FIELD SUPERVISORS:</u>	John Gordon, Doug Rakstad
<u>CHEMICAL COMPANY:</u>	NOV
<u>MUD ENGINEER:</u>	Joe Vaith, Joe Stander, Ken Rockeman
<u>MUD TYPE:</u>	Fresh water in surface hole Diesel invert in vertical/curve; Salt water in lateral
<u>MUD LOSSES:</u>	Invert Mud: 500 bbls, Salt Water: 0 bbls
<u>PROSPECT GEOLOGIST:</u>	Nathan Gabelman
<u>WELLSITE GEOLOGISTS:</u>	G. Wayne Peterson, Michelle Baker, Zachary Moses, Molly Hagstrom
<u>GEOSTEERING SYSTEM:</u>	Sunburst Digital Wellsite Geological System
<u>ROCK SAMPLING:</u>	100' from 4,700' - 8,200' 30' from 8,200' -20,519' (TD)
<u>SAMPLE EXAMINATION:</u>	Binocular microscope & fluoroscope
<u>SAMPLE CUTS:</u>	Trichloroethylene (Carbo-Sol)
<u>GAS DETECTION:</u>	MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph Serial Number(s): ML-134
<u>ELECTRIC LOGS:</u>	n/a
<u>DRILL STEM TESTS:</u>	n/a
<u>DIRECTIONAL DRILLERS:</u>	RPM, Inc. John Gordon, Doug Rakstad, Robert Jervis
<u>MWD:</u>	Ryan Mike McCammond, Ronald Maddalena, Rebekah Hungerford

CASING:

Surface: 13 3/8" 54# J-55 set to 2,175'

Second: 9 5/8" 40# HCL-80 set to 6,162'

Intermediate: 7" 32# P-110 set to 11,161'

KEY OFFSET WELLS:

Oasis Petroleum North America, LLC

Chalmers 5300 31-19H

NW SW Sec. 19 T153N R100W

McKenzie County, ND

Texas Gas Exploration Company

Lindvig 1-35

SE SE Sec. 35 T153N R101W

McKenzie County, ND

Mosbacher Production Company

Verlin Fossum ET AL No. 26-1

NW SE Sec. 26, T153N R101W

McKenzie County, ND

Oasis Petroleum North America, LLC

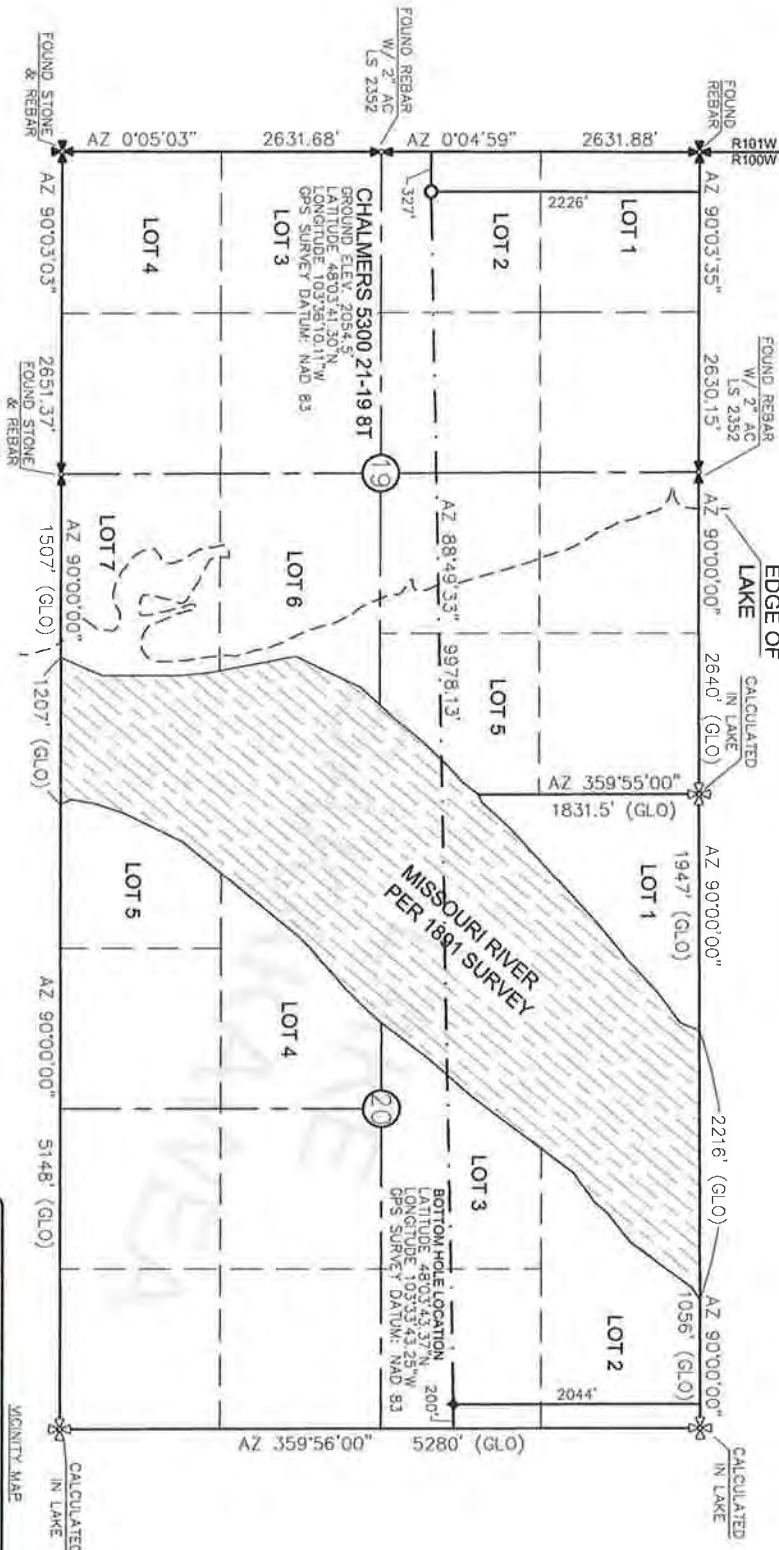
Chalmers 5301 44-24 4T2R

SE SE Sec. 24 T153N R101W

McKenzie County, ND

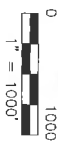
WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
CHALMERS 5300 21-19 8T
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA
2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
EDGE OF LAKE



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, P.L.S. REGISTRATION NUMBER 3880 ON 5/07/14. THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

MONUMENT - RECOVERED
MONUMENT - NOT RECOVERED



STAKED ON 1/29/14
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'
THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY
THAT THIS PLAT CORRECTLY REPRESENTS WORK
PERFORMED BY ME OR UNDER MY SUPERVISION
AND IS TRUE AND CORRECT TO THE BEST OF MY
KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880



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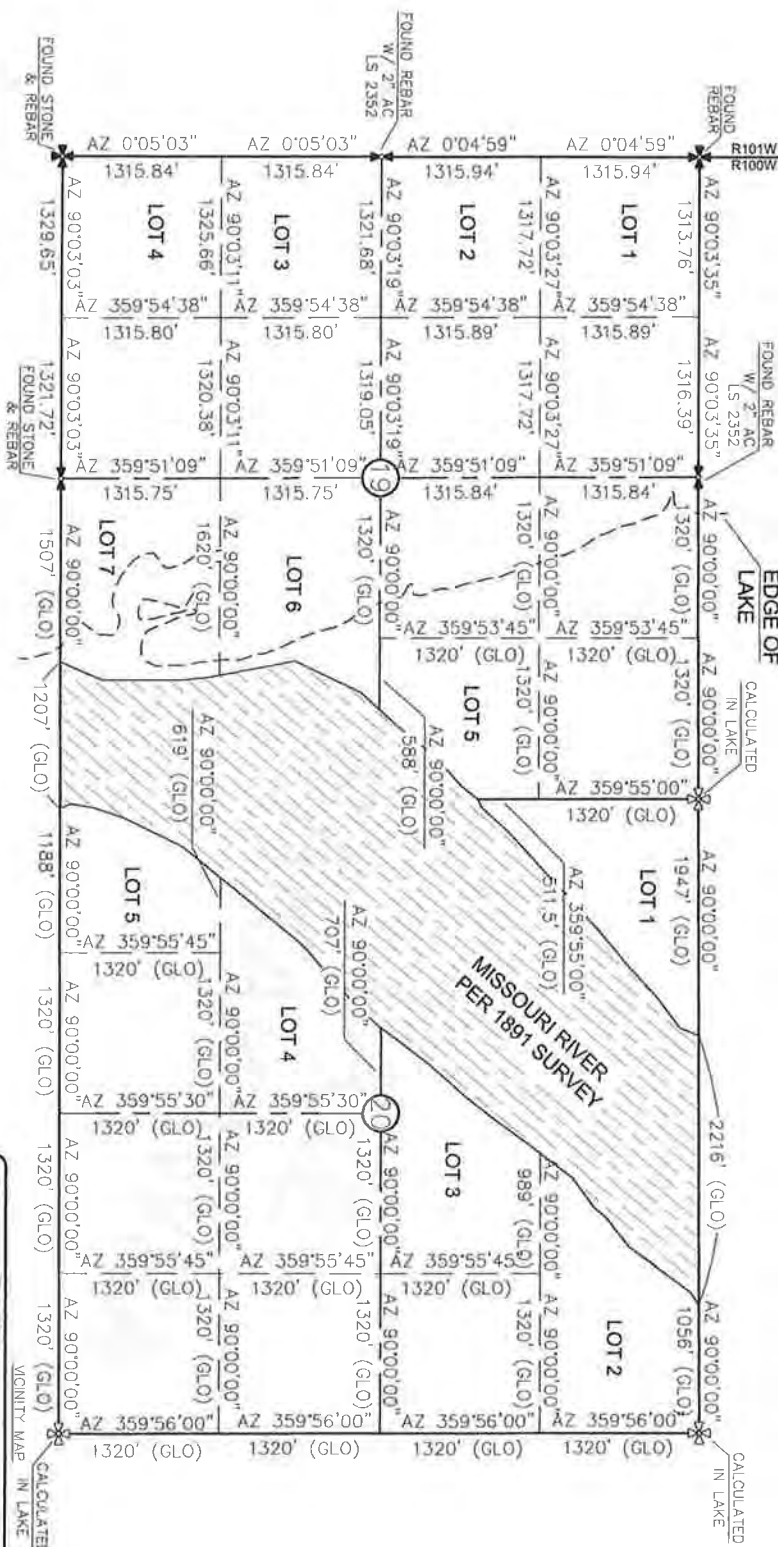
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Skiway, Montana 59270
Ph (406) 433-5817
Fax (406) 433-5618
www.interstateeng.com
Other offices in Salt Lake, North Dakota and San Francisco

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 19, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No: 613-00-202-03
Checked By: D.D.K. Date: JAN 2014

Revision No.	Date	By	Description
REV 1	3/12/14	AM	PG&A WELLS ON PAD
REV 2	4/22/14	AM	WELLS ON PAD/REVISED PAD
REV 3	5/7/14	AM	WELLS ON PAD/REVISED PAD

SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 CHALMERS 5300 21-19 81"
 2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
 SECTIONS 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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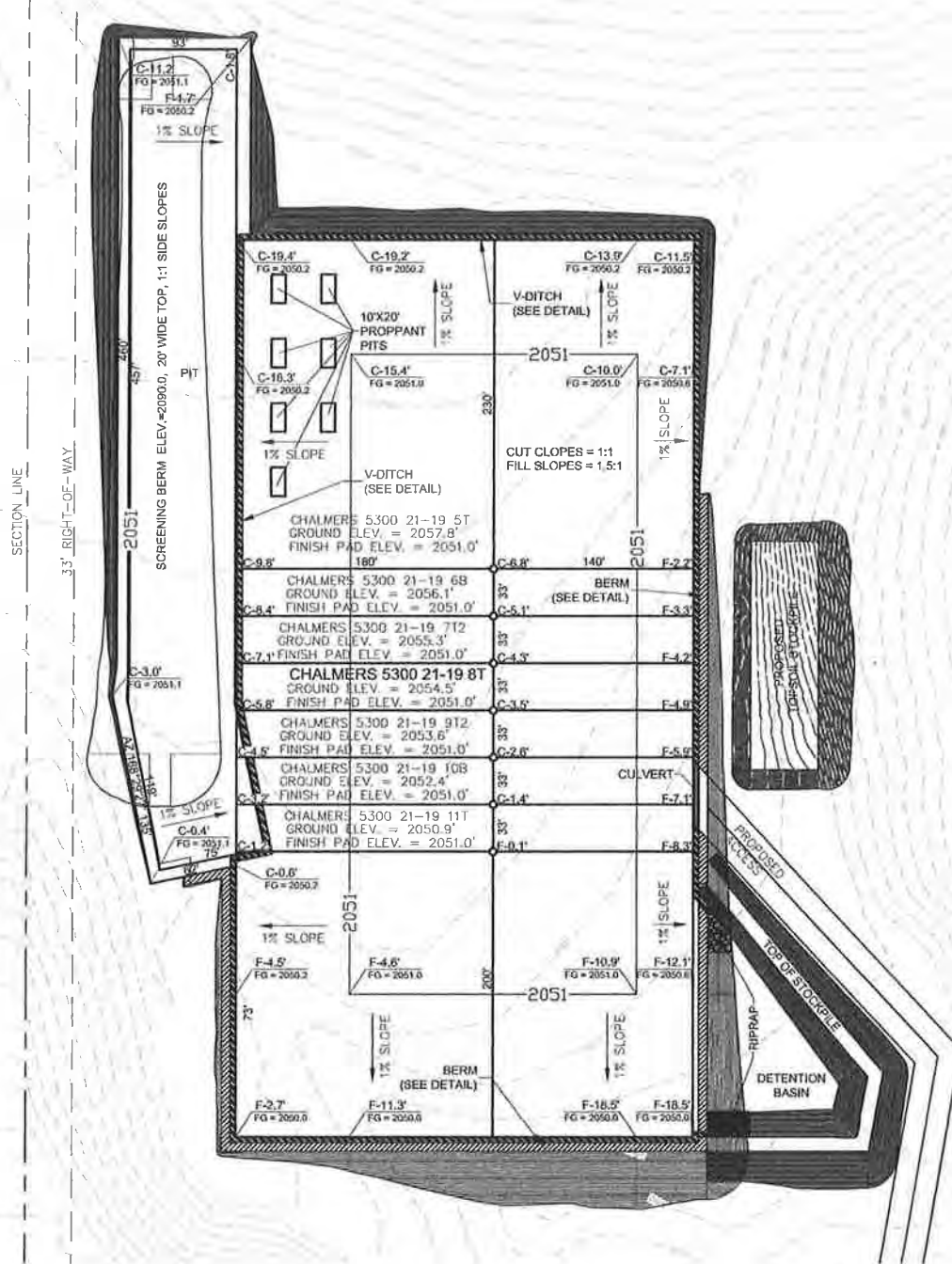
ALL AZIMUTHS ARE BASED ON GPS OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY -0°03'.



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2/8 SHEET NO.		Interstate Engineering, Inc. P.O. Box 648 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5618 www.interstateeng.com Office in Billings, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTIONS 19 & 20, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA		Revision REV. 1 1/18/14 US MOVED WELLS ON PAD REV. 2 4/28/14 NEW MOVED WELLS ON PAD/REMOVED PAD REV. 3 5/2/14 NEW MOVED WELLS ON PAD/REMOVED PAD
			Drawn By: B.H.H. Checked By: D.D.S.	Project No.: 515-06-262.03 Date: JAN 2014	Scale: 1" = 1000'
			Project Name: OASIS PETROLEUM NORTH AMERICA, LLC Project Location: SECTIONS 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA		
			Project Description: SECTION BREAKDOWN		

PAD LAYOUT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5300 21-19 8T"
 2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
 SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE 1: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE 2: Screening berm is to be built after drilling operations are complete.

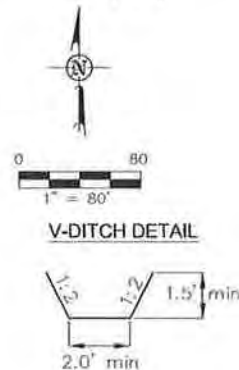
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— Proposed Contours
 - - - Original Contours
 [Hatched Box] BERM
 [Cross-hatched Box] DITCH

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 PAD LAYOUT
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: D.J.H. Project No.: 513-09-282.03
 Checked By: D.D.K. Date: JAN, 2014

Revision	Date	By	Description
REV 1	3/12/14	LS	MOVED WELLS ON PAD
REV 2	4/22/14	LS	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/22/14	LS	MOVED WELLS ON PAD/REVISED PAD

2076

OASIS PETROLEUM NORTH AMERICA, LLC
 CHALMERS 5300 21-19 8T
 2226' FNL/327' FWL
 QUAD LOCATION MAP
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

CHALMERS 5300
 21-19 8T

PROPOSED
 ACCESS ROAD
 OPTION B

CHALMERS
 5300 31-19H

PROPOSED
 ACCESS ROAD
 OPTION B

Indian Creek Bay
 ition Area

PROPOSED
 ACCESS ROAD
 OPTION C

WADE 5300 21-30H

EX. LEASE ROAD

EX. LEASE ROAD

M I S S O U R I



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5/8
 SHEET NO.



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 www.interstateeng.com
 Other Offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-282.03
 Checked By: D.D.K. Date: JAN. 2014

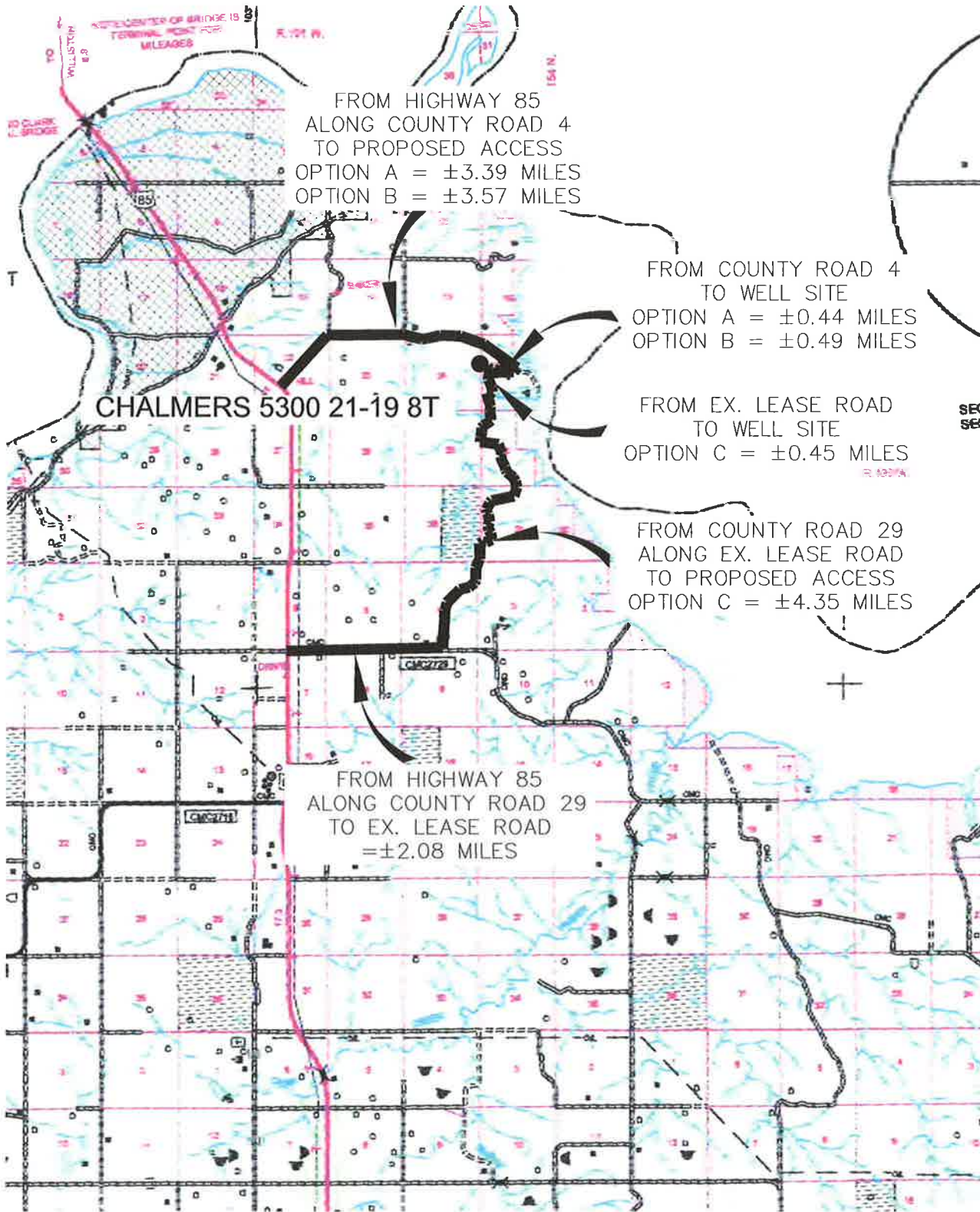
Revision No.	Date	By	Description
REV 1	3/13/14	JUS	MOVED WELLS ON PAD
REV 2	4/22/14	SMH	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/2/14	BBH	MOVED WELLS ON PAD/REVISED PAD

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5300 21-19 8T"

2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

6/8
SHEET NO.



Interstate Engineering, Inc.
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Ph (406) 433-5617
Fax (406) 433-5618
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Other offices in Minnesota, North Dakota and South Dakota

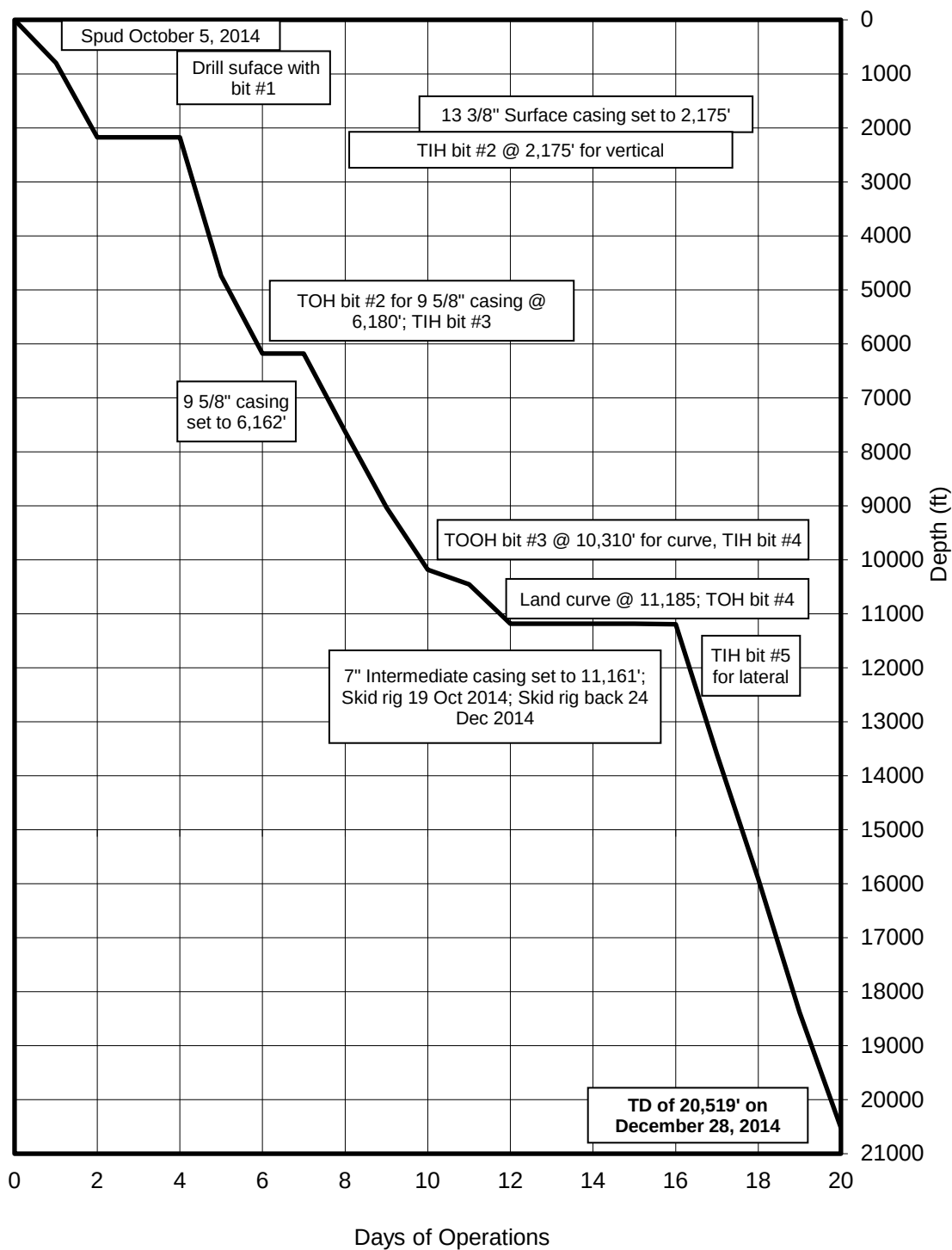
OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 19, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-06-282.03
Checked By: D.D.K. Date: JAN. 2014

Revision No.	Date	By	Description
REV 1	3/22/14	JAS	MOVED WELLS ON PAD
REV 2	4/22/14	SHH	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/2/14	SHH	MOVED WELLS ON PAD/REVISED PAD

TIME VS DEPTH

Oasis Petroleum North America, LLC
Chalmers 5300 21-19 8T



DAILY DRILLING SUMMARY

Day	Date 2014	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
0	10/5	0		-	-	-	-	-	-	-	-	-	Rig released from Chalmers 7T2 @ 13:00 on 10/5/2014. Rig up. Set mats for new hole. Skid rig. Rig up clim assist. Catwalk. Cover old well. Derrick escape.	Surface
1	10/6	800'	800	1	5	50	-	-	1,800	100	100	704	Rig up ground substructure, center stairs. Pre spud checklist. Pre spud safety meeting. Clean rig floor, take bail extensions off, casing elevators, lay down old bit and get ready for BHA, directional work. Rig up MWD main cable. Pick up BHA. Service rig. Rotary drilling 0-800'.	Surface
2	10/7	2,175'	1,375	1	10	50	-	-	3,650	100	100	704	Rotary drilling 800'-2,175'. Circulate and condition. TOH. TIH. Circulate and condition, circulate for 1 hour. TOH Service rig. Lay down BHA, MWD. Bit, motor, UBHO. Rig up to run casing. Rig up casing crew. Hold safety meeting. Run 13 3/8" casing.	Surface
3	10/8	2,175'	0	-	-	-	-	-	-	-	-	-	Run casing 13 3/8". Wash down last joint. Circulate and condition. Held safety meeting. Rig up cement head and pressure test head. Cement. Rig down cementers. Rig down cellar pumps. Install wellheads. Drain invert. Cut off conductor. Cut off casing. Weld on wellhead. Weatherford installed B section. Place BOP in sub. Remove mud cross. Stab BOP. Waiting on Weatherford to torque BOP to wellhead. Service rig. Nipple up BOPS choke line kookey lines, rotating head, mud cross.	Pierre
4	10/9	2,175'	0	-	-	-	-	-	-	-	-	-	Nipple up BOPS. Hook up kookey lines, remove rotating head and spool. Finish choke line. Round up crossovers for test plug. Clean mud tanks. Test BOPS. Rig up flare line. Mofify spool for rotating head with welder. Install rotating head, flow line. Fill line, center stack. Service rig. Install wear bushing. Pick up BHA, bit, motor, 2 monels, UBHO, MWD. TIH.	Pierre
5	10/10	4,741'	2,566	2	16	30	-	85	3500	92	93	651	TIH, and pick up 12 joints of heavy wt. Drilling cement from 2,142'-2,190'. Cement at 2,142', float at 2,146, shoe at 2,175'. Fit test at 2,190', 227 psi for 30 min. drilling and survey from 2,190'-3,061'. Drilling and survey from 3,061'-4,461'. Service rig- grease wash pipe. drilling and survey from 4,461'-4,741'	Greenhorn
6	10/11	6,180'	1,439	2	16	30	-	85	3500	92	93	651	Drill and survey, sliding when needed from 4,741'-5,581'. Service rig. Drill and survey, sliding when needed from 5,581'-6,150'. Circulate and condition- pump bottoms up. Short tip 11 stands. Build volume- weight- circulate bottoms up, pump dry job. TOH. TIH. Drill and survey, sliding when needed from 6,150'-6,180'. Circulate and condition due to lost circulation- circulate bottoms up, pump dry job. TOH. Remove rotating head and install trip nipple. Lay down BHA, bit, motor, two monels, UBHO, MWD.	Swift

DAILY DRILLING SUMMARY

Day	Date 2014	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
7	10/12	6,180'	0	-	-	-	-	-	-	-	-	-	Lay down BHA. Pull wear bushing. Rig up/down to run casing- held safety meeting. Run casing- 9 5/8. Circulate and condition- wait on pub joint. Run casing- make up pub joint and land casing. Rig up/down to run casing- rig down casing crew/ verify landed with weatherford. Circulate and condition. Circulate while waiting on rig crew. Rig up cementers. Hold safety meeting with cementers. Pump cement. rig down cement crew. Lay down landing joint. Take off bell extensions, elevators, install 5" elevators. Install pack off, install wear bushing.	Swift
8	10/13	7,631'	1,451	3	25	60	17	132	3300	78	78	549	Install wear bushing with weatherford. Cut drilling line. Pick up BHA. TIH. Test casing 1500 psi for 30 min. Drilling cement from 6,110'-6,190'; cement at 6,110', float at 6,112', shoe at 6,158'. Fit test 11.5 PPG, 345 psi at 6,190' for 30 min. Drill and survey, sliding when needed from 6,190'-6,697'. SPR #1- 324 psi at 40 strokes at 6,324', function pipes- 5 sec, blinds 6 sec. Rotary drilling sliding as needed 6,697'-7,444'. Service rig. Rotary drilling sliding as needed 7,444'-7,631'.	Opeche
9	10/14	9,031'	1,400	3	15	45	20	132	3500	78	78	549	Rotary drilling sliding as needed 7,631'-8,097'. Service rig. Rotary drilling sliding as needed 8,097'-8,377'. Rotary drilling sliding as needed 8,377'-8,844'. Service rig. Function pipe 6 sec. Rotary drilling sliding as needed 8,844'-9,031'.	Charles
10	10/15	10,182'	1,151	3	43	60	40	132	3700	78	78	549	Rotary drilling sliding as needed 9,031'-9,622'. Service rig. Rotary drilling sliding as needed 9,622'-9,964'. Service rig. Greased wash pipe, top drive, functioned pipes 6 seconds, slow pump rate 30 strokes @ 9,964' 310 PSI. Rotary drilling sliding as needed 9,964'-10,182'.	Lodgepole
11	10/16	10,453'	271	3/4	50	60	33	267	3900	73	73	514	Rotary drilling sliding as needed 10,182'-10,310'. Circulate and condition. TOH. Lay down BHA. Cased hole logs- rig up wire line truck. Held safety meeting, cased hole logs. Multi-shot surveys- rig down wireline crew. Pick up BHA, bit motor, 2 monels, 2 ponys, MWD, UBHO. TIH Test MWD. TIH. Remove trip nipple, install rotating head. TIH. Slide drilling rotating as needed 10,310'-10,453'. Slow pump 40 strokes @ 10,453'.	Lodgepole
12	10/17	11,185'	732	4	50	60	33	267	3900	73	73	514	Slide drilling build curve 10,453'-10,827'. Service rig. Slide drilling build curve 10,827'-11,185'. Circulate and condition. Circulate 30 min. Short trip. Service rig. Circulate and condition. Circulate bottoms up. TOH back reaming. TOH.	Three Forks
13	10/18	11,185'	0	-	-	-	-	-	-	-	-	-	TOH. Lay down BHA. Pull wear bushing. Rig up to run casing. Held safety meeting. Run 7" casing. Pick up CRT. Run casing. Wash last 8 stands down. Circulate cement displace. Circulate and work casing. Put on casing slips, trouble shoot. Rig down CRT. Safety meeting with cementers. Hold safety meeting. Circulate cement displace. Rig up cementers. Cement.	Three Forks

DAILY DRILLING SUMMARY

Day	Date 2014	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
14	10/19	11,185'	0	-	-	-	-	-	-	-	-	-	Cement. Nipple down BOPS. Set casing slips with BOP winches. Install wellhead cap well with Weatherford. Rig released @ 15:00 10/19/2014	Three Forks
15	12/23	11,185'	0	-	-	-	-	-	-	-	-	-	Rig accepted to Chalmers 5300 21-19 8T @ 03:00 12/23/2014	Three Forks
													Rig up floor for drilling. Stretch out counter weights and angel lift. Pre-spud checklist. Test BOPS. Install wear bushing. Cut drilling line. Service rig. Pick up BHA. Install BHA. TIH. Install rotating head. Adjust stack. Change slip dies. TIH, fill pipe. Level rig. Drilling cement, float @ 11,053', shoe @ 11,161'. Drilled 10' new hole to 11,195', fit test.	
16	12/24	11,195'	10	5	20	40	20	264	2400	94	-	330	Rotary drilling sliding as needed from 11,185'-12,283', service top drive.	Three Forks
17	12/25	13,609'	2,414	5	21	40	50	265	3950	86	0	331	Rotary drilling sliding as needed 12,283'-13,609'	Three Forks
18	12/26	15,915'	2,306	5	25	40	51	242	3900	86	0	303	Rotary drilling sliding as needed from 13,609' to 14,746', service rig.	Three Forks
19	12/27	18,380'	2,465	5	25	40	54	226	4000	80	0	282	Rotary drilling sliding as needed from 14,746' to 15,915'	Three Forks
													Rotary drilling sliding as needed from 15,915'-17,147', service top drive, rotary drilling sliding as needed from 17,147'-18,380'	Three Forks
20	12/28	20,519'	2,139	-	25	40	50	209	3950	74	0	261	Rotary drilling sliding as needed from 18,380'-19,370' Service rig. Rotary drilling sliding as needed from 19,370'-20519' TD Lateral. Circulate and condition.	Three Forks

DAILY MUD SUMMARY

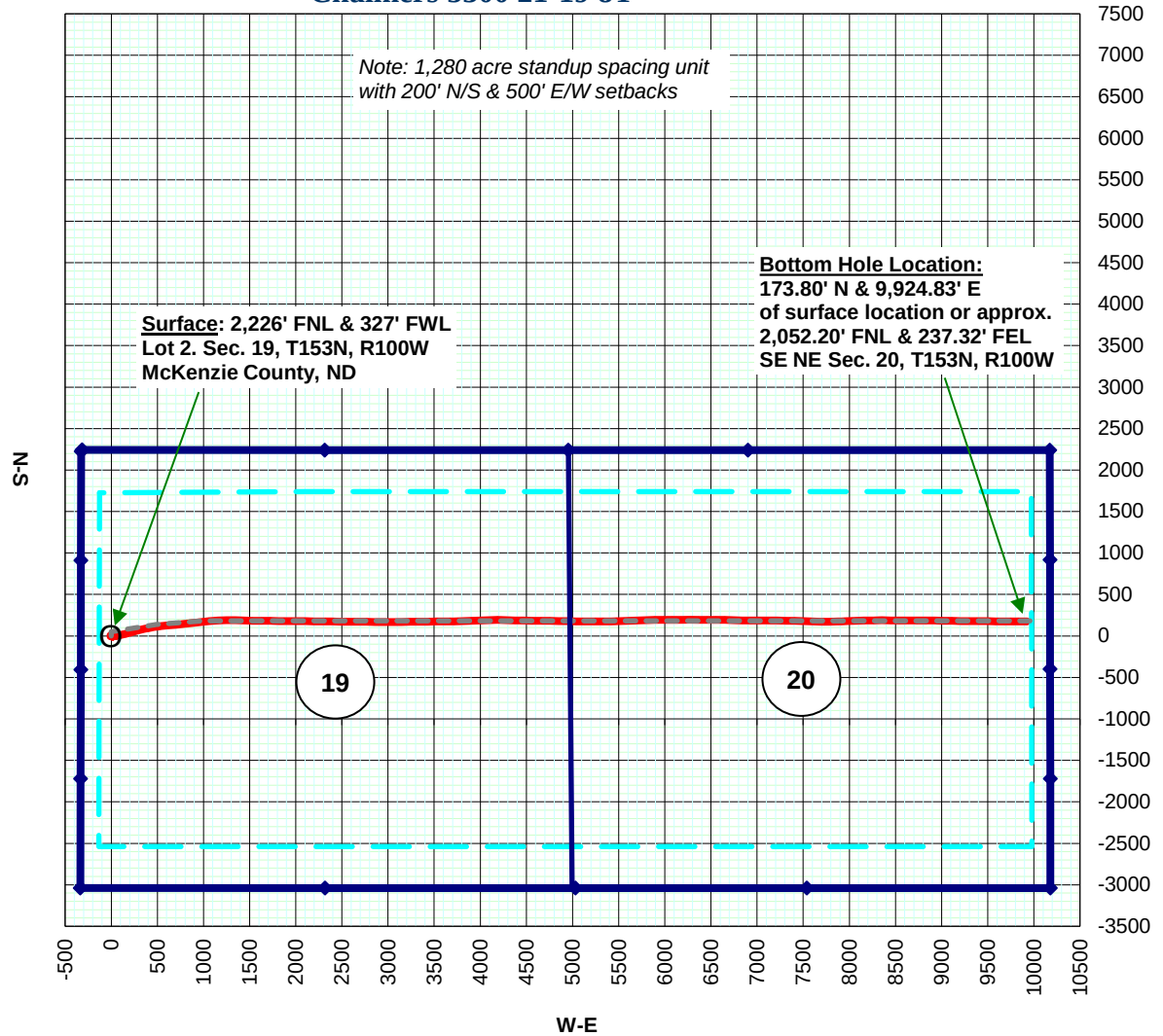
Day	Date 2014	Mud Depth	Drilling Fluid	Mud WT (ppg)	VIS (sec/qt)	PV (cP)	YP (lbs/100 ft2)	Gels (lbs/100 ft ²)	600/300	NAP/ H ₂ O (ratio)	NAP/ H2O (% by vol)	Cake (API/HTHP)	Cor. Solids (%)	Alk	pH	Excess Lime (lb/bbl)	Cl ⁻ (mg/L)	HGS/ LGS (%)	Salinity (ppm)	Electrical Stability	Gain/ Loss (bbls)
0	10/05	-	fresh water	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1	10/06	800'	fresh water	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	10/07	2,175'	fresh water	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	10/08	2,175'	invert	10.15	105	15	6	5/7/-	36/21	76.7/23.3	66/20	3	14	1.4	-	1.8	32k	7.1/5.0	264,320	461	-
4	10/09	2,175'	invert	10.45	57	11	5	4/5/-	27/16	82.4/17.6	64/22	2	13	1.8	-	2.3	35	8.6/4.4	264,320	371	-/83
5	10/10	4,741'	invert	10.45	57	11	5	4/5/-	27/16	82.4/17.6	64/22	2	13	1.8	-	2.3	35	8.6/4.4	264,320	371	-/-
6	10/11	6,180'	invert	10.75	50	17	12	7/11/-	46/29	81.2/18.8	65/15	2	18.8	1.7	-	2.2	20k	8.5/10.3	180,149	694	-/-
7	10/12	6,180'	invert	10.75	50	17	12	7/11/-	46/29	81.2/18.8	65/15	2	18.8	1.7	-	2.2	20k	8.5/10.3	180,149	694	-/183
8	10/13	6,456'	invert	10.25	55	19	8	8/9/-	46/27	79.8/20.2	67/17	2	14.8	2.1	-	2.7	20k	8.5/6.3	162,398	508	-/65
9	10/14	7,707'	invert	10.3	50	16	8	5/9/-	40/24	81.0/19.0	68/16	2	14.7	2.1	-	2.7	21k	8.6/6.1	177,834	508	-/70
10	10/15	9,064'	invert	9.75	47	13	7	5/8/-	33/20	81.6/18.4	71/169	3	11.3	2.3	-	3	28k	6.5/4.8	247,894	739	-/39
11	10/16	10,246'	invert	10.3	48	17	9	7/9/-	43/26	80.7/19.3	67/16	2	14.9	2.6	-	3.4	34k	9.1/5.9	364,320	704	-/244
12	10/17	10,671'	invert	10.25	48	17	10	8/12/-	44/27	82.6/17.4	69/14.5	2	14.6	2.6	-	3.4	33k	8.2/6.4	364,320	742	-/50
13	10/18	11,185'	invert	10.25	48	17	10	8/12/-	44/27	82.6/17.4	69/14.5	2	14.6	2.6	-	3.4	33k	8.2/6.4	364,320	742	-/-
14	10/19	11,185'	invert	10.25	48	17	10	8/12/-	44/27	82.6/17.4	69/14.5	2	14.6	2.6	-	3.4	33k	8.2/6.4	364,320	742	-/-
15	12/23	11,185'	invert	10.25	48	17	10	8/12/-	44/27	82.6/17.4	69/14.5	2	14.6	2.6	-	3.4	33k	8.2/6.4	364,320	742	-/-
16	12/24	11,195'	saltwater	9.5	27	2	1	-	5/3	-	1/93.5	-	8.5	-	9.5	-	139k	0.0/0.3	-	-	-/-
17	12/25	11,427'	saltwater	9.5	27	2	1	-	5/3	-	1/93.5	-	8.5	-	9.5	-	139k	0.0/0.3	-	-	-/-
18	12/26	15,915'	saltwater	9.5	27	2	1	-	5/3	-	0/91.8	-	8.2	-	9	-	131k	0.0/0.4	-	-	-/-
19	12/27	16,175'	saltwater	9.5	27	2	1	-	5/3	-	0/91.8	-	8.2	-	9	-	131k	0.0/0.4	-	-	-/-
20	12/28	20,519'	saltwater	9.5	27	2	1	-	5/3	-	0/91.8	-	8.2	-	9	-	131k	0.0/0.4	-	-	-/-

BOTTOM HOLE ASSEMBLY RECORD

BHA Run	Depth In	Depth Out	Footage	Hours	Accum. Hours	Vert. Dev.	Bit Data								Motor Data					
							Bit #	Size (in.)	Type	Make	Model	Serial #	Jets	Hours	Motor #	Make	Model	Bend	Hours	Rev/Gal
1	0'	2,175'	2,175'	18	24.00	Surface	1	17 1/2	PDC	NOV	BBK	E163630	9x16	18	-	-	-	-	-	
2	2,175'	6,180'	4,005'	32	50.00	Vertical	2	12 1/4	PDC	NOV	A204276	A198097	6x18	32	2	NOV	6/5 5.0	2.12°	32	0.13
3	6,180'	10,310'	4,130'	65	115.00	Vertical	3	8 3/4	PDC	NOV	DS616M	A202739	6x22	65	3	NOV	7/8 5.7	1.50°	65	0.24
4	10,310'	11,185'	875'	23.5	138.50	Curve	4	8 3/4	PDC	Smith	MDI516	JJ2135	5x18	23.5	4	NOV	7/8 5.0	2.38°	23.5	0.52
5	11,185'	20,519'	9,334'	94	232.50	Lateral	5	6	PDC	Smith	Z613	JK1605	2x22, 3x16	94	5	Ryan	6/7 ML 8.0	1.50°	94	0.80

PLAN VIEW

Oasis Petroleum North America, LLC
Chalmers 5300 21-19 8T



Oasis Petroleum North America, LLC

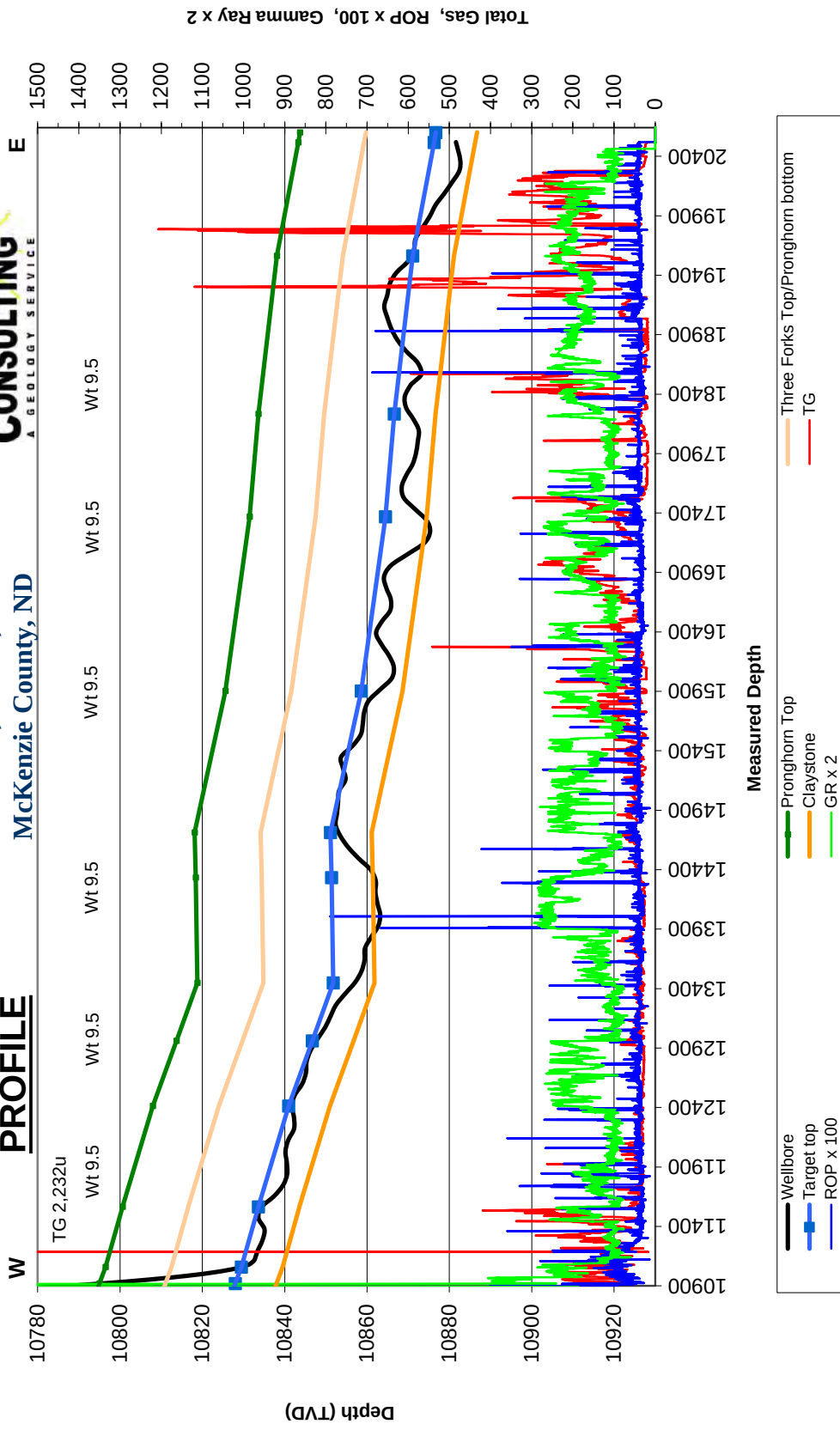
Chalmers 5300 21-19 8T

Lot 2 Sec. 19, T153N, R100W

McKenzie County, ND



PROFILE



FORMATION MARKERS & DIP ESTIMATES

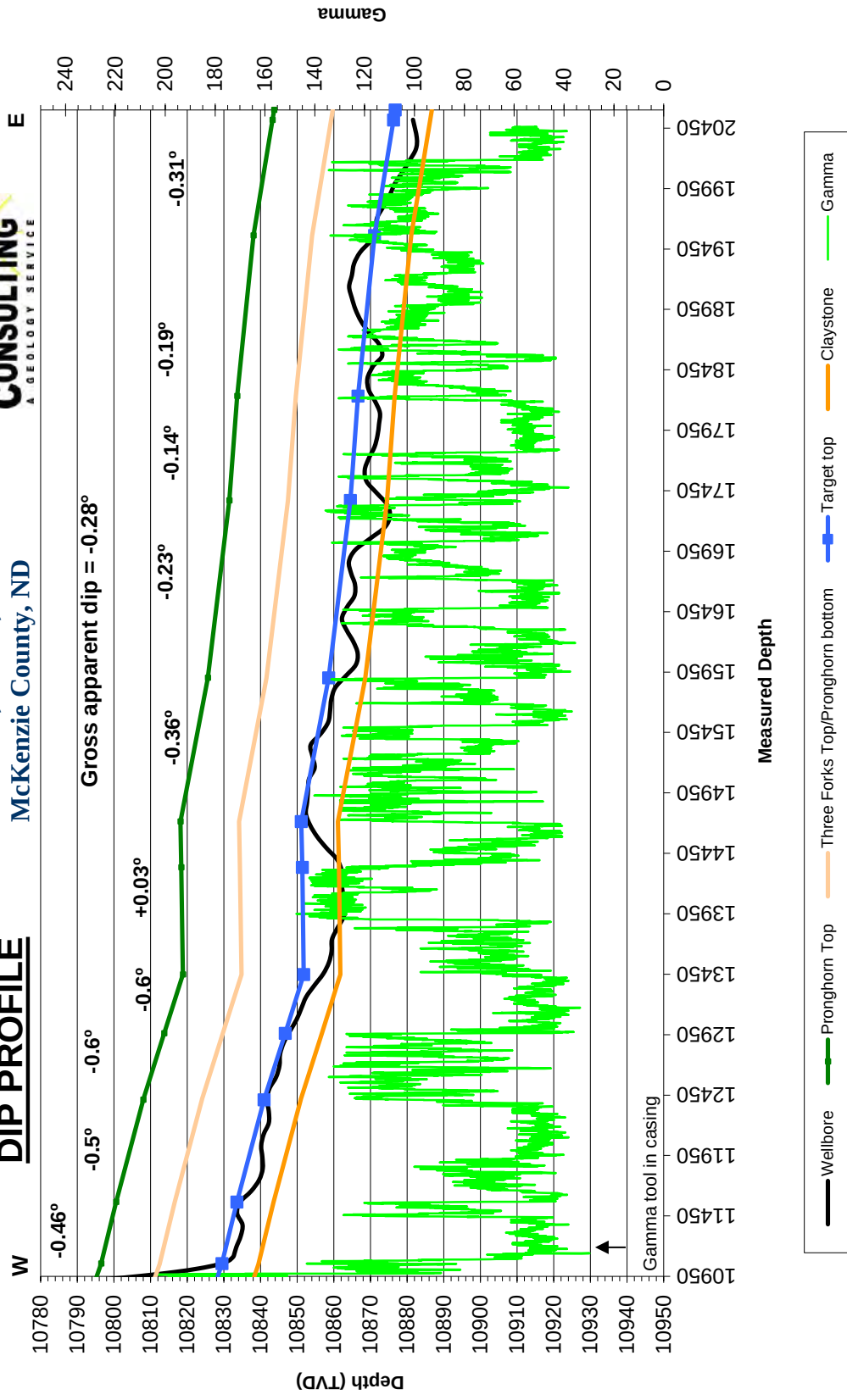
Oasis Petroleum North America, LLC - Chalmers 5300 21-19 8T

Dip Change Points	MD	TVD	TVD diff.	MD diff.	Dip	Dipping up/down	Type of Marker
Marker							
Target entry	11,057'	10,829.50					Gamma
Upper high gamma	11,565'	10,833.58	4.08	508.00	-0.46	Down	Gamma
Upper high gamma	12,412'	10,840.97	7.39	847.00	-0.50	Down	Gamma
Upper high gamma	12,960'	10,846.71	5.74	548.00	-0.60	Down	Gamma
Lower high gamma	13,447'	10,851.80	5.09	487.00	-0.60	Down	Gamma
Claystone	14,448'	10,851.28	-0.52	1001.00	0.03	Up	Gamma
Upper high gamma	14,713'	10,851.14	-0.14	265.00	0.03	Up	Gamma
Upper high gamma	15,903'	10,858.61	7.47	1190.00	-0.36	Down	Gamma
Claystone	17,369'	10,864.50	5.89	1466.00	-0.23	Down	Gamma
Cool gamma	18,232'	10,866.61	2.11	863.00	-0.14	Down	Gamma
Upper high gamma	19,561'	10,871.10	4.49	1329.00	-0.19	Down	Gamma
Projected Final Target Contact	20,516'	10,876.27	5.17	955.00	-0.31	Down	Gamma
Gross Dip							
Initial Target Contact	11,057'	10,829.50					
Projected Final Target Contact	20,516'	10,876.27	46.77	9459.00	-0.28	Down	Projection

Oasis Petroleum North America, LLC
Chalmers 5300 21-19 8T
Lot 2 Sec. 19, T153N, R100W
McKenzie County, ND



DIP PROFILE



<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5300 21-19 8T		
County:	McKenzie	State:	ND
QQ:	Lot 2	Section:	19
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	2226	FN/SL:	N
	327	FE/WL:	W

Kick-off:	10/15/2014
Finish:	12/28/2014
Directional Supervision:	RPM

Date: 1/9/2015

Time: 11:28

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 88.96

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
Tie	147	0.30	295.90	147.00	0.17	-0.35	-0.34	
1	240.00	0.30	298.40	240.00	0.39	-0.78	-0.78	0.01
2	333.00	0.40	273.50	333.00	0.53	-1.32	-1.31	0.19
3	427.00	0.10	334.70	427.00	0.62	-1.68	-1.67	0.39
4	520.00	0.30	337.70	520.00	0.92	-1.81	-1.79	0.22
5	613.00	0.30	347.50	612.99	1.38	-1.96	-1.93	0.06
6	702.00	0.60	288.80	701.99	1.76	-2.45	-2.42	0.58
7	790.00	0.50	326.20	789.99	2.23	-3.10	-3.06	0.41
8	880.00	0.50	276.40	879.98	2.60	-3.71	-3.66	0.47
9	970.00	0.20	304.50	969.98	2.73	-4.23	-4.18	0.37
10	1063.00	0.70	292.90	1062.98	3.04	-4.88	-4.83	0.54
11	1156.00	0.70	290.30	1155.97	3.46	-5.94	-5.88	0.03
12	1250.00	0.50	326.00	1249.97	4.00	-6.71	-6.63	0.44
13	1343.00	0.80	327.20	1342.96	4.88	-7.29	-7.20	0.32
14	1436.00	1.10	347.00	1435.95	6.30	-7.84	-7.72	0.47
15	1530.00	0.70	171.00	1529.95	6.61	-7.95	-7.83	1.91
16	1623.00	1.70	181.60	1622.92	4.67	-7.90	-7.81	1.10
17	1717.00	1.60	197.20	1716.89	2.02	-8.33	-8.29	0.49
18	1810.00	1.60	154.70	1809.85	-0.39	-8.16	-8.16	1.25
19	1903.00	1.80	109.10	1902.82	-2.04	-6.22	-6.26	1.43
20	1997.00	1.00	91.90	1996.79	-2.55	-4.01	-4.05	0.95
21	2198.00	0.90	23.60	2197.76	-1.16	-1.62	-1.64	0.53
22	2261.00	1.20	29.80	2260.75	-0.14	-1.10	-1.10	0.51
23	2354.00	0.40	209.50	2353.75	0.42	-0.77	-0.76	1.72
24	2446.00	0.60	204.50	2445.75	-0.29	-1.13	-1.13	0.22
25	2540.00	0.50	187.80	2539.74	-1.15	-1.39	-1.41	0.20
26	2633.00	0.70	205.40	2632.74	-2.06	-1.69	-1.73	0.29
27	2726.00	0.60	192.00	2725.73	-3.05	-2.03	-2.09	0.19
28	2820.00	0.90	113.60	2819.72	-3.83	-1.46	-1.53	1.04
29	2913.00	1.10	110.40	2912.71	-4.43	0.05	-0.03	0.22
30	3007.00	1.00	106.60	3006.69	-4.98	1.68	1.59	0.13
31	3100.00	0.70	100.30	3099.68	-5.31	3.02	2.92	0.34
32	3193.00	0.90	104.20	3192.67	-5.60	4.28	4.18	0.22
33	3287.00	0.90	104.20	3286.66	-5.96	5.71	5.60	0.00
34	3380.00	1.00	99.00	3379.65	-6.26	7.22	7.11	0.14

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5300 21-19 8T		
County:	McKenzie	State:	ND
QQ:	Lot 2	Section:	19
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	2226	FN/SL:	N
	327	FE/WL:	W

Kick-off:	10/15/2014
Finish:	12/28/2014
Directional Supervision:	RPM

Date: 1/9/2015

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F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 88.96

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
35	3474.00	1.10	110.20	3473.63	-6.70	8.88	8.76	0.24
36	3567.00	0.80	108.50	3566.62	-7.22	10.33	10.20	0.32
37	3660.00	0.90	111.10	3659.61	-7.69	11.63	11.49	0.12
38	3754.00	0.80	111.10	3753.60	-8.19	12.93	12.78	0.11
39	3847.00	1.30	67.60	3846.59	-8.02	14.51	14.37	0.97
40	3940.00	1.20	58.90	3939.56	-7.12	16.32	16.19	0.23
41	4034.00	1.30	55.20	4033.54	-6.00	18.04	17.93	0.14
42	4127.00	1.50	61.40	4126.51	-4.81	19.98	19.89	0.27
43	4221.00	1.10	59.80	4220.49	-3.77	21.84	21.76	0.43
44	4314.00	1.20	55.00	4313.47	-2.76	23.40	23.35	0.15
45	4407.00	1.00	61.30	4406.45	-1.82	24.91	24.88	0.25
46	4500.00	0.90	62.30	4499.44	-1.09	26.27	26.25	0.11
47	4593.00	0.90	52.60	4592.43	-0.30	27.50	27.49	0.16
48	4687.00	0.40	33.70	4686.42	0.42	28.27	28.27	0.57
49	4780.00	0.30	340.90	4779.42	0.92	28.37	28.38	0.35
50	4873.00	0.40	17.80	4872.42	1.46	28.39	28.41	0.26
51	4967.00	0.10	35.10	4966.42	1.84	28.54	28.56	0.33
52	5060.00	0.20	335.60	5059.42	2.05	28.52	28.55	0.19
53	5153.00	0.10	358.20	5152.42	2.28	28.45	28.48	0.12
54	5247.00	0.20	317.70	5246.42	2.48	28.33	28.37	0.15
55	5340.00	0.30	216.70	5339.42	2.41	28.08	28.12	0.42
56	5433.00	0.20	241.80	5432.42	2.14	27.79	27.82	0.16
57	5527.00	0.20	160.00	5526.42	1.91	27.70	27.73	0.28
58	5620.00	0.20	302.40	5619.42	1.84	27.62	27.65	0.41
59	5714.00	0.40	276.30	5713.42	1.96	27.16	27.19	0.25
60	5807.00	0.40	266.30	5806.41	1.98	26.51	26.54	0.07
61	5900.00	0.30	253.30	5899.41	1.89	25.95	25.98	0.14
62	5994.00	0.30	241.60	5993.41	1.70	25.50	25.53	0.07
63	6087.00	0.60	212.10	6086.41	1.17	25.03	25.04	0.40
64	6126.00	0.60	226.10	6125.40	0.86	24.77	24.78	0.37
65	6173.00	0.60	213.70	6172.40	0.48	24.46	24.46	0.28
66	6267.00	0.60	116.70	6266.40	-0.15	24.62	24.62	0.96
67	6360.00	1.00	101.70	6359.39	-0.53	25.85	25.84	0.48
68	6453.00	0.50	127.90	6452.38	-0.95	26.97	26.95	0.64
69	6547.00	0.30	153.80	6546.38	-1.42	27.40	27.37	0.28

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5300 21-19 8T		
County:	McKenzie	State:	ND
QQ:	Lot 2	Section:	19
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	2226	FN/SL:	N
	327	FE/WL:	W

Kick-off:	10/15/2014
Finish:	12/28/2014
Directional Supervision:	RPM

Date: 1/9/2015

Time: 11:28

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 88.96

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
70	6640.00	0.90	175.10	6639.37	-2.37	27.57	27.52	0.68
71	6734.00	1.00	187.10	6733.36	-3.92	27.53	27.46	0.24
72	6827.00	0.20	53.30	6826.36	-4.62	27.56	27.47	1.23
73	6920.00	0.10	338.20	6919.36	-4.45	27.66	27.58	0.21
74	7014.00	0.40	317.50	7013.36	-4.13	27.41	27.33	0.33
75	7107.00	0.70	18.60	7106.35	-3.36	27.37	27.31	0.66
76	7200.00	0.80	20.40	7199.34	-2.21	27.78	27.73	0.11
77	7294.00	0.50	30.60	7293.34	-1.24	28.22	28.19	0.34
78	7387.00	0.50	357.70	7386.33	-0.49	28.41	28.39	0.30
79	7480.00	0.70	311.00	7479.33	0.29	27.96	27.96	0.55
80	7574.00	1.60	30.30	7573.31	1.80	28.19	28.22	1.73
81	7667.00	1.70	30.00	7666.28	4.12	29.54	29.60	0.11
82	7760.00	0.80	14.40	7759.25	5.94	30.39	30.49	1.03
83	7854.00	0.70	17.80	7853.25	7.12	30.73	30.85	0.12
84	7947.00	0.50	332.90	7946.24	8.03	30.71	30.85	0.53
85	8040.00	0.00	359.10	8039.24	8.39	30.53	30.68	0.54
86	8134.00	0.30	216.70	8133.24	8.19	30.38	30.53	0.32
87	8227.00	0.40	91.70	8226.24	7.99	30.56	30.70	0.67
88	8320.00	0.50	92.30	8319.23	7.96	31.29	31.43	0.11
89	8414.00	0.50	86.20	8413.23	7.97	32.11	32.25	0.06
90	8507.00	0.60	101.10	8506.23	7.90	32.99	33.13	0.19
91	8600.00	0.60	116.10	8599.22	7.60	33.91	34.04	0.17
92	8694.00	0.60	94.90	8693.22	7.34	34.84	34.97	0.23
93	8787.00	0.70	108.80	8786.21	7.11	35.86	35.99	0.20
94	8881.00	0.30	155.00	8880.21	6.70	36.51	36.63	0.57
95	8974.00	0.60	132.40	8973.20	6.16	36.97	37.08	0.37
96	9067.00	0.40	191.70	9066.20	5.51	37.27	37.36	0.56
97	9161.00	0.40	160.60	9160.20	4.88	37.31	37.39	0.23
98	9254.00	0.40	160.70	9253.20	4.27	37.52	37.60	0.00
99	9347.00	0.40	222.20	9346.20	3.72	37.41	37.48	0.44
100	9441.00	0.70	201.40	9440.19	2.94	36.98	37.03	0.38
101	9534.00	0.60	186.10	9533.19	1.93	36.72	36.75	0.21
102	9627.00	0.60	99.90	9626.18	1.36	37.15	37.17	0.88
103	9720.00	0.90	95.20	9719.17	1.21	38.36	38.38	0.33
104	9814.00	0.80	119.80	9813.16	0.82	39.66	39.67	0.40

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5300 21-19 8T		
County:	McKenzie	State:	ND
QQ:	Lot 2	Section:	19
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	2226	FN/SL:	N
	327	FE/WL:	W

Kick-off:	10/15/2014
Finish:	12/28/2014
Directional Supervision:	RPM

Date: 1/9/2015

Time: 11:28

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 88.96

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
105	9907.00	0.70	118.70	9906.16	0.22	40.73	40.72	0.11
106	10000.00	0.70	95.40	9999.15	-0.10	41.79	41.78	0.30
107	10094.00	0.50	115.30	10093.14	-0.33	42.73	42.72	0.30
108	10187.00	0.20	181.60	10186.14	-0.67	43.09	43.08	0.49
109	10253.00	0.40	111.00	10252.14	-0.87	43.31	43.28	0.58
110	10300.00	0.20	180.20	10299.14	-1.01	43.46	43.43	0.81
111	10331.00	0.50	115.00	10330.14	-1.12	43.58	43.55	1.46
112	10362.00	3.40	80.80	10361.12	-1.03	44.61	44.59	9.68
113	10393.00	7.50	78.20	10391.97	-0.47	47.50	47.48	13.25
114	10424.00	12.70	76.10	10422.48	0.76	52.79	52.80	16.81
115	10456.00	17.10	75.80	10453.40	2.76	60.77	60.81	13.75
116	10487.00	21.10	75.80	10482.69	5.25	70.60	70.69	12.90
117	10518.00	23.60	75.30	10511.35	8.20	82.02	82.15	8.09
118	10549.00	25.50	77.10	10539.55	11.26	94.53	94.71	6.59
119	10580.00	28.00	78.90	10567.23	14.15	108.17	108.41	8.48
120	10611.00	31.30	78.20	10594.17	17.20	123.20	123.49	10.70
121	10642.00	34.30	75.90	10620.23	20.98	139.56	139.92	10.48
122	10673.00	37.20	74.70	10645.38	25.58	157.07	157.51	9.62
123	10704.00	40.60	73.20	10669.50	30.97	175.78	176.31	11.38
124	10736.00	44.50	72.90	10693.07	37.28	196.47	197.11	12.20
125	10767.00	48.20	73.80	10714.47	43.70	217.96	218.71	12.12
126	10798.00	52.10	74.00	10734.33	50.30	240.82	241.69	12.59
127	10829.00	56.00	73.70	10752.52	57.28	264.92	265.92	12.60
128	10860.00	59.80	73.30	10768.99	64.73	290.09	291.22	12.31
129	10891.00	64.30	73.60	10783.52	72.53	316.34	317.60	14.54
130	10922.00	67.60	75.00	10796.15	80.19	343.59	344.98	11.42
131	10953.00	71.20	76.10	10807.06	87.42	371.68	373.21	12.08
132	10985.00	75.20	77.90	10816.31	94.31	401.52	403.17	13.61
133	11016.00	79.00	79.20	10823.23	100.30	431.13	432.88	12.92
134	11047.00	82.50	80.20	10828.21	105.77	461.23	463.08	11.73
135	11078.00	86.50	81.20	10831.18	110.76	491.68	493.61	13.30
136	11109.00	88.80	81.80	10832.45	115.33	522.31	524.32	7.67
137	11125.00	89.10	82.00	10832.74	117.59	538.15	540.19	2.25
138	11191.00	89.80	84.00	10833.38	125.63	603.65	605.83	3.21
139	11284.00	88.70	83.80	10834.59	135.51	696.11	698.46	1.20

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5300 21-19 8T		
County:	McKenzie	State:	ND
QQ:	Lot 2	Section:	19
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	2226	FN/SL:	N
	327	FE/WL:	W

Kick-off:	10/15/2014
Finish:	12/28/2014
Directional Supervision:	RPM

Date: 1/9/2015

Time: 11:28

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 88.96

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
140	11377.00	90.70	83.30	10835.08	145.96	788.52	791.04	2.22
141	11470.00	91.10	83.60	10833.62	156.57	880.90	883.60	0.54
142	11562.00	88.00	83.10	10834.34	167.22	972.27	975.14	3.41
143	11655.00	88.20	84.00	10837.43	177.66	1064.63	1067.68	0.99
144	11748.00	89.10	86.30	10839.62	185.52	1157.26	1160.44	2.66
145	11841.00	89.70	89.00	10840.59	189.33	1250.17	1253.40	2.97
146	11936.00	90.40	91.80	10840.51	188.67	1345.16	1348.36	3.04
147	12030.00	90.00	91.50	10840.18	185.96	1439.12	1442.25	0.53
148	12125.00	89.20	91.50	10840.84	183.47	1534.08	1537.16	0.84
149	12220.00	89.10	90.60	10842.25	181.73	1629.05	1632.08	0.95
150	12315.00	90.90	90.60	10842.25	180.74	1724.04	1727.04	1.89
151	12409.00	89.70	90.50	10841.76	179.84	1818.04	1821.00	1.28
152	12504.00	89.50	90.50	10842.42	179.01	1913.03	1915.96	0.21
153	12599.00	88.30	90.60	10844.25	178.10	2008.01	2010.91	1.27
154	12694.00	90.60	90.20	10845.16	177.43	2102.99	2105.87	2.46
155	12788.00	89.20	91.10	10845.32	176.37	2196.98	2199.82	1.77
156	12883.00	89.60	90.70	10846.32	174.87	2291.97	2294.76	0.60
157	12978.00	88.40	90.60	10847.98	173.80	2386.94	2389.71	1.27
158	13073.00	89.40	90.60	10849.80	172.80	2481.92	2484.65	1.05
159	13167.00	89.00	90.30	10851.11	172.06	2575.91	2578.61	0.53
160	13262.00	89.40	90.80	10852.44	171.15	2670.89	2673.56	0.67
161	13357.00	87.90	91.10	10854.68	169.58	2765.85	2768.47	1.61
162	13452.00	89.20	90.50	10857.08	168.25	2860.81	2863.39	1.51
163	13546.00	88.90	90.40	10858.64	167.51	2954.79	2957.35	0.34
164	13641.00	90.20	89.60	10859.39	167.51	3049.79	3052.33	1.61
165	13736.00	89.70	89.40	10859.47	168.34	3144.78	3147.32	0.57
166	13831.00	88.50	88.90	10860.96	169.75	3239.76	3242.31	1.37
167	13926.00	89.40	89.20	10862.70	171.33	3334.73	3337.29	1.00
168	14020.00	90.00	88.80	10863.19	172.97	3428.71	3431.29	0.77
169	14115.00	90.80	89.20	10862.53	174.63	3523.70	3526.28	0.94
170	14210.00	89.80	90.00	10862.03	175.29	3618.69	3621.28	1.35
171	14305.00	90.10	88.20	10862.12	176.78	3713.67	3716.27	1.92
172	14399.00	91.40	87.70	10860.89	180.14	3807.60	3810.25	1.48
173	14494.00	92.10	87.20	10857.98	184.37	3902.46	3905.17	0.91
174	14589.00	91.00	88.20	10855.42	188.18	3997.35	4000.11	1.56

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Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5300 21-19 8T		
County:	McKenzie	State:	ND
QQ:	Lot 2	Section:	19
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	2226	FN/SL:	N
	327	FE/WL:	W

Kick-off:	10/15/2014
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Directional Supervision:	RPM

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F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 88.96

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
175	14683.00	91.30	88.20		10853.53	191.13	4091.29	4094.08	0.32
176	14778.00	90.10	90.50		10852.37	192.21	4186.26	4189.06	2.73
177	14873.00	89.50	91.40		10852.70	190.63	4281.25	4284.00	1.14
178	14968.00	90.20	91.90		10852.95	187.90	4376.21	4378.90	0.91
179	15063.00	89.50	91.00		10853.20	185.49	4471.18	4473.81	1.20
180	15157.00	88.60	91.00		10854.76	183.85	4565.15	4567.73	0.96
181	15252.00	92.10	89.70		10854.18	183.27	4660.13	4662.69	3.93
182	15347.00	88.50	91.10		10853.68	182.61	4755.11	4757.64	4.07
183	15442.00	88.10	91.60		10856.50	180.37	4850.04	4852.51	0.67
184	15537.00	89.50	91.70		10858.49	177.64	4944.97	4947.38	1.48
185	15631.00	89.90	91.80		10858.98	174.77	5038.93	5041.27	0.44
186	15726.00	89.70	89.60		10859.31	173.61	5133.92	5136.22	2.33
187	15821.00	89.00	89.60		10860.39	174.27	5228.91	5231.21	0.74
188	15916.00	87.60	90.10		10863.21	174.52	5323.86	5326.15	1.56
189	16010.00	89.00	89.30		10865.99	175.01	5417.82	5420.10	1.72
190	16105.00	90.40	86.70		10866.49	178.33	5512.75	5515.07	3.11
191	16200.00	91.00	86.50		10865.33	183.96	5607.57	5609.99	0.67
192	16294.00	91.20	88.90		10863.53	187.73	5701.47	5703.94	2.56
193	16389.00	90.40	88.40		10862.20	189.97	5796.43	5798.93	0.99
194	16484.00	88.00	88.90		10863.53	192.21	5891.39	5893.91	2.58
195	16579.00	89.50	89.40		10865.60	193.62	5986.36	5988.88	1.66
196	16674.00	90.30	89.60		10865.76	194.44	6081.35	6083.88	0.87
197	16768.00	91.00	89.70		10864.70	195.02	6175.34	6177.87	0.75
198	16863.00	89.70	90.00		10864.12	195.27	6270.34	6272.85	1.40
199	16958.00	88.30	90.40		10865.78	194.94	6365.32	6367.81	1.53
200	17053.00	86.80	90.20		10869.84	194.44	6460.23	6462.70	1.59
201	17148.00	88.30	90.30		10873.90	194.02	6555.14	6557.58	1.58
202	17243.00	90.00	90.50		10875.31	193.36	6650.12	6652.54	1.80
203	17337.00	91.00	92.20		10874.49	191.15	6744.09	6746.45	2.10
204	17432.00	92.40	92.10		10871.67	187.58	6838.98	6841.26	1.48
205	17527.00	90.90	89.90		10868.93	185.93	6933.92	6936.15	2.80
206	17622.00	89.70	89.90		10868.43	186.09	7028.91	7031.13	1.26
207	17716.00	89.00	89.80		10869.50	186.34	7122.91	7125.11	0.75
208	17811.00	89.10	90.40		10871.08	186.17	7217.89	7220.08	0.64
209	17906.00	89.90	91.80		10871.91	184.35	7312.87	7315.01	1.70

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Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5300 21-19 8T		
County:	McKenzie	State:	ND
QQ:	Lot 2	Section:	19
Township:	153	N/S:	N
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Minimum Curvature Method (SPE-3362)

Proposed dir: 88.96

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
210	18001.00	89.60	92.10		10872.32	181.12	7407.81	7409.88	0.45
211	18095.00	90.10	92.30		10872.57	177.51	7501.74	7503.73	0.57
212	18190.00	91.50	92.30		10871.24	173.70	7596.65	7598.55	1.47
213	18285.00	90.60	90.90		10869.50	171.05	7691.60	7693.43	1.75
214	18380.00	89.80	88.50		10869.17	171.54	7786.59	7788.42	2.66
215	18475.00	88.50	88.10		10870.58	174.36	7881.53	7883.40	1.43
216	18506.00	88.10	88.10		10871.50	175.39	7912.50	7914.38	1.29
217	18569.00	88.80	87.80		10873.20	177.64	7975.44	7977.35	1.21
218	18664.00	92.30	87.40		10872.29	181.62	8070.34	8072.30	3.71
219	18759.00	91.00	88.60		10869.55	184.93	8165.23	8167.25	1.86
220	18854.00	91.60	88.40		10867.40	187.42	8260.18	8262.22	0.67
221	18948.00	90.20	90.40		10865.92	188.40	8354.15	8356.20	2.60
222	19043.00	91.00	90.00		10864.93	188.07	8449.15	8451.17	0.94
223	19138.00	89.90	90.30		10864.18	187.82	8544.14	8546.14	1.20
224	19233.00	89.20	90.10		10864.93	187.49	8639.14	8641.12	0.77
225	19328.00	90.10	90.00		10865.51	187.41	8734.13	8736.10	0.95
226	19422.00	87.90	90.30		10867.15	187.16	8828.11	8830.06	2.36
227	19517.00	88.70	91.50		10869.97	185.67	8923.06	8924.96	1.52
228	19612.00	89.60	91.60		10871.37	183.10	9018.01	9019.85	0.95
229	19707.00	89.90	90.90		10871.79	181.03	9112.99	9114.77	0.80
230	19802.00	88.00	90.50		10873.53	179.87	9207.96	9209.71	2.04
231	19896.00	89.90	90.30		10875.25	179.21	9301.94	9303.66	2.03
232	19991.00	88.40	91.10		10876.66	178.05	9396.92	9398.60	1.79
233	20086.00	89.10	90.10		10878.73	177.06	9491.89	9493.54	1.28
234	20181.00	88.60	90.90		10880.64	176.23	9586.86	9588.48	0.99
235	20276.00	89.30	90.40		10882.38	175.15	9681.84	9683.42	0.91
236	20370.00	90.40	90.30		10882.63	174.58	9775.84	9777.40	1.18
237	20519.00	90.40	90.30		10881.59	173.80	9924.83	9926.35	0.00

DEVIATION SURVEYS

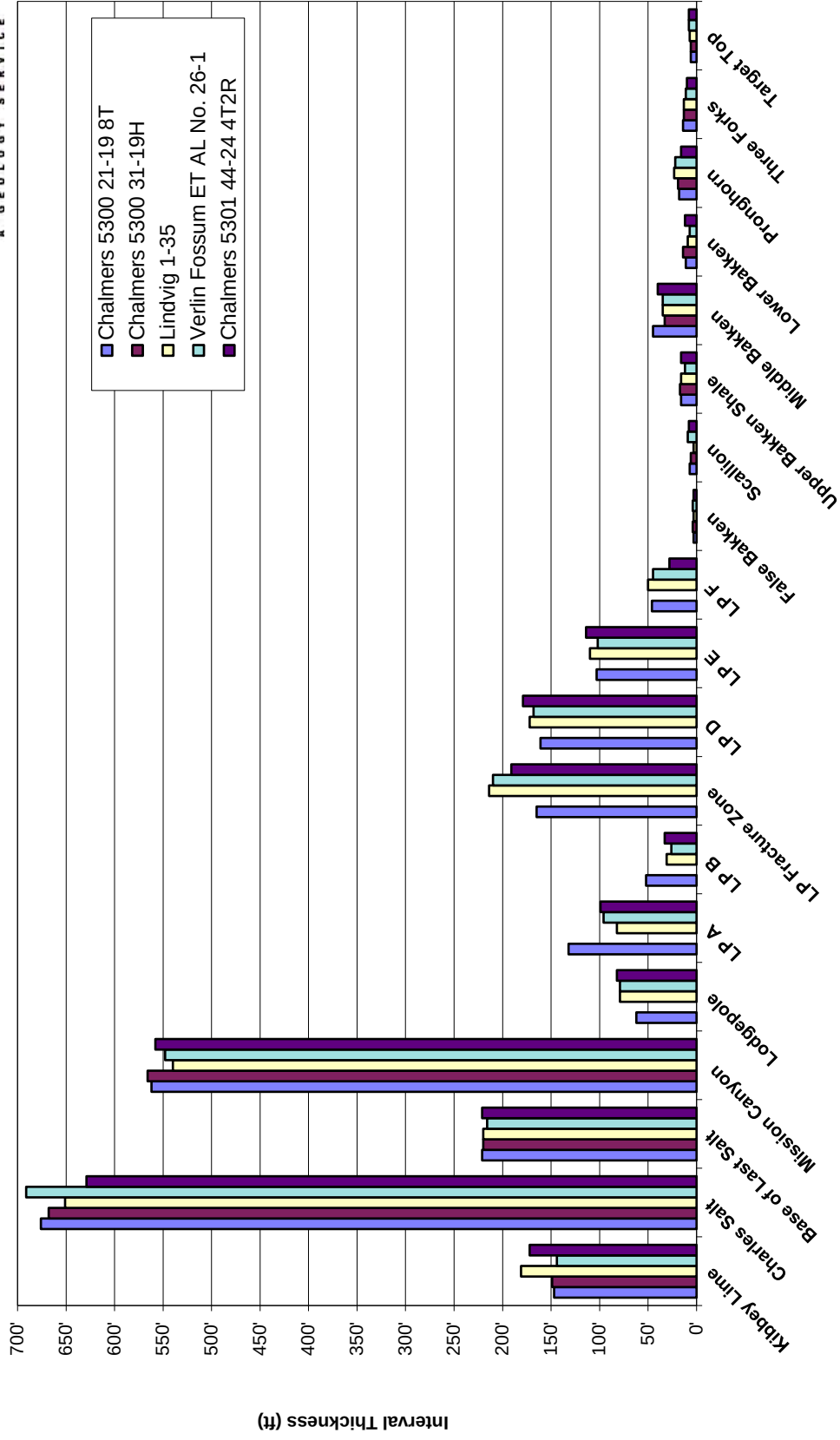
Depth	Inclination	Azimuth
147	0.30	295.90
240	0.30	298.40
333	0.40	273.50
427	0.10	334.70
520	0.30	337.70
613	0.30	347.50
702	0.60	288.80
790	0.50	326.20
880	0.50	276.40
970	0.20	304.50
1063	0.70	292.90
1156	0.70	290.30
1250	0.50	326.00
1343	0.80	327.20
1436	1.10	347.00
1530	0.70	171.00
1623	1.70	181.60
1717	1.60	197.20
1810	1.60	154.70
1903	1.80	109.10
1997	1.00	91.90

CONTROL DATA

Operator: Well Name: Location:	Oasis Petroleum North America, LLC Chalmers 5300 31-19H NW SW Sec. 19 T153N R100W McKenzie County, ND ~1/4 mile S of Chalmers 5300 21-19 8T KB: 1,929'	Texas Gas Exploration Company Lindvig 1-35 SE SE Sec. 35 T153N R101W McKenzie County, ND ~2 1/4 miles SSW of Chalmers 5300 21-19 8T KB: 2,226'	Mosbacher Production Company Verlin Fossum ET AL No. 26-1 NW SE Sec. 26, T153N R101W McKenzie County, ND ~1 3/4 miles SSW of Chalmers 5300 21-19 8T KB: 2,175'	Oasis Petroleum North America, LLC Chalmers 5301 44-24 4T2R SE SE Sec. 24 T153N R101W McKenzie County, ND ~1/2 mile SSW of Chalmers 5300 21-19 8T KB: 2,039'												
Elevation: Formation/ Zone	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target Landing	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target Landing	TVD Top	Datum (MSL)	Interval Thickness	Thickness to Target Landing	TVD Top	Datum (MSL)	Interval Thickness	Thickness to Target Landing
Kibbey Lime	8,243'	-6,314'	149'	2,425'	8,449'	-6,223'	181'	2,439'	8,372'	-6,197'	144'	2,433'	8,266'	-6,091'	172'	2,419'
Charles Salt	8,392'	-6,463'	668'	2,276'	8,630'	-6,404'	651'	2,258'	8,516'	-6,341'	691'	2,289'	8,438'	-6,263'	629'	2,247'
Base of Last Salt	9,060'	-7,131'	220'	1,608'	9,281'	-7,055'	220'	1,607'	9,207'	-7,032'	216'	1,598'	9,067'	-6,892'	221'	1,618'
Mission Canyon	9,280'	-7,351'	566'	1,388'	9,501'	-7,275'	540'	1,387'	9,423'	-7,248'	548'	1,382'	9,288'	-7,113'	558'	1,397'
Lodgepole	9,846'	-7,917'	-	822'	10,041'	-7,815'	79'	847'	9,971'	-7,796'	79'	834'	9,846'	-7,671'	82'	839'
LP A	-	-	-	-	10,120'	-7,894'	82'	768'	10,050'	-7,875'	96'	755'	9,928'	-7,753'	99'	757'
LP B	-	-	-	-	10,202'	-7,976'	31'	686'	10,146'	-7,971'	26'	659'	10,027'	-7,852'	33'	658'
LP Fracture Zone	-	-	-	-	10,233'	-8,007'	214'	655'	10,172'	-7,997'	210'	633'	10,060'	-7,885'	191'	625'
LP D	-	-	-	-	10,447'	-8,221'	172'	441'	10,382'	-8,207'	168'	423'	10,251'	-8,076'	179'	434'
LP E	-	-	-	-	10,619'	-8,393'	110'	269'	10,550'	-8,375'	102'	255'	10,430'	-8,255'	114'	255'
LP F	-	-	-	-	10,729'	-8,503'	50'	159'	10,652'	-8,477'	45'	153'	10,544'	-8,369'	28'	141'
False Bakken	10,556'	-8,627'	4'	112'	10,779'	-8,553'	3'	109'	10,697'	-8,522'	4'	108'	10,572'	-8,397'	3'	113'
Scallion	10,560'	-8,631'	6'	108'	10,782'	-8,556'	3'	106'	10,701'	-8,526'	9'	104'	10,575'	-8,400'	8'	110'
Upper Bakken Shale	10,566'	-8,637'	17'	102'	10,785'	-8,559'	16'	103'	10,710'	-8,535'	12'	95'	10,583'	-8,408'	16'	102'
Middle Bakken	10,583'	-8,654'	33'	85'	10,801'	-8,575'	35'	87'	10,722'	-8,547'	35'	83'	10,599'	-8,424'	40'	86'
Lower Bakken	10,616'	-8,687'	14'	52'	10,836'	-8,610'	9'	52'	10,757'	-8,582'	7'	48'	10,639'	-8,464'	12'	46'
Pronghorn	10,630'	-8,701'	19'	38'	10,845'	-8,619'	23'	43'	10,764'	-8,589'	22'	41'	10,651'	-8,476'	16'	34'
Three Forks	10,649'	-8,720'	13'	19'	10,868'	-8,642'	13'	20'	10,786'	-8,611'	11'	19'	10,667'	-8,492'	10'	18'
Target Top	10,662'	-8,733'	6'	6'	10,881'	-8,655'	7'	7'	10,797'	-8,622'	8'	8'	10,677'	-8,502'	8'	8'
Target Landing	10,668'	-8,739'	4'	0'	10,888'	-8,662'	4'	0'	10,805'	-8,630'	4'	0'	10,685'	-8,510'	4'	0'
Target Base/Claystone	10,672'	-8,743'	1'	-	10,892'	-8,666'	1'	-	10,809'	-8,634'	1'	-	10,689'	-8,514'	-	-
Claystone	10,673'	-8,744'	-	-	10,893'	-8,667'	-	-	10,810'	-8,635'	-	-	-	-	-	-

INTERVAL THICKNESS

Oasis Petroleum North America, LLC - Chalmers 5300 21-19 8T



LANDING PROJECTION

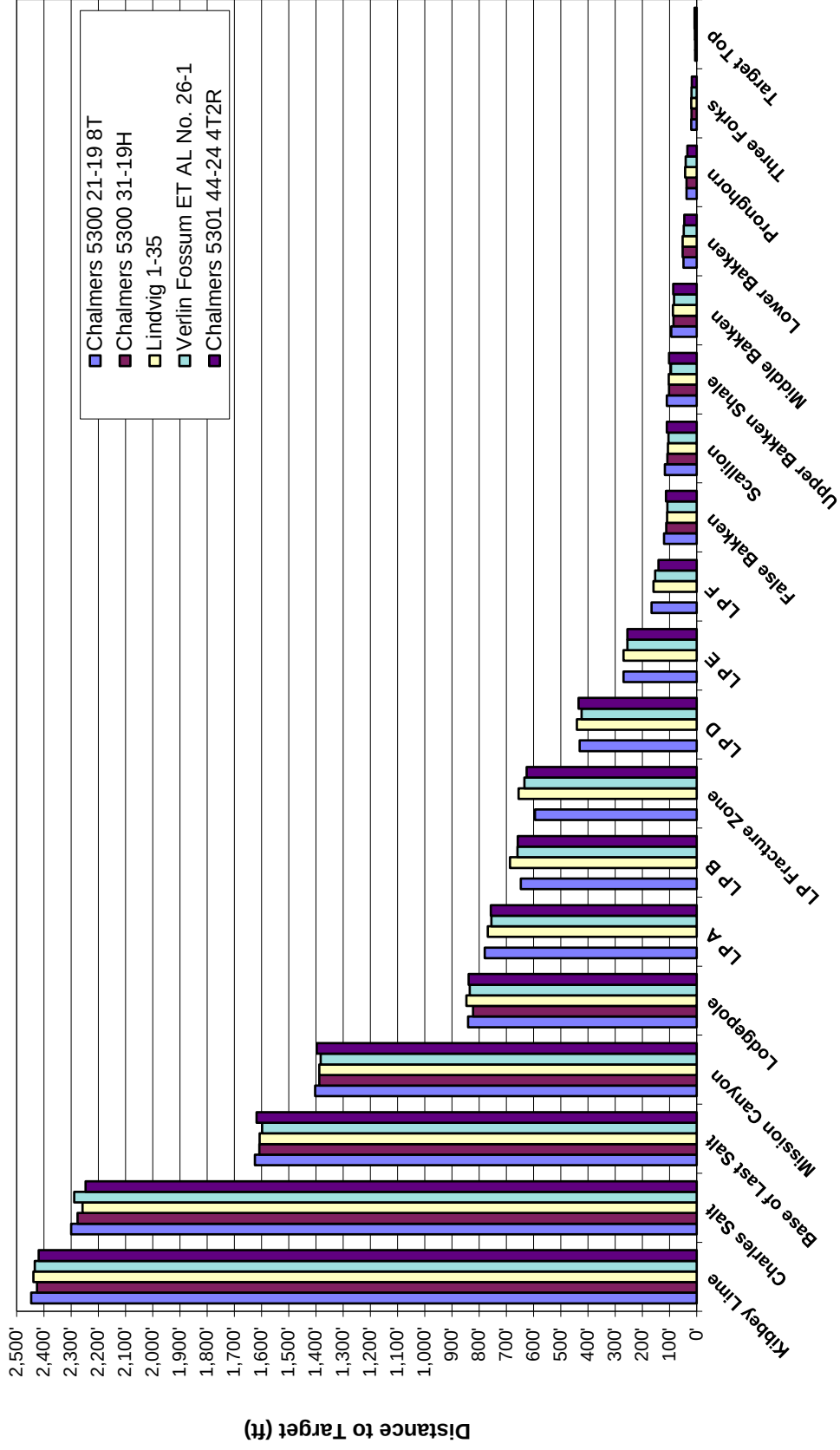
Formation/ Zone:	Proposed Target Landing From:				
	Chalmers 5300 31-19H	Lindvig 1-35	Verlin Fossum ET AL No. 26-1	Chalmers 5301 44-24 4T2R	Average of Offset Wells
Kibbey Lime	10,811'	10,825'	10,819'	10,805'	10,815'
Charles Salt	10,809'	10,791'	10,822'	10,780'	10,801'
Base of Last Salt	10,817'	10,816'	10,807'	10,827'	10,817'
Mission Canyon	10,818'	10,817'	10,812'	10,827'	10,819'
Lodgepole	10,814'	10,839'	10,826'	10,831'	10,828'
LP A	-	10,822'	10,809'	10,811'	10,814'
LP B	-	10,872'	10,845'	10,844'	10,854'
LP Fracture Zone	-	10,893'	10,871'	10,863'	10,876'
LP D	-	10,844'	10,826'	10,837'	10,836'
LP E	-	10,833'	10,819'	10,819'	10,824'
LP F	-	10,826'	10,820'	10,808'	10,818'
False Bakken	10,825'	10,822'	10,821'	10,826'	10,824'
Scallion	10,824'	10,822'	10,820'	10,826'	10,823'
Upper Bakken Shale	10,825'	10,826'	10,818'	10,825'	10,824'
Middle Bakken	10,824'	10,826'	10,822'	10,825'	10,824'
Lower Bakken	10,836'	10,836'	10,832'	10,830'	10,834'
Pronghorn	10,833'	10,838'	10,836'	10,829'	10,834'
Three Forks	10,832'	10,833'	10,832'	10,831'	10,832'
Target Landing	10,833'	10,834'	10,835'	10,835'	10,834'

Current Landing Target (20' below top of Three Forks): **10,833'**

Landing targets are subject to change as new formation tops are available

ISOPACH TO TARGET

Oasis Petroleum North America, LLC - Chalmers 5300 21-19 8T



LITHOLOGY

Oasis Petroleum North America, LLC Chalmers 5300 21-19 7T2

Rig crews caught 100' sample intervals, under the supervision of Sunburst geologists, from 4,700' to 8,200', 30' intervals from 8,200', to the TD of the lateral at 20,519'. Formation tops and lithologic markers have been inserted into the sample descriptions below for reference. Sample descriptions begin in the Belle Fourche Formation prior to the Kibbey Lime. Samples were examined wet and dry under a binocular microscope. Sample fluorescent cuts are masked by invert mud through intermediate casing. Quantifiers in order of increasing abundance are trace, rare, occasional, common and abundant.

Vertical Log Descriptions: **MD / TVD (MSL Datum)**

Drilling in the Belle Fourche Formation [Cretaceous Colorado Group]

4,700-4,800 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

4,800-4,900 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

4,900-5,000 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

Mowry Formation [Cretaceous Dakota Group] **5,045' MD (-2,969')**

5,000-5,100 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

5,100-5,200 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain; rare SILTSTONE: light gray, friable, sub blocky, calcite cement moderately cemented

5,200-5,300 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

5,300-5,400 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

Dakota Formation [Dakota Group] **5,484' MD (-3,408')**

5,400-5,500 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain; rare SILTY SANDSTONE: light gray, very fine grained, friable-loose, sub rounded, vitreous, moderately sorted, calcite cement moderately cemented

5,500-5,600 SILTSTONE: light gray, light brown, friable, sub blocky, calcite cement moderately cemented; rare SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

5,600-5,700 SILTSTONE: light gray, light brown, friable, sub blocky, calcite cement moderately cemented; rare SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

5,700-5,800 SILTSTONE: light gray, light brown, friable, sub blocky, calcite cement moderately cemented; rare SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

5,800-5,900 SILTY SANDSTONE: light gray, very fine grained, friable-loose, sub rounded, vitreous, moderately sorted, calcite cement moderately cemented; rare SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

Swift Formation [Jurassic] **5,982' (-3,906')**

5,900-6,000 SHALE: light blue-green, common medium gray-brown, firm, earthy, calcareous, common disseminated pyrite, no visible porosity, no visible oil stain

6,000-6,100 SHALE: light blue-green, common medium gray-brown, firm, earthy, calcareous, common disseminated pyrite, no visible porosity, no visible oil stain

6,100-6,150 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

6,150-6,200 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

6,200-6,300 SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain; trace SILTY SANDSTONE: light gray, very fine grained, friable-loose, sub rounded, vitreous, moderately sorted, calcite cement moderately cemented

6,300-6,400 LIMESTONE: mudstone, off white, light gray, very fine crystalline, friable, dense, earthy, possible intercrystalline porosity, no visible oil stain; trace SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

Rierdon Formation [Jurassic] **6,485' MD / 6,484' TVD (-4,408')**

6,400-6,500 LIMESTONE: mudstone, dark-medium gray-brown, very fine crystalline, friable, dense, earthy, possible intercrystalline porosity, no visible oil stain; trace SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

6,500-6,600 LIMESTONE: mudstone, off white, light gray, very fine crystalline, friable, dense, earthy, possible intercrystalline porosity, no visible oil stain; trace SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

6,600-6,700 LIMESTONE: mudstone, off white, light gray, very fine crystalline, friable, dense, earthy, possible intercrystalline porosity, no visible oil stain; trace SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

6,700-6,800 LIMESTONE: mudstone, off white, light gray, very fine crystalline, friable, dense, earthy, possible intercrystalline porosity, no visible oil stain; trace SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

6,800-6,900 ANHYDRITE: milky white, microcrystalline, soft-friable, massive, amorphous, no visible porosity, no visible oil stain; occasional LIMESTONE: mudstone, light gray-cream, microcrystalline, friable-firm, dense, earthy-chalky, no visible porosity, no visible oil stain

Dunham Salt Member [Piper Formation] **6,901' MD / 6,900' TVD (-4,824')**

Dunham Salt Base [Piper Formation] **6,916' MD / 6,915' TVD (-4,839')**

6,900-7,000 LIMESTONE: mudstone, off white, light gray, very fine crystalline, friable, dense, earthy, possible intercrystalline porosity, no visible oil stain; rare SHALE: dark-medium gray, firm, sub blocky, earthy, calcareous, trace disseminated pyrite, no visible porosity, no visible oil stain

7,000-7,100 SILTSTONE: orange-red brown, friable, sub blocky, calcareous cement moderately cemented, possible intergranular porosity, no visible oil stain; ANHYDRITE: milky white, microcrystalline, soft-friable, massive, amorphous, no visible porosity, no visible oil stain

7,100-7,200 SILTSTONE: orange-red brown, friable, sub blocky, calcareous cement moderately cemented, possible intergranular porosity, no visible oil stain; ANHYDRITE: milky white, microcrystalline, soft-friable, massive, amorphous, no visible porosity, no visible oil stain

Pine Salt Member [Spearfish Formation] **7,266' MD / 7,265' TVD (-5,189')**

Pine Salt Base [Spearfish Formation] **7,296' MD / 7,295' TVD (-5,219')**

7,200-7,300 SALT: translucent, milky, microcrystalline, hard, massive, crystalline texture; SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, possible intergranular porosity, no visible oil stain; ANHYDRITE: milky white, microcrystalline, soft-friable, massive, amorphous, no visible porosity, no visible oil stain

7,300-7,400 SALT: translucent, milky, microcrystalline, hard, massive, crystalline texture; SILTSTONE: orange-red brown, friable, sub blocky, calcareous cement moderately cemented, possible intergranular porosity, no visible oil stain

Opeche Salt Member [Opeche Formation] **7,371' MD / 7,370' TVD (-5,294')**

7,300-7,400 SALT: translucent, milky, microcrystalline, hard, massive, crystalline texture; SILTSTONE: orange-red brown, friable, sub blocky, calcareous cement moderately cemented, possible intergranular porosity, no visible oil stain

Opeche Salt Base [Opeche Formation] **7,427' MD / 7,426' TVD (-5,350')**

7,400-7,500 SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain; trace SALT: translucent, milky, microcrystalline, hard, massive, crystalline texture

7,500-7,600 SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain; ANHYDRITE: milky white, microcrystalline, soft-friable, massive, amorphous, no visible porosity, no visible oil stain

Amsden Formation [Pennsylvanian Minnelusa Group] **7,670' MD / 7,669' TVD (-5,593')**

7,600-7,700 DOLOMITE: mudstone, light pink-off white, trace red, microcrystalline, soft-friable, laminated, crystalline-micro sucrosic, calcareous, no visible porosity, no visible oil stain

7,700-7,800 DOLOMITE: mudstone, light pink-off white, trace red, microcrystalline, soft-friable, laminated, crystalline-micro sucrosic, calcareous, no visible porosity, no visible oil stain; occasional SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain; rare SHALE: dark gray, friable, sub blocky, earthy no visible porosity, no visible oil stain

Tyler Formation [Pennsylvanian Minnelusa Group] 7,838' MD / 7,837' TVD (-5,761')

7,800-7,900 SHALE: dark gray, friable, sub blocky, earthy, no visible porosity, no visible oil stain; rare SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain

7,900-8,000 SHALE: dark gray, friable, sub blocky, earthy, no visible porosity, no visible oil stain; rare SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain

Otter [Mississippian Big Snowy Group] 8,041' MD / 8,040' TVD (-5,964')

8,000-8,100 SHALE: dark gray, firm, sub blocky, earthy no visible porosity, no visible oil stain; rare SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain

8,100-8,200 SHALE: light gray-gray, red gray, friable, sub blocky, earthy, no visible porosity, no visible oil stain; LIMESTONE: mudstone-wackestone, light gray, microcrystalline, friable, dense, earthy, no visible porosity, no visible oil stain

8,200-8,240 LIMESTONE: mudstone-wackestone, light gray, microcrystalline, friable, dense, earthy, no visible porosity, no visible oil stain; common SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain

8,240-8,270 LIMESTONE: mudstone-wackestone, light gray, microcrystalline, friable, dense, earthy, no visible porosity, no visible oil stain; common SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain

8,270-8,300 SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain; rare SHALE: dark gray, firm, sub blocky, earthy no visible porosity, no visible oil stain

8,300-8,330 SILTSTONE: orange-red brown, friable-firm, sub blocky, calcareous cement moderately cemented, trace clear quartz grains, possible intergranular porosity, no visible oil stain; rare SHALE: dark gray, firm, sub blocky, earthy no visible porosity, no visible oil stain

8,330-8,360 SILTSTONE: dark orange-light brown, tan, soft, sub blocky, calcite cement, poorly cemented; rare ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: tan-off white, very fine grained, sub rounded, moderately sorted, calcite cement, poorly cemented

Kibbey Lime [Kibbey Formation Mississippian Big Snowy Group] 8,387' MD / 8,386' TVD (-6,310')

8,360-8,390 ANHYDRITE: off white, soft, amorphous texture; rare SILTSTONE: dark orange-light brown, tan, pink, soft, sub blocky, calcite cement, poorly cemented, no visible porosity

8,390-8,420 Limestone: mudstone, light-medium gray-gray brown, micro crystalline, firm-hard, argillaceous in part, dense, chalky texture, no visible porosity; rare ANHYDRITE: off white, soft, amorphous texture; trace SILTSTONE: dark orange-light brown, tan, pink, soft, sub blocky, calcite cement, poorly cemented

8,420-8,450 SILTSTONE: dark orange-light brown, tan, pink, soft, sub blocky, calcite cement, poorly cement; trace SILTY SANDSTONE: tan-off white, very fine grained, sub rounded, moderately sorted, calcite cement, poorly cemented

8,450-8,480 SILTSTONE: dark orange-light brown, tan, pink, soft, sub blocky, calcite cement, poorly cement; trace SILTY SANDSTONE: tan-off white, very fine grained, sub rounded, moderately sorted, calcite cement, poorly cemented

8,480-8,510 SILTSTONE: dark orange-light brown, tan, soft, sub blocky, calcite cement, poorly cement; trace SILTY SANDSTONE: tan-off white, very fine grained, sub rounded, moderately sorted, calcite cement, poorly cemented

Charles Formation [Mississippian Madison Group] 8,534' MD / 8,533' TVD (-6,457')

8,510-8,540 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline; trace Limestone: mudstone, off white, gray, rare tan, fine crystalline, firm, laminated, crystalline-chalky texture, no visible porosity, no visible oil stain; trace SILTSTONE and SILTY SANDSTONE: as above

8,540-8,570 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline; trace ARGILLACEOUS Limestone: mudstone-wackestone, light-medium brown, tan, rare light-medium gray, rare gray tan, micro crystalline, friable, earthy

8,570-8,600 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline

8,600-8,630 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline

8,630-8,660 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline

8,660-8,690 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline

8,690-8,720 ANHYDRITE: off white, soft, amorphous texture; occasional ARGILLACEOUS Limestone: mudstone-wackestone, light-medium brown, tan, rare light-medium gray, rare gray tan, micro crystalline, friable, earthy

8,720-8,750 ARGILLACEOUS Limestone: mudstone-wackestone, light-medium brown, tan, rare light-medium gray, rare gray tan, micro crystalline, friable, earthy; rare SALT: as above; trace ANHYDRITE: off white, soft, amorphous texture

8,750-8,780 SALT: as above; occasional Limestone: mudstone, gray, off white, rare cream-tan, very fine crystalline, firm, laminated, crystalline-chalky texture, possible intercrystalline porosity, no visible oil stain

8,780-8,810 SALT: as above; occasional Limestone: mudstone, gray, off white, rare cream-tan, very fine crystalline, firm, laminated, crystalline-chalky texture, possible intercrystalline porosity, no visible oil stain

8,810-8,840 Limestone: mudstone-wackestone, tan, cream, light brown, very fine crystalline, firm, laminated, crystalline, rare intercrystalline porosity, occasional even-spotty light-medium brown oil stain; trace DOLOMITE: medium-light brown, micro crystalline, firm, crystalline, occasional intercrystalline porosity, common medium-light brown even oil stain; trace: ANHYDRITE: off white, cream, soft, micro crystalline, anhedral, earthy

8,840-8,870 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline

8,870-8,900 ANHYDRITE: off white, cream, soft, microcrystalline, anhedral, earthy; rare LIMESTONE: mudstone-wackestone, tan, cream, light brown, very fine crystalline, firm, laminated, crystalline, rare intercrystalline porosity, occasional spotty light-medium brown oil stain; trace SALT: as above

8,900-8,930 LIMESTONE: mudstone, light brown, light gray brown, off white, microcrystalline, friable, laminated, earthy, trace intercrystalline porosity, rare even-spotty light-medium brown oil stain; common DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional even-spotty light-medium brown oil stain; trace ANHYDRITE: off white, cream-light orange, soft, microcrystalline, anhedral, earthy

8,930-8,960 LIMESTONE: mudstone, light brown, light gray brown, off white, microcrystalline, friable, laminated, earthy, trace intercrystalline porosity, rare even-spotty light-medium brown oil stain; common DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional even-spotty light-medium brown oil stain; trace ANHYDRITE: off white, cream, soft, microcrystalline, anhedral, earthy

8,960-8,990 LIMESTONE: mudstone, light brown, light gray brown, off white, microcrystalline, friable, laminated, earthy, trace intercrystalline porosity, rare even-spotty light-medium brown oil stain; common DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional even-spotty light-medium brown oil stain; trace ANHYDRITE: off white, cream, soft, microcrystalline, anhedral, earthy

8,990-9,020 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline

9,020-9,050 ANHYDRITE: off white, cream-light orange, soft, microcrystalline, anhedral, earthy; occasional DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional spotty light-medium brown oil stain; trace SALT: as above

9,050-9,080 ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous; occasional DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional spotty light-medium brown oil stain; rare LIMESTONE: mudstone, light brown, light gray brown, off white, microcrystalline, friable, laminated, earthy, trace intercrystalline porosity, trace spotty light-medium brown oil stain

9,080-9,110 LIMESTONE: mudstone, light brown, light gray brown, off white, microcrystalline, friable, laminated, earthy, trace intercrystalline porosity, rare spotty light brown oil stain; occasional DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional spotty light brown oil stain

9,110-9,140 LIMESTONE: mudstone, light brown, light gray brown, off white, microcrystalline, friable, laminated, earthy, trace intercrystalline porosity, rare spotty light brown oil stain; occasional DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional spotty light brown oil stain; rare ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,140-9,170 LIMESTONE: mudstone, light brown, light gray brown, off white, microcrystalline, friable, laminated, earthy, trace intercrystalline porosity, trace spotty light brown oil stain; occasional DOLOMITE: mudstone, light brown, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, trace spotty light brown oil stain; rare ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,170-9,200 SALT: clear-translucent, rarely frosted, crystalline, firm, euhedral, crystalline

Base Last Salt [Charles Formation]

9.210' MD / 9.209' TVD (-7.133')

9,200-9,230 ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous; DOLOMITE: mudstone, light brown, light gray brown, rare light gray, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional spotty light brown oil stain

9,230-9,260 ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous; DOLOMITE: mudstone, light brown, light gray brown, rare light gray, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional spotty light brown oil stain

9,260-9,290 DOLOMITE: mudstone, light brown, light gray, light gray brown, microcrystalline, friable-firm, laminated, earthy trace intercrystalline porosity, occasional spotty light brown oil stain; ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,290-9,320 LIMESTONE: mudstone, light brown-brown, microcrystalline, firm, earthy-crystalline texture, trace intercrystalline porosity, trace spotty light brown oil stain; rare ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,320-9,350 LIMESTONE: mudstone, light gray, light gray brown, rare light brown, firm, earthy-crystalline texture, trace intercrystalline porosity, trace disseminated pyrite, trace spotty light brown oil stain; trace ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,350-9,380 LIMESTONE: mudstone, gray-light gray, gray brown, rare light brown, firm-friable, earthy-crystalline texture, trace intercrystalline porosity, trace disseminated pyrite, no visible oil stain; trace ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,380-9,410 LIMESTONE: mudstone, gray-light gray, gray brown, rare light brown, firm-friable, earthy-crystalline texture, trace intercrystalline porosity, trace disseminated pyrite, no visible oil stain; trace ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,410-9,440 LIMESTONE: mudstone, light brown-off white, light gray brown, trace gray, firm-friable, earthy-crystalline texture, possible intercrystalline porosity, trace disseminated pyrite, argillaceous in part, no visible oil stain; trace ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

Mission Canyon Formation [Mississippian Madison Group]

9.431' MD / 9.430' TVD (-7.354')

9,440-9,470 LIMESTONE: mudstone, light brown-brown, gray brown, trace gray, firm-friable, earthy-crystalline texture, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; trace ANHYDRITE: off white, cream, soft, microcrystalline, massive, earthy-amorphous

9,470-9,500 LIMESTONE: mudstone, light gray, light brown, gray brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,500-9,530 ARGILLACEOUS LIMESTONE: mudstone, light gray, occasional medium gray, rare gray tan, rare off white, trace dark gray, firm-friable, crystalline-chalky texture, trace disseminated pyrite, possible intercrystalline porosity, trace light brown spotty oil stain

9,530-9,560 ARGILLACEOUS LIMESTONE: mudstone, light gray, occasional medium gray, rare gray tan, rare off white, trace dark gray, firm-friable, crystalline-chalky texture, trace disseminated pyrite, no visible porosity, no visible oil stain; LIMESTONE: mudstone, light-gray, rare off white, trace dark gray, trace brown, friable-firm, dense, earthy, possible intercrystalline porosity, trace light brown spotty oil stain

9,560-9,590 Limestone: mudstone, light gray, light brown, gray brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,590-9,620 ARGILLACEOUS LIMESTONE: mudstone, light gray, occasional medium gray, rare gray tan, rare off white, trace dark gray, firm-friable, crystalline-chalky texture, trace disseminated pyrite, trace fossil fragments, trace light brown spotty oil stain

9,620-9,650 Limestone: mudstone, light gray, light brown, gray brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,650-9,680 Limestone: mudstone, light gray, light brown, gray brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,680-9,710 Limestone: mudstone, gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,710-9,740 Limestone: mudstone, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,740-9,770 Limestone: mudstone, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,770-9,800 Limestone: mudstone, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,800-9,830 DOLOMITE: tan-light brown gray, off white, microcrystalline to fine crystalline, rare intercrystalline porosity, argillaceous in part, trace light brown spotty oil stain; rare LIMESTONE: mudstone, cream-tan, gray, trace off white, micro crystalline, friable-firm, dense, massive, trace laminated, occasional Algal laminated, earthy, trace calcite, trace pyrite, no visible porosity, trace dead oil stain

9,830-9,860 Limestone: mudstone, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,860-9,890 Limestone: mudstone, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, trace spotty light brown oil stain

9,890-9,920 ARGILLACEOUS LIMESTONE: mudstone, light gray, occasional medium gray, rare gray tan, rare off white, trace dark gray, firm-friable, crystalline-chalky texture, trace disseminated pyrite, no visible porosity, no visible oil stain; LIMESTONE: mudstone, light-gray, rare off white, trace dark gray, trace brown, friable-firm, dense, earthy, possible intercrystalline porosity, trace light brown spotty oil stain

9,920-9,950 ARGILLACEOUS LIMESTONE: mudstone, light gray, occasional medium gray, rare gray tan, rare off white, trace dark gray, firm-friable, crystalline-chalky texture, trace disseminated pyrite, no visible porosity, no visible oil stain; LIMESTONE: mudstone, light-gray, rare off white, trace dark gray, trace brown, friable-firm, dense, earthy

9,950-9,980 Limestone: mudstone, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, no visible porosity, no visible oil stain

Lodgepole Formation [Mississippian Madison Group] **9,993' MD / 9,992' TVD (-7,916')**

9,980-10,010 Limestone: mudstone, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, trace fossil fragments, no visible porosity, no visible oil stain

10,010-10,040 Limestone: mudstones, light gray-brown, trace dark gray, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,040-10,070 Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,070-10,100 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,100-10,130 Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,130-10,160 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,160-10,190 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,190-10,220 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,220-10,250 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,250-10,280 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,280-10,310 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,310-10,340 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite

10,340-10,370 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite

10,370-10,400 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy, rarely crystalline texture, trace disseminated pyrite

10,400-10,430 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,430-10,460 ARGILLACEOUS Limestone: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,460-10,490 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,490-10,520 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,520-10,550 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,550-10,580 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,580-10,610 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,610-10,640 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,640-10,670 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,670-10,700 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,700-10,730 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

10,730-10,760 ARGILLACEOUS LIMESTONE: mudstone, light gray-gray, gray brown, trace light brown, firm-friable, earthy-crystalline texture, trace disseminated pyrite

False Bakken Member [Lodgepole Formation] 10.766' MD / 10.713' TVD (-8.637')

10,760-10,790 SHALE: black, black gray, hard, splintery, smooth, pyritic, carbonaceous, fracture porosity; trace LIMESTONE: as above

Scallion [Lodgepole Formation] 10.770' MD / 10.716' TVD (-8.640')

10,760-10,790 SHALE: black, black gray, hard, splintery, smooth, pyritic, carbonaceous, fracture porosity; trace LIMESTONE: as above

Upper Bakken Shale [Bakken Formation] 10.782' MD / 10.723' TVD (-8.647')

10,760-10,790 SHALE: black, black gray, hard, splintery, smooth, pyritic, carbonaceous, fracture porosity; trace LIMESTONE: as above

Middle Bakken Member [Bakken Formation] 10.808' MD / 10.739' TVD (-8.663')

10,790-10,820 SILTY SANDSTONE: light gray brown, light brown, trace light gray, very fine grained, friable sub rounded, smooth, moderately sorted, calcite cement moderately cemented, trace disseminated and nodular pyrite, fair intercrystalline porosity, occasional light brown spotty oil stain

10,820-10,850 SILTY SANDSTONE: light gray brown, light brown, trace light gray, very fine grained, friable sub rounded, smooth, moderately sorted, calcite cement moderately cemented, trace disseminated and nodular pyrite, fair intercrystalline porosity, occasional light brown spotty oil stain

10,850-10,880 SILTY SANDSTONE: light gray brown, light brown, trace light gray, very fine grained, friable sub rounded, smooth, moderately sorted, calcite cement moderately cemented, trace disseminated and nodular pyrite, fair intercrystalline porosity, occasional light brown spotty oil stain

Lower Bakken Shale [Bakken Formation] 10,892' MD / 10,784' TVD (-8,708')

10,880-10,910 SHALE: black, black gray, hard, splintery, smooth, pyritic, carbonaceous, fracture porosity; trace SILTY SANDSTONE: light gray brown, light brown, trace light gray, very fine grained, friable sub rounded, smooth, moderately sorted, calcite cement moderately cemented, trace disseminated and nodular pyrite, fair intercrystalline porosity, occasional light brown spotty oil stain

Pronghorn [Bakken Formation] 10,918' MD / 10,795' TVD (-8,719')

10,910-10,940 SILTSTONE: dark gray, trace gray black, friable-firm, sub blocky-sub splintery, moderately dolomite cemented, trace disseminated and nodular pyrite, trace spotty light brown oil stain

10,940-10,970 SILTSTONE: dark gray, trace gray black, friable-firm, sub blocky-sub splintery, moderately dolomite cemented, trace disseminated and nodular pyrite, trace spotty light brown oil stain; common DOLOMITE: mudstone, light brown-gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, possible intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

Three Forks [Devonian] 10,973' MD / 10,813' TVD (-8,737')

10,970-11,000 DOLOMITE: mudstone, light brown-light brown gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,000-11,030 DOLOMITE: mudstone, light brown-light brown gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,030-11,060 DOLOMITE: mudstone, light brown-light brown gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,060-11,090 DOLOMITE: mudstone, light brown-light brown gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,090-11,120 DOLOMITE: mudstone, light brown-light brown gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,120-11,150 DOLOMITE: mudstone, light brown-light brown gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,150-11,185 DOLOMITE: mudstone, light brown-light brown gray, tan-cream, trace pink, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,185-11,210 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, microcrystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain

11,210-11,240 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,240-11,270 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,270-11,300 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,300-11,330 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,330-11,360 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light

11,600-11,630 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,870-11,900 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,900-11,930 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,930-11,960 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,960-11,990 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

11,990-12,020 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,020-12,050 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,050-12,080 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,080-12,110 DOLOMITE: mudstone, light brown-tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,110-12,140 DOLOMITE: mudstone, light tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; trace SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,140-12,170 DOLOMITE: mudstone, light tan, occasional cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; trace SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,170-12,200 DOLOMITE: mudstone, light tan, common cream, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; trace SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,200-12,230 DOLOMITE: mudstone, light-medium tan, common cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; trace SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,230-12,260 DOLOMITE: mudstone, light-medium tan, common cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,260-12,290 DOLOMITE: mudstone, light-medium tan, common cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,290-12,320 DOLOMITE: mudstone, light tan, common cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,320-12,350 DOLOMITE: mudstone, light tan, common cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,350-12,380 DOLOMITE: mudstone, light tan, common cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,380-12,410 DOLOMITE: mudstone, light tan, common cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace even light brown oil stain; rare SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,410-12,440 DOLOMITE: mudstone, light-medium tan, common light brown, occasional cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; occasional SHALE: light green-light gray green, firm, sub blocky, earthy,

occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

12,440-12,470 DOLOMITE: mudstone, light-medium tan, common light brown, occasional cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, common spotty trace even light brown oil stain; occasional SHALE: light green-light gray green, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

12,470-12,500 DOLOMITE: mudstone, light-medium brown, tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; occasional SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

12,500-12,530 DOLOMITE: mudstone, light-medium brown, tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; occasional SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

12,530-12,560 DOLOMITE: mudstone, light-medium tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, common spotty trace even light brown oil stain; abundant SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

12,560-12,590 DOLOMITE: mudstone, light-medium tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; abundant SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

12,590-12,620 DOLOMITE: mudstone, light-medium tan, common light brown, fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; abundant SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,620-12,650 DOLOMITE: mudstone, light-medium tan, common light brown, fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; abundant SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,650-12,680 DOLOMITE: mudstone, light-medium tan, common light brown, fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; abundant SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,680-12,710 DOLOMITE: mudstone, medium tan, common light brown, off white, fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; common SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,710-12,740 DOLOMITE: mudstone, medium tan, common light brown, off white, fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; common SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,740-12,770 DOLOMITE: mudstone, tan, common light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; common SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

12,770-12,800 DOLOMITE: mudstone, tan, common light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; common SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

12,800-12,830 DOLOMITE: mudstone, tan, common light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, occasional spotty trace even light brown oil stain; common SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

12,830-12,860 DOLOMITE: mudstone, light tan, common light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,860-12,890 DOLOMITE: mudstone, light tan, common light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,890-12,920 DOLOMITE: mudstone, light tan, common light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,920-12,950 DOLOMITE: mudstone, light tan, off white, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,950-12,980 DOLOMITE: mudstone, light tan, off white, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

12,980-13,010 DOLOMITE: mudstone, light tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace

SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,010-13,040 DOLOMITE: mudstone, light tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,040-13,070 DOLOMITE: mudstone, light tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,070-13,100 DOLOMITE: mudstone, light tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,100-13,130 DOLOMITE: mudstone, light tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, trace disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,130-13,160 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,160-13,190 DOLOMITE: mudstone, tan, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,190-13,220 DOLOMITE: mudstone, tan, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,220-13,250 DOLOMITE: mudstone, tan, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,250-13,280 DOLOMITE: mudstone, tan, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light green-light gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,550-13,580 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro
sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain;

trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,580-13,610 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,610-13,640 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,640-13,670 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,670-13,700 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,700-13,730 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,730-13,760 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,760-13,790 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,790-13,820 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,820-13,850 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,850-13,880 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,880-13,910 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

13,910-13,940 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain; common DOLOMITE: as above

13,940-13,970 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain; rare DOLOMITE: as above

13,970-14,000 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain; trace DOLOMITE: as above

14,000-14,030 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,030-14,060 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,060-14,090 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,090-14,120 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,120-14,150 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,150-14,180 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,180-14,210 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,210-14,240 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,240-14,270 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,270-14,300 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,300-14,330 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,330-14,360 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain

14,360-14,390 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

14,390-14,420 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

14,420-14,450 DOLOMITE: mudstone, tan, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

14,450-14,480 DOLOMITE: mudstone, tan-light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

14,480-14,510 DOLOMITE: mudstone, tan-light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse-streaming cut fluorescence

14,510-14,540 DOLOMITE: mudstone, tan-light brown, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

14,540-14,570 DOLOMITE: mudstone, off white, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

14,570-14,600 DOLOMITE: mudstone, off white, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

14,600-14,630 DOLOMITE: mudstone, off white, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, even trace light brown oil stain; occasional

SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

14,630-14,660 DOLOMITE: mudstone, off white, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, even trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

14,660-14,690 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, even trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow diffuse cut fluorescence

14,690-14,720 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, even trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

14,720-14,750 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, even trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

14,750-14,780 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown oil stain; abundant SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

14,780-14,810 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown oil stain; abundant SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

14,810-14,840 DOLOMITE: mudstone, tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, even trace light brown oil stain; abundant SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

14,840-14,870 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace light brown oil stain; abundant SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

14,870-14,900 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above; moderately yellow streaming cut fluorescence

14,900-14,930 DOLOMITE: mudstone, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above; moderately yellow streaming cut fluorescence

14,930-14,960 DOLOMITE: mudstone, light tan-light brown, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above; pale yellow streaming cut fluorescence

14,960-14,990 DOLOMITE: mudstone, light tan-light brown, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above; pale yellow streaming cut fluorescence

14,990-15,020 DOLOMITE: mudstone, light tan-light brown, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above, occasional disseminated pyrite; pale yellow streaming cut fluorescence

15,020-15,050 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above, occasional disseminated pyrite; pale yellow streaming cut fluorescence

15,050-15,080 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above, occasional disseminated pyrite; pale yellow streaming cut fluorescence

15,080-15,110 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; pale yellow streaming cut fluorescence

15,110-15,140 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; pale yellow streaming cut fluorescence

15,140-15,170 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above, trace disseminated pyrite; pale yellow diffuse-streaming cut fluorescence

15,170-15,200 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above, trace disseminated pyrite; pale yellow diffuse-streaming cut fluorescence

15,200-15,230 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above, trace disseminated pyrite; pale yellow streaming cut fluorescence

15,230-15,260 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above, trace disseminated pyrite; pale yellow diffuse-streaming cut fluorescence

15,260-15,290 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above, trace disseminated pyrite; pale yellow diffuse-streaming cut fluorescence

15,290-15,320 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above, trace disseminated pyrite; pale yellow diffuse-streaming cut fluorescence

15,320-15,350 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above; pale yellow diffuse-streaming cut fluorescence

15,350-15,380 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above; pale yellow streaming cut fluorescence

15,380-15,410 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; abundant SHALE: as above; pale yellow streaming cut fluorescence

15,410-15,440 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,440-15,470 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,470-15,500 DOLOMITE: mudstone, tan, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,500-15,530 DOLOMITE: mudstone, tan, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,530-15,560 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,560-15,590 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, trace spotty light brown oil stain; rare

SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,590-15,620 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,620-15,650 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,650-15,680 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,680-15,710 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,710-15,740 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,740-15,770 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,770-15,800 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,800-15,830 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,830-15,860 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,860-15,890 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,890-15,920 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,920-15,950 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,950-15,980 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

15,980-16,010 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

16,010-16,040 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

16,040-16,070 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

16,070-16,100 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

16,100-16,130 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

16,130-16,160 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace

16,700-16,730 DOLOMITE: mudstone, light brown-tan, occasional cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace

17,000-17,030 DOLOMITE: mudstone, light tan, cream, rare off white, very fine crystalline, firm, laminated, micro sucrosic, occasional disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,030-17,060 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,060-17,090 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,090-17,120 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,120-17,150 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,150-17,180 DOLOMITE: mudstone, light tan, cream, trace light brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,180-17,210 DOLOMITE: mudstone, light tan, cream, trace light brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,210-17,240 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain; trace DOLOMITE: as above; pale yellow streaming cut fluorescence

17,240-17,270 CLAYSTONE: light gray, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,270-17,300 CLAYSTONE: light gray blue, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,300-17,330 CLAYSTONE: light gray blue, common gray brown, trace off white-white, firm, sub blocky, earthy, common disseminated pyrite, no visible porosity, no visible oil stain; trace DOLOMITE: as above; pale yellow streaming cut fluorescence

17,330-17,360 DOLOMITE: mudstone, light tan, cream, trace light brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace CLAYSTONE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,360-17,390 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace CLAYSTONE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,390-17,420 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,420-17,450 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

17,450-17,480 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

17,480-17,510 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; trace SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

17,510-17,540 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; common SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

17,540-17,570 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; common SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

17,570-17,600 DOLOMITE: mudstone, light tan, cream, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; common SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

17,600-17,630 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty light brown oil stain; rare SHALE:

light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

17,630-17,660 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, occasional disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,660-17,690 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,690-17,720 DOLOMITE: mudstone, light tan, cream, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,720-17,750 DOLOMITE: mudstone, light tan, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,750-17,780 DOLOMITE: mudstone, light tan, off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,780-17,810 DOLOMITE: mudstone, light brown-tan, off white, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,810-17,840 DOLOMITE: mudstone, light brown-tan, off white, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,840-17,870 DOLOMITE: mudstone, light brown-tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,870-17,900 DOLOMITE: mudstone, light brown, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, rare medium gray, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

17,900-17,930 DOLOMITE: mudstone, light tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, rare medium grained, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

17,930-17,960 DOLOMITE: mudstone, light tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, rare medium grained, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

17,960-17,990 DOLOMITE: mudstone, light tan-brown, cream, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, rare medium grained, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

17,990-18,020 DOLOMITE: mudstone, light tan-brown, cream, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, rare medium grained, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,020-18,050 DOLOMITE: mudstone, light tan, cream, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,050-18,080 DOLOMITE: mudstone, light tan, cream, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,080-18,110 DOLOMITE: mudstone, light tan, off white, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,110-18,140 DOLOMITE: mudstone, light tan, off white, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,140-18,170 DOLOMITE: mudstone, light tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,170-18,200 DOLOMITE: mudstone, light tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare

SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,200-18,230 DOLOMITE: mudstone, light tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,230-18,260 DOLOMITE: mudstone, light tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,260-18,290 DOLOMITE: mudstone, light tan, trace light gray, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,290-18,320 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,320-18,350 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,350-18,380 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,380-18,410 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,410-18,440 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,440-18,470 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,470-18,500 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,500-18,530 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,530-18,560 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,560-18,590 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,590-18,620 DOLOMITE: mudstone, light gray, light brown-tan, occasional off white, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slightly yellow streaming cut fluorescence

18,620-18,650 DOLOMITE: mudstone, light brown-tan, rare light gray, trace off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately yellow streaming cut fluorescence

18,650-18,680 DOLOMITE: mudstone, light gray, light brown-tan, occasional off white, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slightly yellow streaming cut fluorescence

18,680-18,710 DOLOMITE: mudstone, light gray, light brown-tan, occasional off white, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slow yellow streaming cut fluorescence

18,710-18,740 DOLOMITE: mudstone, light gray, light brown-tan, occasional off white, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slow yellow streaming cut fluorescence

18,740-18,770 DOLOMITE: mudstone, light gray, light brown-tan, occasional off white, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity,

19,010-19,040 DOLOMITE: mudstone, light gray, light brown-tan, occasional off white, trace light gray brown, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slow yellow streaming cut fluorescence

19,310-19,340 DOLOMITE: mudstone, light gray-brown, tan, occasional off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown

oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slow yellow streaming cut fluorescence

19,340-19,370 DOLOMITE: mudstone, light gray-brown, tan, occasional off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slow pale yellow streaming cut fluorescence

19,370-19,400 DOLOMITE: mudstone, light gray-brown, tan, occasional off white, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slow pale yellow streaming cut fluorescence

19,400-19,430 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; slow pale yellow streaming cut fluorescence

19,430-19,460 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,460-19,490 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, rare spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,490-19,520 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, rare disseminated pyrite, occasional intercrystalline porosity, spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,520-19,550 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,550-19,580 DOLOMITE: mudstone, light gray-brown, tan, trace cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,580-19,610 DOLOMITE: mudstone, light gray-brown, tan, trace cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,610-19,640 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, spotty trace light brown oil stain; occasional SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,640-19,670 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,670-19,700 DOLOMITE: mudstone, light gray, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,700-19,730 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, trace spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately pale yellow streaming cut fluorescence

19,730-19,760 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately pale yellow streaming cut fluorescence

19,760-19,790 DOLOMITE: mudstone, light gray-brown, tan, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately pale yellow streaming cut fluorescence

19,790-19,820 DOLOMITE: mudstone, light gray-brown, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; moderately pale yellow streaming cut fluorescence

19,820-19,850 DOLOMITE: mudstone, light gray-brown, tan, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,850-19,880 DOLOMITE: mudstone, light gray-brown, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,880-19,910 DOLOMITE: mudstone, light gray-brown, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain;

common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,910-19,940 DOLOMITE: mudstone, light gray-brown, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; common SHALE: light gray blue, firm, sub blocky, earthy, trace disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,940-19,970 DOLOMITE: mudstone, light gray-brown, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

19,970-20,000 DOLOMITE: mudstone, light gray-brown, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

20,000-20,030 DOLOMITE: mudstone, light gray-brown, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

20,030-20,060 DOLOMITE: mudstone, light tan-brown, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

20,060-20,090 DOLOMITE: mudstone, light tan-brown, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

20,090-20,120 DOLOMITE: mudstone, light tan, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty trace light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

20,120-20,150 DOLOMITE: mudstone, light tan, off white, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, rare disseminated pyrite, trace spotty light brown oil stain; trace SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow streaming cut fluorescence

20,150-20,180 DOLOMITE: mudstone, light tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; pale yellow diffuse-streaming cut fluorescence

20,180-20,210 DOLOMITE: mudstone, light tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, possible intergranular porosity, no visible oil stain; slow pale yellow diffuse-streaming cut fluorescence

20,210-20,240 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; slow pale yellow diffuse-streaming cut fluorescence

20,240-20,270 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; slow pale yellow diffuse cut fluorescence

20,270-20,300 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence

20,300-20,330 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence

20,330-20,360 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence

20,360-20,390 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence

20,390-20,420 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence

20,420-20,450 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence

20,450-20,480 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare

SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence

20,480-20,519 DOLOMITE: mudstone, tan, off white, cream, very fine crystalline, firm, laminated, micro sucrosic, occasional intercrystalline porosity, occasional disseminated pyrite, trace spotty light brown oil stain; rare
SHALE: light gray blue, firm, sub blocky, earthy, rare disseminated pyrite, possible intergranular porosity, no visible oil stain; pale yellow diffuse cut fluorescence



RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Wednesday, December 31, 2014

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Chalmers 5300 21-19 8T
McKenzie, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc.. Other information required by your office is as follows:

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of</i>	<i>TD Straight Line Projection</i>
Mike McCammond	MWD Operator	O.H.	0'	11125'	10/05/14	10/17/14	MWD	11125'
Mike McCammond	MWD Operator	O.H.	11125'	20456'	12/23/14	12/26/14	MWD	20519'

If any other information is required please contact the undersigned at the letterhead address or phone number.

Douglas Hudson
Well Planner



RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Directional Services, Inc.

19510 Oil Center Blvd.

Houston, Texas 77073

Bus: 281.443.1414

Fax: 281.443.1676

Friday, October 17, 2014

State of North Dakota

County of McKenzie

Subject: **Survey Certification Letter**

Survey Company: **Ryan Directional Services, Inc.**

Job Number: **8158**

Survey Job Type: **Ryan MWD**

Customer: **Oasis Petroleum**

Well Name: **Chalmers 5300 21-19 8T**

Rig Name: **Nabors B-22**

Surface: **48 3' 41.300 N / 103 36' 10.110 W**

A.P.I. No: **33-053-06021**

Location: **McKenzie, ND**

RKB Height: **2076'**

Distance to Bit: **60'**

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of</i>	<i>TD Straight Line Projection</i>
Mike McCammond	MWD Supervisor	OH	0'	11185'	10/05/14	10/17/14	MWD	11185'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

Mike McCammond

MWD Supervisor

Ryan Directional Services, Inc.



RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Directional Services, Inc.

19510 Oil Center Blvd.

Houston, Texas 77073

Bus: 281.443.1414

Fax: 281.443.1676

Sunday, December 28, 2014

State of North Dakota

County of McKenzie

Subject: **Survey Certification Letter**

Survey Company: Ryan Directional Services, Inc.

Job Number: 8417

Survey Job Type: Ryan MWD

Customer: Oasis Petroleum

Well Name: Chalmers 5300 21-19 8T

Rig Name: Nabors B-22

Surface: 48 3' 41.300 N / 103 36' 10.110 W

A.P.I. No: 33-053-06021

Location: McKenzie, ND

RKB Height: 2076'

Distance to Bit: 63'

Surveyor Name	Surveyor Title	Borehole Number	Start Depth	End Depth	Start Date	End Date	Type of	TD Straight
								Line Projection
Mike McCammond	MWD Supervisor	OH	11191'	20456'	12/23/14	12/28/14	MWD	20519'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

Mike McCammond

MWD Supervisor

Ryan Directional Services, Inc.

Report #: 1

Date: _____

**RYAN DIRECTIONAL SERVICES, INC.**

A NABORS COMPANY

Ryan Job # 8158

SURVEY REPORT

Customer: **Oasis Petroleum**
 Well Name: **Chalmers 5300 21-19 8T**
 Rig #: **Nabors B-22**
 API #: **33-053-06021**
 Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **M. McCammond / R. Maddalena**
 Directional Drillers: **RPM**
 Survey Corrected To: **True North**
 Vertical Section Direction: **88.95**
 Total Correction: **8.17**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	147	0.30	295.90	64.00	147.00	-0.34	0.17	-0.35	0.20
2	240	0.30	298.40	66.00	240.00	-0.77	0.39	-0.78	0.01
3	333	0.40	273.50	66.00	333.00	-1.31	0.53	-1.32	0.19
4	427	0.10	334.70	68.00	427.00	-1.67	0.62	-1.68	0.39
5	520	0.30	337.70	68.00	519.99	-1.79	0.92	-1.81	0.22
6	613	0.30	347.50	69.00	612.99	-1.93	1.38	-1.95	0.06
7	702	0.60	288.80	66.00	701.99	-2.41	1.76	-2.44	0.58
8	790	0.50	326.20	69.00	789.99	-3.05	2.23	-3.09	0.41
9	880	0.50	276.40	71.00	879.98	-3.65	2.60	-3.70	0.47
10	970	0.20	304.50	75.00	969.98	-4.17	2.73	-4.22	0.37
11	1063	0.70	292.90	75.00	1062.98	-4.82	3.04	-4.88	0.54
12	1156	0.70	290.30	78.00	1155.97	-5.87	3.46	-5.94	0.03
13	1250	0.50	326.00	78.00	1249.97	-6.63	4.00	-6.70	0.44
14	1343	0.80	327.20	82.00	1342.96	-7.19	4.88	-7.28	0.32
15	1436	1.10	347.00	78.00	1435.95	-7.72	6.30	-7.83	0.47
16	1530	0.70	171.00	77.00	1529.94	-7.82	6.61	-7.95	1.91
17	1623	1.70	181.60	80.00	1622.92	-7.81	4.67	-7.90	1.10
18	1717	1.60	197.20	82.00	1716.88	-8.29	2.02	-8.32	0.49
19	1810	1.60	154.70	82.00	1809.85	-8.16	-0.39	-8.15	1.25
20	1903	1.80	109.10	82.00	1902.81	-6.25	-2.04	-6.22	1.43
21	1997	1.00	91.90	86.00	1996.79	-4.05	-2.55	-4.00	0.95
22	2198	0.90	23.60	80.00	2197.76	-1.64	-1.17	-1.62	0.53
23	2261	1.20	29.80	87.00	2260.75	-1.09	-0.14	-1.09	0.51
24	2354	0.40	209.50	93.00	2353.75	-0.76	0.42	-0.77	1.72
25	2446	0.60	204.50	98.00	2445.74	-1.13	-0.29	-1.13	0.22
26	2540	0.50	187.80	100.00	2539.74	-1.41	-1.15	-1.39	0.20
27	2633	0.70	205.40	104.00	2632.74	-1.72	-2.06	-1.68	0.29
28	2726	0.60	192.00	107.00	2725.73	-2.08	-3.05	-2.03	0.19
29	2820	0.90	113.60	111.00	2819.72	-1.53	-3.83	-1.46	1.04
30	2913	1.10	110.40	113.00	2912.71	-0.03	-4.43	0.05	0.22
31	3007	1.00	106.60	116.00	3006.69	1.59	-4.98	1.68	0.13
32	3100	0.70	100.30	118.00	3099.68	2.92	-5.32	3.02	0.34
33	3193	0.90	104.20	120.00	3192.67	4.18	-5.60	4.29	0.22
34	3287	0.90	104.20	122.00	3286.66	5.61	-5.96	5.72	0.00
35	3380	1.00	99.00	123.00	3379.65	7.11	-6.27	7.23	0.14
36	3474	1.10	110.20	125.00	3473.63	8.76	-6.71	8.88	0.24
37	3567	0.80	108.50	127.00	3566.62	10.20	-7.22	10.34	0.32
38	3660	0.90	111.10	127.00	3659.61	11.49	-7.69	11.63	0.12
39	3754	0.80	111.10	129.00	3753.60	12.78	-8.19	12.94	0.11
40	3847	1.30	67.60	132.00	3846.59	14.37	-8.02	14.52	0.97
41	3940	1.20	58.90	132.00	3939.56	16.19	-7.12	16.33	0.23
42	4034	1.30	55.20	134.00	4033.54	17.93	-6.00	18.04	0.14
43	4127	1.50	61.40	136.00	4126.51	19.89	-4.82	19.98	0.27
44	4221	1.10	59.80	138.00	4220.49	21.77	-3.77	21.84	0.43
45	4314	1.20	55.00	138.00	4313.47	23.35	-2.77	23.41	0.15
46	4407	1.00	61.30	140.00	4406.45	24.88	-1.82	24.92	0.25
47	4500	0.90	62.30	141.00	4499.44	26.25	-1.09	26.28	0.11
48	4593	0.90	52.60	143.00	4592.43	27.49	-0.30	27.50	0.16
49	4687	0.40	33.70	145.00	4686.42	28.28	0.42	28.27	0.57
50	4780	0.30	340.90	145.00	4779.42	28.38	0.92	28.37	0.35
51	4873	0.40	17.80	147.00	4872.42	28.41	1.46	28.39	0.26
52	4967	0.10	35.10	147.00	4966.42	28.57	1.84	28.54	0.33
53	5060	0.20	335.60	149.00	5059.42	28.55	2.05	28.52	0.19
54	5153	0.10	358.20	150.00	5152.42	28.49	2.28	28.45	0.12
55	5247	0.20	317.70	152.00	5246.42	28.38	2.48	28.34	0.15
56	5340	0.30	216.70	154.00	5339.42	28.12	2.41	28.08	0.42
57	5433	0.20	241.80	156.00	5432.42	27.83	2.13	27.79	0.16
58	5527	0.20	160.00	156.00	5526.42	27.74	1.90	27.71	0.28
59	5620	0.20	302.40	158.00	5619.42	27.65	1.84	27.62	0.41
60	5714	0.40	276.30	158.00	5713.41	27.19	1.96	27.16	0.25

Report #: **1**

Date: _____

**RYAN DIRECTIONAL SERVICES, INC.**

A NABORS COMPANY

Ryan Job # **8158****SURVEY REPORT**

Customer: **Oasis Petroleum**
 Well Name: **Chalmers 5300 21-19 8T**
 Rig #: **Nabors B-22**
 API #: **33-053-06021**
 Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **M. McCammond / R. Maddalena**
 Directional Drillers: **RPM**
 Survey Corrected To: **True North**
 Vertical Section Direction: **88.95**
 Total Correction: **8.17**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
61	5807	0.40	266.30	159.00	5806.41	26.54	1.98	26.51	0.07
62	5900	0.30	253.30	161.00	5899.41	25.99	1.89	25.96	0.14
63	5994	0.30	241.60	163.00	5993.41	25.53	1.70	25.50	0.07
64	6087	0.60	212.10	161.00	6086.41	25.05	1.17	25.03	0.40
65	6126	0.60	226.10	152.00	6125.40	24.79	0.85	24.77	0.37
66	6173	0.60	213.70	129.00	6172.40	24.47	0.48	24.46	0.28
67	6267	0.60	116.70	134.00	6266.40	24.62	-0.15	24.63	0.96
68	6360	1.00	101.70	140.00	6359.39	25.84	-0.53	25.86	0.48
69	6453	0.50	127.90	143.00	6452.38	26.95	-0.95	26.97	0.64
70	6547	0.30	153.80	147.00	6546.38	27.37	-1.42	27.40	0.28
71	6640	0.90	175.10	149.00	6639.37	27.53	-2.37	27.57	0.68
72	6734	1.00	187.10	152.00	6733.36	27.46	-3.92	27.54	0.24
73	6827	0.20	53.30	156.00	6826.36	27.48	-4.63	27.57	1.23
74	6920	0.10	338.20	159.00	6919.36	27.58	-4.45	27.67	0.21
75	7014	0.40	317.50	163.00	7013.36	27.33	-4.13	27.41	0.33
76	7107	0.70	18.60	165.00	7106.35	27.31	-3.36	27.38	0.66
77	7200	0.80	20.40	168.00	7199.34	27.74	-2.21	27.78	0.11
78	7294	0.50	30.60	170.00	7293.34	28.19	-1.24	28.22	0.34
79	7387	0.50	357.70	174.00	7386.33	28.40	-0.49	28.41	0.30
80	7480	0.70	311.00	176.00	7479.33	27.97	0.29	27.97	0.55
81	7574	1.60	30.30	177.00	7573.31	28.22	1.80	28.19	1.73
82	7667	1.70	30.00	179.00	7666.28	29.61	4.12	29.54	0.11
83	7760	0.80	14.40	179.00	7759.25	30.49	5.94	30.39	1.03
84	7854	0.70	17.80	181.00	7853.24	30.85	7.12	30.73	0.12
85	7947	0.50	332.90	183.00	7946.24	30.86	8.02	30.72	0.53
86	8040	0.00	359.10	185.00	8039.24	30.68	8.39	30.53	0.54
87	8134	0.30	216.70	186.00	8133.24	30.53	8.19	30.39	0.32
88	8227	0.40	91.70	188.00	8226.24	30.71	7.98	30.56	0.67
89	8320	0.50	92.30	190.00	8319.23	31.44	7.96	31.29	0.11
90	8414	0.50	86.20	192.00	8413.23	32.25	7.97	32.11	0.06
91	8507	0.60	101.10	188.00	8506.23	33.14	7.90	33.00	0.19
92	8600	0.60	116.10	186.00	8599.22	34.05	7.59	33.91	0.17
93	8694	0.60	94.90	188.00	8693.22	34.97	7.33	34.84	0.23
94	8787	0.70	108.80	192.00	8786.21	35.99	7.11	35.87	0.20
95	8881	0.30	155.00	194.00	8880.21	36.63	6.70	36.51	0.57
96	8974	0.60	132.40	195.00	8973.20	37.08	6.15	36.98	0.37
97	9067	0.40	191.70	197.00	9066.20	37.37	5.51	37.27	0.56
98	9161	0.40	160.60	199.00	9160.20	37.40	4.88	37.31	0.23
99	9254	0.40	160.70	201.00	9253.20	37.60	4.26	37.53	0.00
100	9347	0.40	222.20	203.00	9346.19	37.48	3.72	37.42	0.44
101	9441	0.70	201.40	201.00	9440.19	37.04	2.94	36.99	0.38
102	9534	0.60	186.10	204.00	9533.18	36.76	1.93	36.73	0.21
103	9627	0.60	99.90	206.00	9626.18	37.18	1.36	37.16	0.88
104	9720	0.90	95.20	208.00	9719.17	38.38	1.21	38.36	0.33
105	9814	0.80	119.80	212.00	9813.16	39.68	0.81	39.67	0.40
106	9907	0.70	118.70	213.00	9906.15	40.73	0.22	40.73	0.11
107	10000	0.70	95.40	215.00	9999.15	41.78	-0.11	41.79	0.30
108	10094	0.50	115.30	217.00	10093.14	42.72	-0.34	42.74	0.30
109	10187	0.20	181.60	217.00	10186.14	43.08	-0.67	43.10	0.49
110	10253	0.40	111.00	219.00	10252.14	43.29	-0.87	43.31	0.58
111	10300	0.20	180.20	186.00	10299.14	43.44	-1.01	43.46	0.81
112	10331	0.50	115.00	188.00	10330.14	43.56	-1.12	43.59	1.46
113	10362	3.40	80.80	186.00	10361.12	44.59	-1.03	44.62	9.68
114	10393	7.50	78.20	186.00	10391.97	47.49	-0.47	47.51	13.25
115	10424	12.70	76.10	190.00	10422.48	52.80	0.76	52.80	16.81
116	10456	17.10	75.80	190.00	10453.40	60.82	2.76	60.78	13.75
117	10487	21.10	75.80	190.00	10482.69	70.69	5.25	70.61	12.90
118	10518	23.60	75.30	190.00	10511.35	82.16	8.19	82.02	8.09
119	10549	25.50	77.10	192.00	10539.55	94.72	11.26	94.53	6.59
120	10580	28.00	78.90	192.00	10567.23	108.42	14.15	108.18	8.48

Report #: **1**

Date: _____

**RYAN DIRECTIONAL SERVICES, INC.**

A NABORS COMPANY

Ryan Job # **8158****SURVEY REPORT**

Customer: **Oasis Petroleum**
 Well Name: **Chalmers 5300 21-19 8T**
 Rig #: **Nabors B-22**
 API #: **33-053-06021**
 Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **M. McCammond / R. Maddalena**
 Directional Drillers: **RPM**
 Survey Corrected To: **True North**
 Vertical Section Direction: **88.95**
 Total Correction: **8.17**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
121	10611	31.30	78.20	194.00	10594.17	123.50	17.20	123.20	10.70
122	10642	34.30	75.90	194.00	10620.22	139.92	20.97	139.56	10.48
123	10673	37.20	74.70	194.00	10645.38	157.52	25.58	157.08	9.62
124	10704	40.60	73.20	195.00	10669.50	176.32	30.97	175.78	11.38
125	10736	44.50	72.90	194.00	10693.07	197.12	37.28	196.47	12.20
126	10767	48.20	73.80	192.00	10714.47	218.73	43.70	217.96	12.12
127	10798	52.10	74.00	192.00	10734.33	241.70	50.29	240.82	12.59
128	10829	56.00	73.70	192.00	10752.52	265.93	57.27	264.92	12.60
129	10860	59.80	73.30	192.00	10768.99	291.23	64.73	290.10	12.31
130	10891	64.30	73.60	194.00	10783.52	317.62	72.53	316.34	14.54
131	10922	67.60	75.00	195.00	10796.15	345.00	80.18	343.59	11.42
132	10953	71.20	76.10	194.00	10807.06	373.22	87.42	371.69	12.08
133	10985	75.20	77.90	195.00	10816.31	403.19	94.31	401.53	13.61
134	11016	79.00	79.20	195.00	10823.23	432.90	100.30	431.14	12.92
135	11047	82.50	80.20	197.00	10828.21	463.10	105.77	461.24	11.73
136	11078	86.50	81.20	199.00	10831.18	493.63	110.75	491.68	13.30
137	11109	88.80	81.80	199.00	10832.45	524.34	115.33	522.31	7.67
138	11125	89.10	82.00	199.00	10832.74	540.22	117.59	538.15	2.25
139	11191	89.80	84.00	222.00	10833.38	605.85	125.63	603.65	3.21
140	11284	88.70	83.80	222.00	10834.59	698.48	135.51	696.12	1.20
141	11377	90.70	83.30	221.00	10835.08	791.06	145.96	788.52	2.22
142	11470	91.10	83.60	222.00	10833.62	883.62	156.56	880.90	0.54
143	11562	88.00	83.10	222.00	10834.34	975.17	167.22	972.27	3.41
144	11655	88.20	84.00	222.00	10837.43	1067.71	177.66	1064.63	0.99
145	11748	89.10	86.30	222.00	10839.62	1160.47	185.52	1157.26	2.66
146	11841	89.70	89.00	224.00	10840.59	1253.43	189.33	1250.17	2.97
147	11936	90.40	91.80	226.00	10840.51	1348.39	188.67	1345.16	3.04
148	12030	90.00	91.50	228.00	10840.18	1442.29	185.96	1439.12	0.53
149	12125	89.20	91.50	230.00	10840.84	1537.19	183.47	1534.08	0.84
150	12220	89.10	90.60	233.00	10842.25	1632.11	181.73	1629.06	0.95
151	12315	90.90	90.60	233.00	10842.25	1727.07	180.74	1724.05	1.89
152	12409	89.70	90.50	235.00	10841.76	1821.03	179.84	1818.04	1.28
153	12504	89.50	90.50	237.00	10842.42	1915.99	179.01	1913.03	0.21
154	12599	88.30	90.60	239.00	10844.25	2010.94	178.09	2008.01	1.27
155	12694	90.60	90.20	239.00	10845.16	2105.90	177.43	2103.00	2.46
156	12788	89.20	91.10	240.00	10845.32	2199.85	176.37	2196.99	1.77
157	12883	89.60	90.70	242.00	10846.32	2294.79	174.87	2291.97	0.60
158	12978	88.40	90.60	242.00	10847.98	2389.73	173.80	2386.95	1.27
159	13073	89.40	90.60	242.00	10849.80	2484.67	172.80	2481.92	1.05
160	13167	89.00	90.30	244.00	10851.11	2578.63	172.06	2575.91	0.53
161	13262	89.40	90.80	244.00	10852.44	2673.59	171.15	2670.90	0.67
162	13357	87.90	91.10	246.00	10854.68	2768.50	169.58	2765.86	1.61
163	13452	89.20	90.50	246.00	10857.08	2863.42	168.25	2860.81	1.51
164	13546	88.90	90.40	248.00	10858.64	2957.37	167.51	2954.80	0.34
165	13641	90.20	89.60	248.00	10859.39	3052.35	167.51	3049.79	1.61
166	13736	89.70	89.40	249.00	10859.47	3147.34	168.34	3144.79	0.57
167	13831	88.50	88.90	249.00	10860.96	3242.33	169.75	3239.76	1.37
168	13926	89.40	89.20	249.00	10862.70	3337.31	171.33	3334.73	1.00
169	14020	90.00	88.80	249.00	10863.19	3431.31	172.97	3428.72	0.77
170	14115	90.80	89.20	251.00	10862.53	3526.31	174.62	3523.70	0.94
171	14210	89.80	90.00	253.00	10862.03	3621.30	175.29	3618.69	1.35
172	14305	90.10	88.20	251.00	10862.12	3716.29	176.78	3713.68	1.92
173	14399	91.40	87.70	251.00	10860.89	3810.27	180.14	3807.61	1.48
174	14494	92.10	87.20	253.00	10857.98	3905.19	184.37	3902.47	0.91
175	14589	91.00	88.20	251.00	10855.41	4000.13	188.18	3997.35	1.56
176	14683	91.30	88.20	253.00	10853.53	4094.10	191.13	4091.29	0.32
177	14778	90.10	90.50	255.00	10852.37	4189.09	192.21	4186.27	2.73
178	14873	89.50	91.40	255.00	10852.70	4284.03	190.63	4281.25	1.14
179	14968	90.20	91.90	257.00	10852.95	4378.92	187.90	4376.21	0.91
180	15063	89.50	91.00	257.00	10853.20	4473.83	185.49	4471.18	1.20

Report #: **1**

Date: _____

**RYAN DIRECTIONAL SERVICES, INC.**

A NABORS COMPANY

Ryan Job # **8158****SURVEY REPORT**

Customer: **Oasis Petroleum**
 Well Name: **Chalmers 5300 21-19 8T**
 Rig #: **Nabors B-22**
 API #: **33-053-06021**
 Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **M. McCammond / R. Maddalena**
 Directional Drillers: **RPM**
 Survey Corrected To: **True North**
 Vertical Section Direction: **88.95**
 Total Correction: **8.17**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
181	15157	88.60	91.00	257.00	10854.75	4567.75	183.85	4565.15	0.96
182	15252	92.10	89.70	257.00	10854.17	4662.71	183.27	4660.13	3.93
183	15347	88.50	91.10	247.00	10853.68	4757.66	182.61	4755.11	4.07
184	15442	88.10	91.60	257.00	10856.50	4852.53	180.37	4850.04	0.67
185	15537	89.50	91.70	257.00	10858.48	4947.40	177.64	4944.98	1.48
186	15631	89.90	91.80	257.00	10858.98	5041.29	174.76	5038.93	0.44
187	15726	89.70	89.60	257.00	10859.31	5136.24	173.60	5133.92	2.33
188	15821	89.00	89.60	258.00	10860.39	5231.23	174.27	5228.91	0.74
189	15916	87.60	90.10	260.00	10863.20	5326.17	174.52	5323.87	1.56
190	16010	89.00	89.30	258.00	10865.99	5420.12	175.01	5417.82	1.72
191	16105	90.40	86.70	258.00	10866.49	5515.09	178.32	5512.75	3.11
192	16200	91.00	86.50	258.00	10865.33	5610.00	183.96	5607.58	0.67
193	16294	91.20	88.90	258.00	10863.53	5703.96	187.73	5701.48	2.56
194	16389	90.40	88.40	258.00	10862.20	5798.95	189.97	5796.44	0.99
195	16484	88.00	88.90	260.00	10863.53	5893.93	192.20	5891.40	2.58
196	16579	89.50	89.40	260.00	10865.60	5988.90	193.61	5986.36	1.66
197	16674	90.30	89.60	260.00	10865.76	6083.90	194.44	6081.36	0.87
198	16768	91.00	89.70	262.00	10864.70	6177.88	195.02	6175.35	0.75
199	16863	89.70	90.00	260.00	10864.12	6272.87	195.27	6270.34	1.40
200	16958	88.30	90.40	262.00	10865.77	6367.83	194.93	6365.32	1.53
201	17053	86.80	90.20	262.00	10869.84	6462.71	194.44	6460.23	1.59
202	17148	88.30	90.30	262.00	10873.90	6557.60	194.02	6555.14	1.58
203	17243	90.00	90.50	262.00	10875.31	6652.55	193.36	6650.13	1.80
204	17337	91.00	92.20	262.00	10874.49	6746.46	191.15	6744.09	2.10
205	17432	92.40	92.10	262.00	10871.67	6841.27	187.58	6838.98	1.48
206	17527	90.90	89.90	262.00	10868.93	6936.16	185.93	6933.92	2.80
207	17622	89.70	89.90	260.00	10868.43	7031.15	186.09	7028.92	1.26
208	17716	89.00	89.80	262.00	10869.50	7125.13	186.34	7122.91	0.75
209	17811	89.10	90.40	262.00	10871.08	7220.10	186.17	7217.90	0.64
210	17906	89.90	91.80	262.00	10871.90	7315.02	184.35	7312.87	1.70
211	18001	89.60	92.10	264.00	10872.32	7409.89	181.12	7407.82	0.45
212	18095	90.10	92.30	264.00	10872.57	7503.74	177.51	7501.75	0.57
213	18190	91.50	92.30	264.00	10871.24	7598.56	173.70	7596.66	1.47
214	18285	90.60	90.90	264.00	10869.50	7693.44	171.04	7691.60	1.75
215	18380	89.80	88.50	262.00	10869.17	7788.43	171.54	7786.59	2.66
216	18475	88.50	88.10	266.00	10870.58	7883.41	174.36	7881.54	1.43
217	18506	88.10	88.10	264.00	10871.50	7914.39	175.39	7912.51	1.29
218	18569	88.80	87.80	262.00	10873.20	7977.36	177.64	7975.44	1.21
219	18664	92.30	87.40	264.00	10872.29	8072.31	181.62	8070.34	3.71
220	18759	91.00	88.60	264.00	10869.55	8167.26	184.93	8165.24	1.86
221	18854	91.60	88.40	266.00	10867.40	8262.23	187.42	8260.18	0.67
222	18948	90.20	90.40	264.00	10865.92	8356.21	188.40	8354.16	2.60
223	19043	91.00	90.00	262.00	10864.93	8451.18	188.07	8449.15	0.94
224	19138	89.90	90.30	264.00	10864.18	8546.15	187.82	8544.15	1.20
225	19233	89.20	90.10	267.00	10864.93	8641.13	187.49	8639.14	0.77
226	19328	90.10	90.00	267.00	10865.51	8736.11	187.41	8734.14	0.95
227	19422	87.90	90.30	267.00	10867.15	8830.07	187.16	8828.12	2.36
228	19517	88.70	91.50	267.00	10869.96	8924.97	185.67	8923.06	1.52
229	19612	89.60	91.60	269.00	10871.37	9019.86	183.10	9018.02	0.95
230	19707	89.90	90.90	269.00	10871.79	9114.78	181.03	9112.99	0.80
231	19802	88.00	90.50	269.00	10873.53	9209.71	179.87	9207.96	2.04
232	19896	89.90	90.30	269.00	10875.25	9303.66	179.21	9301.94	2.03
233	19991	88.40	91.10	269.00	10876.66	9398.61	178.05	9396.92	1.79
234	20086	89.10	90.10	269.00	10878.73	9493.54	177.05	9491.89	1.28
235	20181	88.60	90.90	269.00	10880.64	9588.49	176.23	9586.87	0.99
236	20276	89.30	90.40	267.00	10882.38	9683.43	175.15	9681.84	0.91
237	20370	90.40	90.30	269.00	10882.63	9777.40	174.57	9775.84	1.18
238	20456	90.40	90.80	269.00	10882.03	9863.36	173.75	9861.83	0.58
Projection	20519	90.40	90.80	PTB	10881.59	9926.33	172.87	9924.83	0.00

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28636

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date July 29, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☒ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Change casing

Well Name and Number Chalmers 5300 21-19 8T					
Footages	Qtr-Qtr	Section	Township	Range	
2226 F N L 326 F W L	LOT2	19	153 N	100 W	
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests permission to make the following changes to the above referenced well:

- Surface casing changed to 13 3/8" and depth changed to 2,126'
- Contingency 9 5/8" casing added
- 7' casing changed to all 32#

Attached are revised drill plan, well summary, directional plan and plot

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Regulatory Assistant	Date July 29, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 8-15-14	
By <i>Heather K. Ebel</i>	
Title Petroleum Resource Specialist	

**Oasis Petroleum
Well Summary
Chalmers 5300 21-19 8T
Sec. 19 T153N R100W
McKenzie County, North Dakota**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
13-3/8"	0' to 2,126'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,126'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.14	2730 / 2.74	514 / 2.57	

API Rating & Safety Factor

- a) Collapse pressure based on full casing evacuation with 9 ppg fluid on backside (2126' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2126' setting depth).
- c) Tension based on string weight in 9 ppg fluid at 2126' TVD plus 100k# overpull. (Buoyed weight equals 100k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **629 sks** (325 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCL₂, 4% D079 Extender, and 2% D053 Expanding Agent.

Tail Slurry: **374 sks** (77 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25 lb/sk Lost Circulation Control Agent, and .25% CaCL₂.

**Oasis Petroleum
Well Summary
Chalmers 5300 21-19 8T
Sec. 19 T153N R100W
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' - 6000'	40	HCL-80	LTC	8.835"	8.75"***	5,450	7,270	9,090

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6000'	9-5/8", 40#, HCL-80, LTC, 8rd	3090 / 3.96*	5750 / 1.23	837 / 2.75

API Rating & Safety Factor

- d) Collapse pressure based on 11.5ppg fluid on backside and 9ppg fluid inside of casing.
- e) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- f) Tension based on string weight in 10 ppg fluid, (217k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 9-5/8" casing set in an 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): 20 bbls Chem wash

Lead Slurry: 540 sks (280 bbls) Conventional system with 75 lb/sk cement, 0.5lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti foam, and 0.4% fluid loss

Tail Slurry: 373 sks (77 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti foam, and 0.1% retarder

**Oasis Petroleum
Well Summary
Chalmers 5300 21-19 8T
Sec. 19 T153N R100W
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' - 11106'	32	HCP-110	LTC	6.094"	6.000"***	6730	8970	9870

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 11106'	7", 32#, P-110, LTC, 8rd	11820 / 2.10*	12460 / 1.28	897 / 2.23
6741' - 9209'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.28**	12460 / 1.30	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,818' TVD.
- c. Based on string weight in 10 ppg fluid, 301k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls** Saltwater
 40 bbls Weighted MudPush Express

Lead Slurry: **219 sks** (86 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 Extender, 3.0% KCl, 3.0% D154 Extender, 0.3% D208 Viscosifier, 0.07% Retarder, 0.2% Anti Foam, 0.5lb/sk D130 LCM

Tail Slurry: **614 sks** (168 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% Retarder, 0.2% Fluid Loss, 0.2% Anti Foam, 0.5 lb/sk LCM

**Oasis Petroleum
Well Summary
Chalmers 5300 21-19 8T
Sec. 19 T153N R100W
McKenzie County, North Dakota**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Torque
4-1/2"	10291' - 20582'	13.5	P-110	BTC	3.920"	3.795"	2270

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
10291' - 20582'	4-1/2", 13.5 lb, P-110, BTC	10670 / 1.98	12410 / 1.28	443 / 2.02

API Rating & Safety Factor

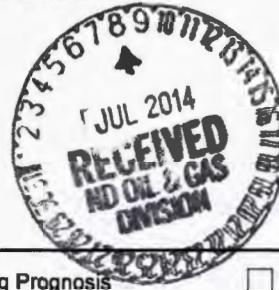
- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10876' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10876' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 119k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28636

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date July 7, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	location revision

Well Name and Number Chalmers 5300 21-19 8T					
Footages	Qtr-Qtr	Section	Township	Range	
2226 F N L 326 F W L	LOT2	19	153 N	100 W	
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

We respectfully request permission to revise the APD issued for this well to include the following change:
The surface location will change to 2226' FNL & 327' FWL, Lot 19-153N-100W. (This modification was made to accommodate the drilling rigs)

Valid plats are attached to replace those within the approved permit.

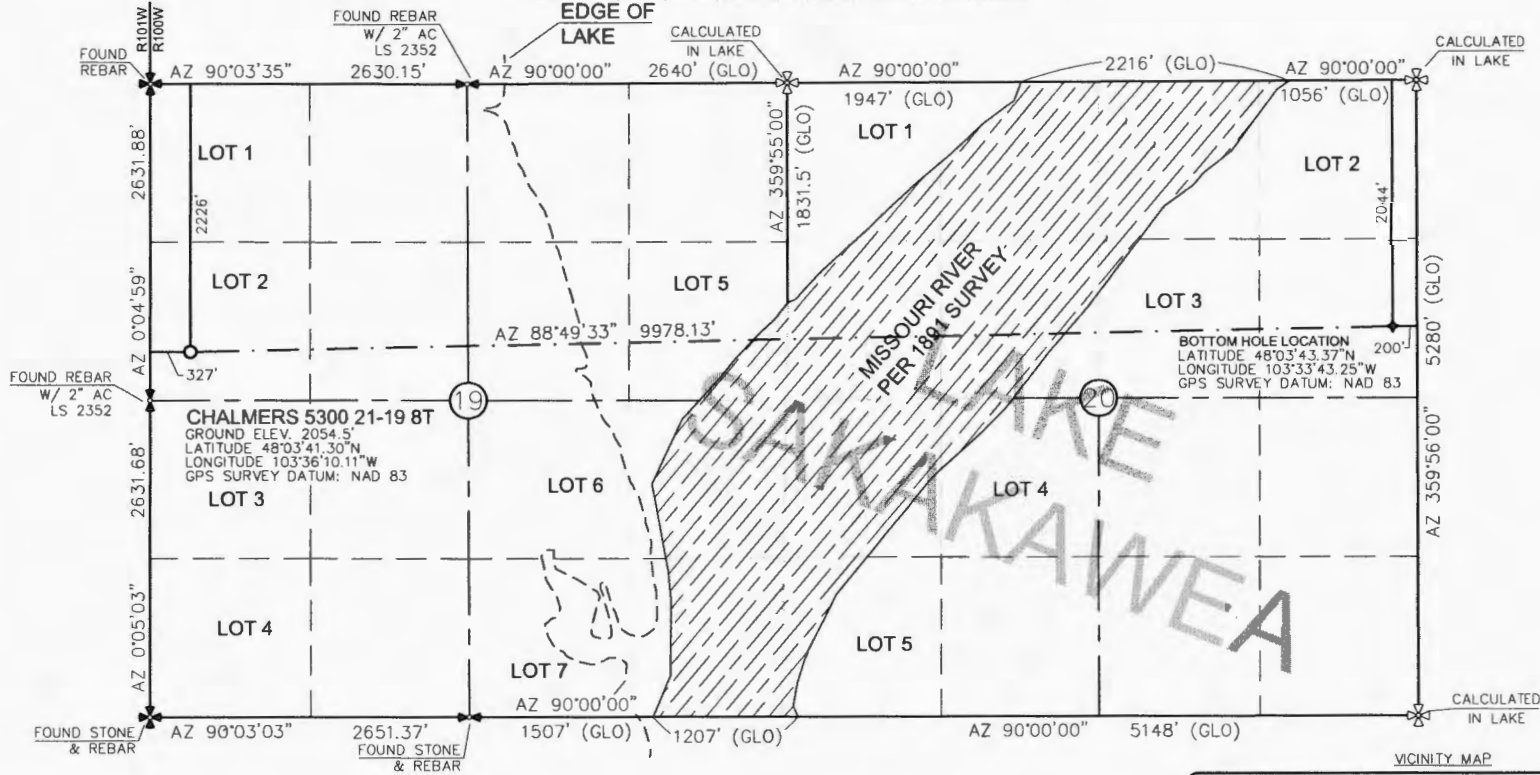
The following statements remain true:
Notice has been provided to the owner of any permanently occupied dwelling within 1320 feet.
This well is not located within 500 feet of an occupied dwelling.

Attached is the new well plat, drill plan, directional plan, and directional plot.

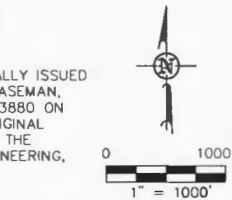
Company Oasis Petroleum Inc.		Telephone Number (281) 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Regulatory Assistant	Date July 7, 2014		
Email Address hmcowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 7/8/14	
By <i>Anthony E. White</i>	
Title Petroleum Resource Specialist	

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5300 21-19 8T"
 2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
 SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED
 AND SEALED BY DARYL D. KASEMAN,
 PLS, REGISTRATION NUMBER 3880 ON
 5/07/14 AND THE ORIGINAL
 DOCUMENTS ARE STORED AT THE
 OFFICES OF INTERSTATE ENGINEERING,
 INC.

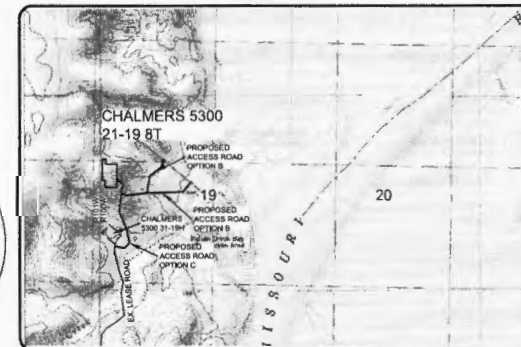
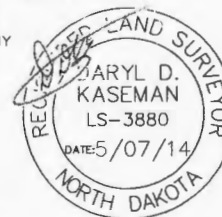


⛶ - MONUMENT - RECOVERED
 ⛶ - MONUMENT - NOT RECOVERED

STAKED ON 1/29/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
 REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY
 THAT THIS PLAT CORRECTLY REPRESENTS WORK
 PERFORMED BY ME OR UNDER MY SUPERVISION
 AND IS TRUE AND CORRECT TO THE BEST OF MY
 KNOWLEDGE AND BELIEF.

[Signature]
 DARYL D. KASEMAN LS-3880



© 2014, INTERSTATE ENGINEERING, INC.

Revision	Date	By	Description
REV 1	07/24/14	JK	200' WIDE WELLS ON PAD
REV 2	02/27/14	JK	WIDEN WELLS ON PAD/REMOVE PAD
REV 3	02/14/14	JK	WIDEN WELLS ON PAD/REMOVE PAD

OASIS PETROLEUM NORTH AMERICA, LLC	
WELL LOCATION PLAT	
SECTION 19, T153N, R100W	
MCKENZIE COUNTY, NORTH DAKOTA	
Drawn By: J.D.K.	Project No.: 813492-2013
Checked By: J.D.K.	Date: JAN 2014

Interstate Engineering, Inc.
 P.O. Box 948
 425 East Main Street
 Bismarck, ND 58070
 Ph: (408) 433-5617
 Fax: (408) 433-5618
 www.interstateeng.com
 Other offices in Colorado, New Mexico and South Dakota



1/8
 SHEET NO.



Azimuths to True North
Magnetic North: 8.17°

Magnetic Field
Strength: 56490.5nT
Dip Angle: 72.96°
Date: 2/17/2014
Model: IGRF200510



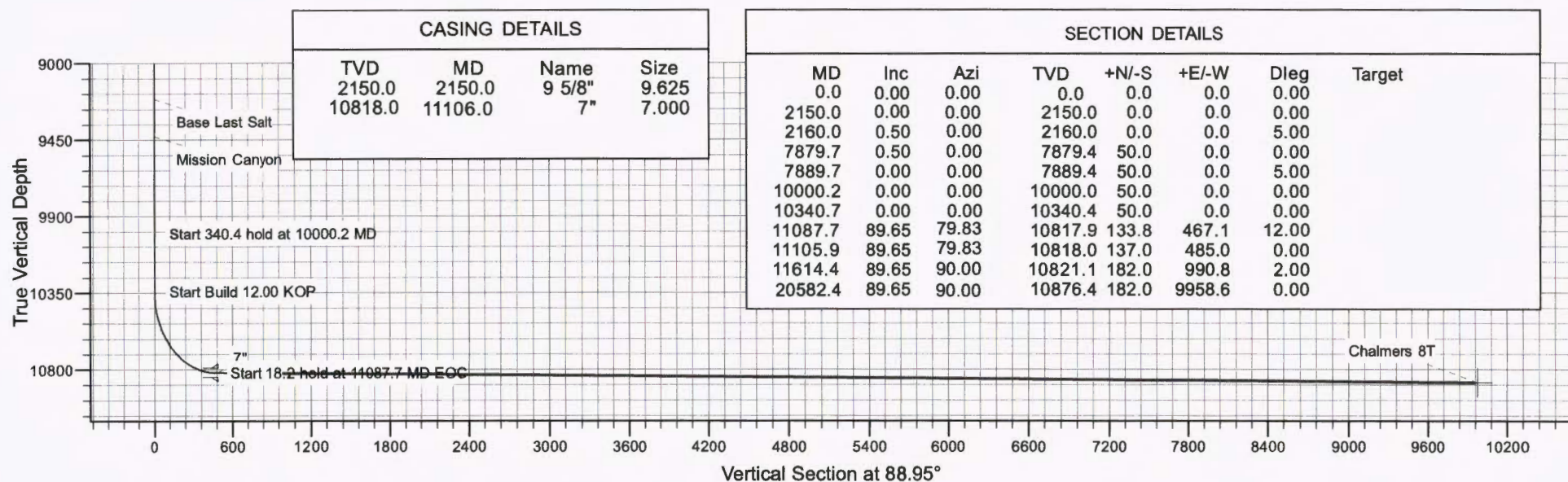
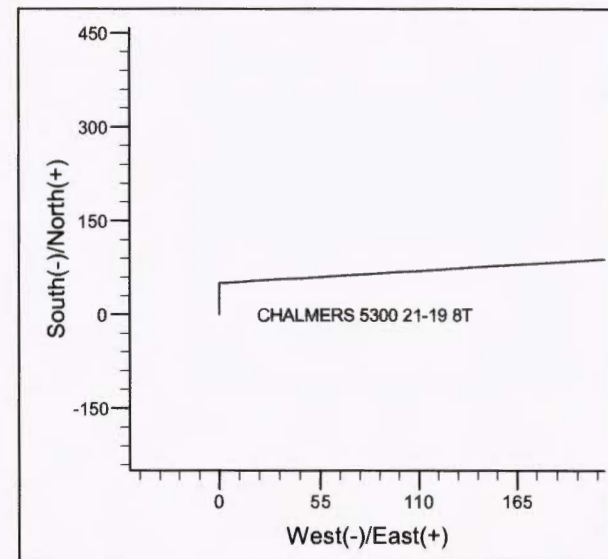
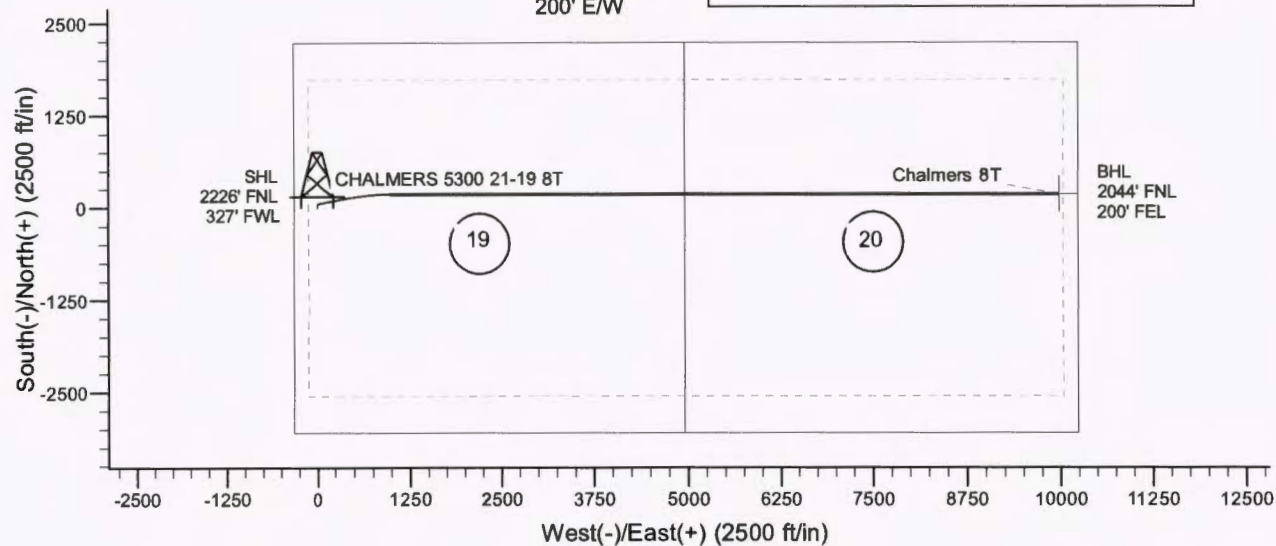
Project: Indian Hills
Site: 153N-100W-19/20
Well: CHALMERS 5300 21-19 8T
Wellbore: CHALMERS 5300 21-19 8T
Design: Plan #1

Setbacks
500' N/S
200' E/W

SITE DETAILS: 153N-100W-19/20

Well Centre Latitude: 48° 3' 41.300 N
Longitude: 103° 36' 10.110 W

Positional Uncertainty: 0.0
Convergence: -2.31
Local North: True



DRILLING PLAN									
OPERATOR		Oasis Petroleum				COUNTY/STATE		McKenzie Co., ND	
WELL NAME		Chalmers 5300 21-19 8T				RIG		B 25	
WELL TYPE		Horizontal Three Forks							
LOCATION		SW NW 19-153N-100W				Surface Location (survey plat):		2225' FNL 327' FWL	
EST. T.D.		20,582'				GROUND ELEV:		2,046'	
		TOTAL LATERAL: 9,478'				KB ELEV:		2,071'	
MARKER		TVD		Subsea TVD		LOGS:			
Pierre		NDIC MAP		2,021		OH Logs: Request a Sundry for an Open Hole Log Waiver: Oasis Chalmers 5300 31-19H 1,850' to S sec 19 153N 100W			
Greenhorn				4,624		S sec 19 153N 100W			
Mowry				5,029		CBL/GR: Above top of cement/GR to base of casing			
Dakota				5,417		MWD GR: KOP to lateral TD			
Rierdon				8,463		DEVIATION: Surf: 3 deg. max., 1 deg / 100'; srvy every 500'			
Dunham Salt				8,891		Prod: 5 deg. max., 1 deg / 100'; srvy every 100'			
Dunham Salt Base				8,960					
Pine Salt				7,257					
Pine Salt Base				7,290					
Opeche Salt				7,351					
Opeche Salt Base				7,426					
Amsden				7,662					
Tyler				7,828					
Otter/Base Minnelusa				8,032		DST'S: None planned			
Kibbey Lime				8,384					
Charles Salt				8,534		CORES: None planned			
Base Last Salt				9,209					
Mission Canyon				9,429					
Lodgepole				9,993					
False Bakken				10,706					
Upper Bakken Shale				10,716		MUDLOGGING: Two-Man: Begin 200' above Kibbey			
Middle Bakken				10,732		30' samples in curve and lateral			
Lower Bakken Shale				10,766					
Pronghorn				10,780					
Threeforks				10,799					
Threeforks(Top of Target)				10,811					
Threeforks(Base of Target)				10,822					
Claystone				10,822		BOP: 11" 5000 psi blind, pipe & annular			
Est. Dip Rate: 12 deg									
Max. Anticipated BHP: 4880									
Surface Formation: Glacial till									
MUD:		Interval		Type		WT		Vis	
Surface:		0' -		2,150'		FW		28-32	
Intermediate:		2,150' -		11,106'		Invert		40-50	
Lateral:		11,106' -		20,582'		Salt Water		28-32	
CASING:		Size		Wt ppf		Hole		Depth	
Surface:		9-5/8"		36#		13-1/2"		2,150'	
Intermediate:		7"		32#		8-3/4"		11,106'	
Production Liner:		4.5"		13.5#		6"		20,582'	
								TOL @ 10,291'	
PROBABLE PLUGS, IF REQ'D:									
OTHER:									
Surface:		MD		TVD		FNL/FSL		FEL/FWL	
KOP:		2,150'		2,150'		2226 FNL		327 FWL	
EOC:		10,341'		10,341'		2176 FNL		327 FWL	
Casing Point:		11,088'		10,818'		2092 FNL		793 FWL	
Three Forks Lateral TD:		11,106'		10,818'		2089 FNL		811 FWL	
		20,582'		10,876'		2044 FNL		200 FEL	
								S-T-R	
								SEC. 19 T153N R100W	
								SEC. 19 T153N R100W	
								SEC. 19 T153N R100W	
								SEC. 19 T153N R100W	
								SEC. 20 T153N R100W	
								AZI	
								Survey Company:	
								Build Rate: 12 deg /100'	
								79.8	
								79.8	
								90.0	
Comments:									
Request a Sundry for an Open Hole Log Waiver: Oasis Chalmers 5300 31-19H 1,850' to S sec 19 153N 100W									
No frac string planned									
35 packers and 25 sleeves planned 3.6MM lbs 30% ceramic									
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.									
68334-30-5 (Primary Name: Fuel oil, diesel) 68476-34-6 (Primary Name: Fuel oil, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)									
68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)									
									
Geology: N. Gabelman 2/4/2014 Engineering: 2/4/2014									

Oasis

Indian Hills

153N-100W-19/20

CHALMERS 5300 21-19 8T

CHALMERS 5300 21-19 8T

Plan: Plan #1

Standard Planning Report

11 June, 2014

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-19/20			
Site Position:		Northing:	402,777.74 ft	Latitude: 48° 3' 44.270 N
From:	Lat/Long	Easting:	1,209,962.51 ft	Longitude: 103° 36' 10.700 W
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence: -2.31 °

Well	CHALMERS 5300 21-19 8T			
Well Position	+N/-S	-300.9 ft	Northing:	402,475.42 ft
	+E/-W	40.1 ft	Easting:	1,209,990.43 ft
Position Uncertainty		0.0 ft	Wellhead Elevation:	Ground Level: 2,046.0 ft

Wellbore	CHALMERS 5300 21-19 8T				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	2/17/2014	8.17	72.96	56,491

Design	Plan #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W	Direction
	(ft)	(ft)	(ft)	(°)
	0.0	0.0	0.0	88.95

Plan Sections										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,150.0	0.00	0.00	2,150.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,160.0	0.50	0.00	2,160.0	0.0	0.0	5.00	5.00	0.00	0.00	
7,879.7	0.50	0.00	7,879.4	50.0	0.0	0.00	0.00	0.00	0.00	
7,889.7	0.00	0.00	7,889.4	50.0	0.0	5.00	-5.00	0.00	180.00	
10,000.2	0.00	0.00	10,000.0	50.0	0.0	0.00	0.00	0.00	0.00	
10,340.7	0.00	0.00	10,340.4	50.0	0.0	0.00	0.00	0.00	0.00	
11,087.7	89.65	79.83	10,817.9	133.8	467.1	12.00	12.00	0.00	79.83	
11,105.9	89.65	79.83	10,818.0	137.0	485.0	0.00	0.00	0.00	0.00	
11,614.4	89.65	90.00	10,821.1	182.0	990.8	2.00	0.00	2.00	90.05	
20,582.4	89.65	90.00	10,876.4	182.0	9,958.6	0.00	0.00	0.00	0.00	

Oasis Petroleum

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-19/20
Well: CHALMERS 5300 21-19 8T
Wellbore: CHALMERS 5300 21-19 8T
Design: Plan #1

Local Co-ordinate Reference:
TVD Reference:
MD Reference:
North Reference:
Survey Calculation Method:

Well CHALMERS 5300 21-19 8T
WELL @ 2071.0ft (Original Well Elev)
WELL @ 2071.0ft (Original Well Elev)
True
Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,021.0	0.00	0.00	2,021.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
2,150.0	0.00	0.00	2,150.0	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 5.00 - 9 5/8"									
2,160.0	0.50	0.00	2,160.0	0.0	0.0	0.0	5.00	5.00	0.00
Start 5719.7 hold at 2160.0 MD									
2,200.0	0.50	0.00	2,200.0	0.4	0.0	0.0	0.00	0.00	0.00
2,300.0	0.50	0.00	2,300.0	1.3	0.0	0.0	0.00	0.00	0.00
2,400.0	0.50	0.00	2,400.0	2.1	0.0	0.0	0.00	0.00	0.00
2,500.0	0.50	0.00	2,500.0	3.0	0.0	0.1	0.00	0.00	0.00
2,600.0	0.50	0.00	2,600.0	3.9	0.0	0.1	0.00	0.00	0.00
2,700.0	0.50	0.00	2,700.0	4.8	0.0	0.1	0.00	0.00	0.00
2,800.0	0.50	0.00	2,800.0	5.6	0.0	0.1	0.00	0.00	0.00
2,900.0	0.50	0.00	2,900.0	6.5	0.0	0.1	0.00	0.00	0.00
3,000.0	0.50	0.00	3,000.0	7.4	0.0	0.1	0.00	0.00	0.00
3,100.0	0.50	0.00	3,100.0	8.2	0.0	0.2	0.00	0.00	0.00
3,200.0	0.50	0.00	3,200.0	9.1	0.0	0.2	0.00	0.00	0.00
3,300.0	0.50	0.00	3,300.0	10.0	0.0	0.2	0.00	0.00	0.00
3,400.0	0.50	0.00	3,400.0	10.9	0.0	0.2	0.00	0.00	0.00
3,500.0	0.50	0.00	3,499.9	11.7	0.0	0.2	0.00	0.00	0.00
3,600.0	0.50	0.00	3,599.9	12.6	0.0	0.2	0.00	0.00	0.00
3,700.0	0.50	0.00	3,699.9	13.5	0.0	0.2	0.00	0.00	0.00
3,800.0	0.50	0.00	3,799.9	14.4	0.0	0.3	0.00	0.00	0.00
3,900.0	0.50	0.00	3,899.9	15.2	0.0	0.3	0.00	0.00	0.00
4,000.0	0.50	0.00	3,999.9	16.1	0.0	0.3	0.00	0.00	0.00
4,100.0	0.50	0.00	4,099.9	17.0	0.0	0.3	0.00	0.00	0.00
4,200.0	0.50	0.00	4,199.9	17.8	0.0	0.3	0.00	0.00	0.00
4,300.0	0.50	0.00	4,299.9	18.7	0.0	0.3	0.00	0.00	0.00
4,400.0	0.50	0.00	4,399.9	19.6	0.0	0.4	0.00	0.00	0.00
4,500.0	0.50	0.00	4,499.9	20.5	0.0	0.4	0.00	0.00	0.00
4,600.0	0.50	0.00	4,599.9	21.3	0.0	0.4	0.00	0.00	0.00
4,624.1	0.50	0.00	4,624.0	21.5	0.0	0.4	0.00	0.00	0.00
Greenhorn									

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,700.0	0.50	0.00	4,699.9	22.2	0.0	0.4	0.00	0.00	0.00
4,800.0	0.50	0.00	4,799.9	23.1	0.0	0.4	0.00	0.00	0.00
4,900.0	0.50	0.00	4,899.9	24.0	0.0	0.4	0.00	0.00	0.00
5,000.0	0.50	0.00	4,999.9	24.8	0.0	0.5	0.00	0.00	0.00
5,029.1	0.50	0.00	5,029.0	25.1	0.0	0.5	0.00	0.00	0.00
Mowry									
5,100.0	0.50	0.00	5,099.9	25.7	0.0	0.5	0.00	0.00	0.00
5,200.0	0.50	0.00	5,199.9	26.6	0.0	0.5	0.00	0.00	0.00
5,300.0	0.50	0.00	5,299.9	27.4	0.0	0.5	0.00	0.00	0.00
5,400.0	0.50	0.00	5,399.9	28.3	0.0	0.5	0.00	0.00	0.00
5,417.1	0.50	0.00	5,417.0	28.5	0.0	0.5	0.00	0.00	0.00
Dakota									
5,500.0	0.50	0.00	5,499.9	29.2	0.0	0.5	0.00	0.00	0.00
5,600.0	0.50	0.00	5,599.9	30.1	0.0	0.5	0.00	0.00	0.00
5,700.0	0.50	0.00	5,699.9	30.9	0.0	0.6	0.00	0.00	0.00
5,800.0	0.50	0.00	5,799.9	31.8	0.0	0.6	0.00	0.00	0.00
5,900.0	0.50	0.00	5,899.9	32.7	0.0	0.6	0.00	0.00	0.00
6,000.0	0.50	0.00	5,999.9	33.6	0.0	0.6	0.00	0.00	0.00
6,100.0	0.50	0.00	6,099.8	34.4	0.0	0.6	0.00	0.00	0.00
6,200.0	0.50	0.00	6,199.8	35.3	0.0	0.6	0.00	0.00	0.00
6,300.0	0.50	0.00	6,299.8	36.2	0.0	0.7	0.00	0.00	0.00
6,400.0	0.50	0.00	6,399.8	37.0	0.0	0.7	0.00	0.00	0.00
6,463.2	0.50	0.00	6,463.0	37.6	0.0	0.7	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	0.00	6,499.8	37.9	0.0	0.7	0.00	0.00	0.00
6,600.0	0.50	0.00	6,599.8	38.8	0.0	0.7	0.00	0.00	0.00
6,700.0	0.50	0.00	6,699.8	39.7	0.0	0.7	0.00	0.00	0.00
6,800.0	0.50	0.00	6,799.8	40.5	0.0	0.7	0.00	0.00	0.00
6,891.2	0.50	0.00	6,891.0	41.3	0.0	0.8	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.50	0.00	6,899.8	41.4	0.0	0.8	0.00	0.00	0.00
6,960.2	0.50	0.00	6,960.0	41.9	0.0	0.8	0.00	0.00	0.00
Dunham Salt Base									
7,000.0	0.50	0.00	6,999.8	42.3	0.0	0.8	0.00	0.00	0.00
7,100.0	0.50	0.00	7,099.8	43.2	0.0	0.8	0.00	0.00	0.00
7,200.0	0.50	0.00	7,199.8	44.0	0.0	0.8	0.00	0.00	0.00
7,257.2	0.50	0.00	7,257.0	44.5	0.0	0.8	0.00	0.00	0.00
Pine Salt									
7,290.2	0.50	0.00	7,290.0	44.8	0.0	0.8	0.00	0.00	0.00
Pine Salt Base									
7,300.0	0.50	0.00	7,299.8	44.9	0.0	0.8	0.00	0.00	0.00
7,351.2	0.50	0.00	7,351.0	45.3	0.0	0.8	0.00	0.00	0.00
Opeche Salt									
7,400.0	0.50	0.00	7,399.8	45.8	0.0	0.8	0.00	0.00	0.00
7,426.2	0.50	0.00	7,426.0	46.0	0.0	0.8	0.00	0.00	0.00
Opeche Salt Base									
7,500.0	0.50	0.00	7,499.8	46.6	0.0	0.9	0.00	0.00	0.00
7,600.0	0.50	0.00	7,599.8	47.5	0.0	0.9	0.00	0.00	0.00
7,662.2	0.50	0.00	7,662.0	48.1	0.0	0.9	0.00	0.00	0.00
Amsden									
7,700.0	0.50	0.00	7,699.8	48.4	0.0	0.9	0.00	0.00	0.00
7,800.0	0.50	0.00	7,799.8	49.3	0.0	0.9	0.00	0.00	0.00
7,828.2	0.50	0.00	7,828.0	49.5	0.0	0.9	0.00	0.00	0.00
Tyler									

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,879.7	0.50	0.00	7,879.4	50.0	0.0	0.9	0.00	0.00	0.00
Start Drop -5.00									
7,889.7	0.00	0.00	7,889.4	50.0	0.0	0.9	5.00	-5.00	0.00
Start 2110.6 hold at 7889.7 MD									
7,900.0	0.00	0.00	7,899.8	50.0	0.0	0.9	0.00	0.00	0.00
8,000.0	0.00	0.00	7,999.8	50.0	0.0	0.9	0.00	0.00	0.00
8,032.2	0.00	0.00	8,032.0	50.0	0.0	0.9	0.00	0.00	0.00
Otter/Base Minnelusa									
8,100.0	0.00	0.00	8,099.8	50.0	0.0	0.9	0.00	0.00	0.00
8,200.0	0.00	0.00	8,199.8	50.0	0.0	0.9	0.00	0.00	0.00
8,300.0	0.00	0.00	8,299.8	50.0	0.0	0.9	0.00	0.00	0.00
8,384.2	0.00	0.00	8,384.0	50.0	0.0	0.9	0.00	0.00	0.00
Kibbey Lime									
8,400.0	0.00	0.00	8,399.8	50.0	0.0	0.9	0.00	0.00	0.00
8,500.0	0.00	0.00	8,499.8	50.0	0.0	0.9	0.00	0.00	0.00
8,534.2	0.00	0.00	8,534.0	50.0	0.0	0.9	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,599.8	50.0	0.0	0.9	0.00	0.00	0.00
8,700.0	0.00	0.00	8,699.8	50.0	0.0	0.9	0.00	0.00	0.00
8,800.0	0.00	0.00	8,799.8	50.0	0.0	0.9	0.00	0.00	0.00
8,900.0	0.00	0.00	8,899.8	50.0	0.0	0.9	0.00	0.00	0.00
9,000.0	0.00	0.00	8,999.8	50.0	0.0	0.9	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	50.0	0.0	0.9	0.00	0.00	0.00
9,200.0	0.00	0.00	9,199.8	50.0	0.0	0.9	0.00	0.00	0.00
9,209.2	0.00	0.00	9,209.0	50.0	0.0	0.9	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	50.0	0.0	0.9	0.00	0.00	0.00
9,400.0	0.00	0.00	9,399.8	50.0	0.0	0.9	0.00	0.00	0.00
9,429.2	0.00	0.00	9,429.0	50.0	0.0	0.9	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	50.0	0.0	0.9	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	50.0	0.0	0.9	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	50.0	0.0	0.9	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	50.0	0.0	0.9	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	50.0	0.0	0.9	0.00	0.00	0.00
9,993.2	0.00	0.00	9,993.0	50.0	0.0	0.9	0.00	0.00	0.00
Lodgepole									
10,000.2	0.00	0.00	10,000.0	50.0	0.0	0.9	0.00	0.00	0.00
Start 340.4 hold at 10000.2 MD									
10,100.0	0.00	0.00	10,099.8	50.0	0.0	0.9	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	50.0	0.0	0.9	0.00	0.00	0.00
10,300.0	0.00	0.00	10,299.8	50.0	0.0	0.9	0.00	0.00	0.00
10,340.7	0.00	0.00	10,340.4	50.0	0.0	0.9	0.00	0.00	0.00
Start Build 12.00 KOP									
10,350.0	1.12	79.83	10,349.8	50.0	0.1	1.0	12.00	12.00	0.00
10,375.0	4.12	79.83	10,374.8	50.2	1.2	2.1	12.00	12.00	0.00
10,400.0	7.12	79.83	10,399.6	50.7	3.6	4.6	12.00	12.00	0.00
10,425.0	10.12	79.83	10,424.3	51.3	7.3	8.3	12.00	12.00	0.00
10,450.0	13.12	79.83	10,448.8	52.2	12.3	13.2	12.00	12.00	0.00
10,475.0	16.12	79.83	10,473.0	53.3	18.5	19.5	12.00	12.00	0.00
10,500.0	19.12	79.83	10,496.8	54.7	25.9	26.9	12.00	12.00	0.00
10,525.0	22.12	79.83	10,520.2	56.2	34.6	35.6	12.00	12.00	0.00
10,550.0	25.12	79.83	10,543.1	58.0	44.5	45.5	12.00	12.00	0.00
10,575.0	28.12	79.83	10,565.5	60.0	55.5	56.6	12.00	12.00	0.00

Oasis Petroleum

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-19/20
Well: CHALMERS 5300 21-19 8T
Wellbore: CHALMERS 5300 21-19 8T
Design: Plan #1

Local Co-ordinate Reference:
TVD Reference:
MD Reference:
North Reference:
Survey Calculation Method:

Well CHALMERS 5300 21-19 8T
WELL @ 2071.0ft (Original Well Elev)
WELL @ 2071.0ft (Original Well Elev)
True
Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,600.0	31.12	79.83	10,587.2	62.1	67.6	68.8	12.00	12.00	0.00
10,625.0	34.12	79.83	10,608.3	64.5	80.9	82.1	12.00	12.00	0.00
10,650.0	37.12	79.83	10,628.6	67.1	95.2	96.4	12.00	12.00	0.00
10,675.0	40.12	79.83	10,648.1	69.8	110.6	111.9	12.00	12.00	0.00
10,700.0	43.12	79.83	10,666.8	72.8	126.9	128.2	12.00	12.00	0.00
10,725.0	46.12	79.83	10,684.6	75.9	144.2	145.6	12.00	12.00	0.00
10,750.0	49.12	79.83	10,701.4	79.1	162.4	163.8	12.00	12.00	0.00
10,757.0	49.96	79.83	10,706.0	80.1	167.7	169.1	12.00	12.00	0.00
False Bakken									
10,772.9	51.87	79.83	10,716.0	82.2	179.8	181.2	12.00	12.00	0.00
Upper Bakken Shale									
10,775.0	52.12	79.83	10,717.3	82.5	181.4	182.9	12.00	12.00	0.00
10,799.8	55.09	79.83	10,732.0	86.1	201.0	202.6	12.00	12.00	0.00
Middle Bakken									
10,800.0	55.12	79.83	10,732.1	86.1	201.2	202.8	12.00	12.00	0.00
10,825.0	58.12	79.83	10,745.9	89.8	221.8	223.4	12.00	12.00	0.00
10,850.0	61.12	79.83	10,758.5	93.6	243.0	244.7	12.00	12.00	0.00
10,866.0	63.04	79.83	10,766.0	96.1	256.9	258.6	12.00	12.00	0.00
Lower Bakken Shale									
10,875.0	64.12	79.83	10,770.0	97.5	264.8	266.6	12.00	12.00	0.00
10,899.1	67.02	79.83	10,780.0	101.4	286.5	288.3	12.00	12.00	0.00
Pronghorn									
10,900.0	67.12	79.83	10,780.3	101.5	287.3	289.1	12.00	12.00	0.00
10,925.0	70.12	79.83	10,789.4	105.6	310.2	312.0	12.00	12.00	0.00
10,950.0	73.12	79.83	10,797.3	109.8	333.5	335.5	12.00	12.00	0.00
10,955.9	73.83	79.83	10,799.0	110.8	339.1	341.0	12.00	12.00	0.00
Threeforks									
10,975.0	76.12	79.83	10,804.0	114.1	357.2	359.3	12.00	12.00	0.00
11,000.0	79.12	79.83	10,809.3	118.4	381.3	383.4	12.00	12.00	0.00
11,009.4	80.25	79.83	10,811.0	120.0	390.4	392.5	12.00	12.00	0.00
Threeforks(Top of Target)									
11,025.0	82.12	79.83	10,813.4	122.7	405.5	407.7	12.00	12.00	0.00
11,050.0	85.12	79.83	10,816.2	127.1	430.0	432.3	12.00	12.00	0.00
11,075.0	88.12	79.83	10,817.6	131.5	454.6	456.9	12.00	12.00	0.00
11,087.7	89.65	79.83	10,817.9	133.8	467.1	469.5	12.00	12.00	0.00
Start 18.2 hold at 11087.7 MD EOC									
11,105.9	89.65	79.83	10,818.0	137.0	485.0	487.4	0.00	0.00	0.00
Start DLS 2.00 TFO 90.05 Csg Pt									
11,106.0	89.65	79.83	10,818.0	137.0	485.1	487.5	0.00	0.00	0.00
7"									
11,200.0	89.65	81.71	10,818.6	152.1	577.8	580.5	2.00	0.00	2.00
11,300.0	89.65	83.71	10,819.2	164.8	677.0	679.9	2.00	0.00	2.00
11,400.0	89.65	85.71	10,819.8	174.0	776.6	779.7	2.00	0.00	2.00
11,500.0	89.65	87.71	10,820.4	179.7	876.4	879.6	2.00	0.00	2.00
11,600.0	89.65	89.71	10,821.0	182.0	976.4	979.6	2.00	0.00	2.00
11,614.4	89.65	90.00	10,821.1	182.0	990.8	994.0	2.00	0.00	2.00
Start 8984.3 hold at 11614.4 MD									
11,700.0	89.65	90.00	10,821.7	182.0	1,076.4	1,079.5	0.00	0.00	0.00
11,800.0	89.65	90.00	10,822.3	182.0	1,176.4	1,179.5	0.00	0.00	0.00
11,900.0	89.65	90.00	10,822.9	182.0	1,276.4	1,279.5	0.00	0.00	0.00
12,000.0	89.65	90.00	10,823.5	182.0	1,376.4	1,379.5	0.00	0.00	0.00
12,100.0	89.65	90.00	10,824.1	182.0	1,476.4	1,479.5	0.00	0.00	0.00
12,200.0	89.65	90.00	10,824.7	182.0	1,576.4	1,579.4	0.00	0.00	0.00
12,300.0	89.65	90.00	10,825.4	182.0	1,676.4	1,679.4	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-19/20
Well: CHALMERS 5300 21-19 8T
Wellbore: CHALMERS 5300 21-19 8T
Design: Plan #1

Local Co-ordinate Reference: Well CHALMERS 5300 21-19 8T
TVD Reference: WELL @ 2071.0ft (Original Well Elev)
MD Reference: WELL @ 2071.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
12,400.0	89.65	90.00	10,826.0	182.0	1,776.4	1,779.4	0.00	0.00	0.00
12,500.0	89.65	90.00	10,826.6	182.0	1,876.4	1,879.4	0.00	0.00	0.00
12,600.0	89.65	90.00	10,827.2	182.0	1,976.4	1,979.4	0.00	0.00	0.00
12,700.0	89.65	90.00	10,827.8	182.0	2,076.4	2,079.4	0.00	0.00	0.00
12,800.0	89.65	90.00	10,828.4	182.0	2,176.4	2,179.3	0.00	0.00	0.00
12,900.0	89.65	90.00	10,829.1	182.0	2,276.4	2,279.3	0.00	0.00	0.00
13,000.0	89.65	90.00	10,829.7	182.0	2,376.4	2,379.3	0.00	0.00	0.00
13,100.0	89.65	90.00	10,830.3	182.0	2,476.4	2,479.3	0.00	0.00	0.00
13,200.0	89.65	90.00	10,830.9	182.0	2,576.4	2,579.3	0.00	0.00	0.00
13,300.0	89.65	90.00	10,831.5	182.0	2,676.4	2,679.2	0.00	0.00	0.00
13,400.0	89.65	90.00	10,832.1	182.0	2,776.4	2,779.2	0.00	0.00	0.00
13,500.0	89.65	90.00	10,832.8	182.0	2,876.4	2,879.2	0.00	0.00	0.00
13,600.0	89.65	90.00	10,833.4	182.0	2,976.4	2,979.2	0.00	0.00	0.00
13,700.0	89.65	90.00	10,834.0	182.0	3,076.4	3,079.2	0.00	0.00	0.00
13,800.0	89.65	90.00	10,834.6	182.0	3,176.4	3,179.2	0.00	0.00	0.00
13,900.0	89.65	90.00	10,835.2	182.0	3,276.4	3,279.1	0.00	0.00	0.00
14,000.0	89.65	90.00	10,835.8	182.0	3,376.4	3,379.1	0.00	0.00	0.00
14,100.0	89.65	90.00	10,836.5	182.0	3,476.3	3,479.1	0.00	0.00	0.00
14,200.0	89.65	90.00	10,837.1	182.0	3,576.3	3,579.1	0.00	0.00	0.00
14,300.0	89.65	90.00	10,837.7	182.0	3,676.3	3,679.1	0.00	0.00	0.00
14,400.0	89.65	90.00	10,838.3	182.0	3,776.3	3,779.0	0.00	0.00	0.00
14,500.0	89.65	90.00	10,838.9	182.0	3,876.3	3,879.0	0.00	0.00	0.00
14,600.0	89.65	90.00	10,839.5	182.0	3,976.3	3,979.0	0.00	0.00	0.00
14,700.0	89.65	90.00	10,840.2	182.0	4,076.3	4,079.0	0.00	0.00	0.00
14,800.0	89.65	90.00	10,840.8	182.0	4,176.3	4,179.0	0.00	0.00	0.00
14,900.0	89.65	90.00	10,841.4	182.0	4,276.3	4,278.9	0.00	0.00	0.00
15,000.0	89.65	90.00	10,842.0	182.0	4,376.3	4,378.9	0.00	0.00	0.00
15,100.0	89.65	90.00	10,842.6	182.0	4,476.3	4,478.9	0.00	0.00	0.00
15,200.0	89.65	90.00	10,843.2	182.0	4,576.3	4,578.9	0.00	0.00	0.00
15,300.0	89.65	90.00	10,843.9	182.0	4,676.3	4,678.9	0.00	0.00	0.00
15,400.0	89.65	90.00	10,844.5	182.0	4,776.3	4,778.9	0.00	0.00	0.00
15,500.0	89.65	90.00	10,845.1	182.0	4,876.3	4,878.8	0.00	0.00	0.00
15,600.0	89.65	90.00	10,845.7	182.0	4,976.3	4,978.8	0.00	0.00	0.00
15,700.0	89.65	90.00	10,846.3	182.0	5,076.3	5,078.8	0.00	0.00	0.00
15,800.0	89.65	90.00	10,846.9	182.0	5,176.3	5,178.8	0.00	0.00	0.00
15,900.0	89.65	90.00	10,847.6	182.0	5,276.3	5,278.8	0.00	0.00	0.00
16,000.0	89.65	90.00	10,848.2	182.0	5,376.3	5,378.7	0.00	0.00	0.00
16,100.0	89.65	90.00	10,848.8	182.0	5,476.3	5,478.7	0.00	0.00	0.00
16,200.0	89.65	90.00	10,849.4	182.0	5,576.3	5,578.7	0.00	0.00	0.00
16,300.0	89.65	90.00	10,850.0	182.0	5,676.3	5,678.7	0.00	0.00	0.00
16,400.0	89.65	90.00	10,850.6	182.0	5,776.3	5,778.7	0.00	0.00	0.00
16,500.0	89.65	90.00	10,851.3	182.0	5,876.3	5,878.6	0.00	0.00	0.00
16,600.0	89.65	90.00	10,851.9	182.0	5,976.3	5,978.6	0.00	0.00	0.00
16,700.0	89.65	90.00	10,852.5	182.0	6,076.3	6,078.6	0.00	0.00	0.00
16,800.0	89.65	90.00	10,853.1	182.0	6,176.3	6,178.6	0.00	0.00	0.00
16,900.0	89.65	90.00	10,853.7	182.0	6,276.3	6,278.6	0.00	0.00	0.00
17,000.0	89.65	90.00	10,854.3	182.0	6,376.3	6,378.6	0.00	0.00	0.00
17,100.0	89.65	90.00	10,855.0	182.0	6,476.3	6,478.5	0.00	0.00	0.00
17,200.0	89.65	90.00	10,855.6	182.0	6,576.3	6,578.5	0.00	0.00	0.00
17,300.0	89.65	90.00	10,856.2	182.0	6,676.3	6,678.5	0.00	0.00	0.00
17,400.0	89.65	90.00	10,856.8	182.0	6,776.3	6,778.5	0.00	0.00	0.00
17,500.0	89.65	90.00	10,857.4	182.0	6,876.3	6,878.5	0.00	0.00	0.00
17,600.0	89.65	90.00	10,858.0	182.0	6,976.3	6,978.4	0.00	0.00	0.00
17,700.0	89.65	90.00	10,858.7	182.0	7,076.3	7,078.4	0.00	0.00	0.00
17,800.0	89.65	90.00	10,859.3	182.0	7,176.3	7,178.4	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
17,900.0	89.65	90.00	10,859.9	182.0	7,276.3	7,278.4	0.00	0.00	0.00
18,000.0	89.65	90.00	10,860.5	182.0	7,376.3	7,378.4	0.00	0.00	0.00
18,100.0	89.65	90.00	10,861.1	182.0	7,476.3	7,478.4	0.00	0.00	0.00
18,200.0	89.65	90.00	10,861.7	182.0	7,576.3	7,578.3	0.00	0.00	0.00
18,300.0	89.65	90.00	10,862.4	182.0	7,676.3	7,678.3	0.00	0.00	0.00
18,400.0	89.65	90.00	10,863.0	182.0	7,776.3	7,778.3	0.00	0.00	0.00
18,500.0	89.65	90.00	10,863.6	182.0	7,876.3	7,878.3	0.00	0.00	0.00
18,600.0	89.65	90.00	10,864.2	182.0	7,976.3	7,978.3	0.00	0.00	0.00
18,700.0	89.65	90.00	10,864.8	182.0	8,076.3	8,078.2	0.00	0.00	0.00
18,800.0	89.65	90.00	10,865.4	182.0	8,176.3	8,178.2	0.00	0.00	0.00
18,900.0	89.65	90.00	10,866.1	182.0	8,276.3	8,278.2	0.00	0.00	0.00
19,000.0	89.65	90.00	10,866.7	182.0	8,376.3	8,378.2	0.00	0.00	0.00
19,100.0	89.65	90.00	10,867.3	182.0	8,476.3	8,478.2	0.00	0.00	0.00
19,200.0	89.65	90.00	10,867.9	182.0	8,576.3	8,578.1	0.00	0.00	0.00
19,300.0	89.65	90.00	10,868.5	182.0	8,676.3	8,678.1	0.00	0.00	0.00
19,400.0	89.65	90.00	10,869.1	182.0	8,776.2	8,778.1	0.00	0.00	0.00
19,500.0	89.65	90.00	10,869.8	182.0	8,876.2	8,878.1	0.00	0.00	0.00
19,600.0	89.65	90.00	10,870.4	182.0	8,976.2	8,978.1	0.00	0.00	0.00
19,700.0	89.65	90.00	10,871.0	182.0	9,076.2	9,078.1	0.00	0.00	0.00
19,800.0	89.65	90.00	10,871.6	182.0	9,176.2	9,178.0	0.00	0.00	0.00
19,900.0	89.65	90.00	10,872.2	182.0	9,276.2	9,278.0	0.00	0.00	0.00
20,000.0	89.65	90.00	10,872.8	182.0	9,376.2	9,378.0	0.00	0.00	0.00
20,100.0	89.65	90.00	10,873.5	182.0	9,476.2	9,478.0	0.00	0.00	0.00
20,200.0	89.65	90.00	10,874.1	182.0	9,576.2	9,578.0	0.00	0.00	0.00
20,300.0	89.65	90.00	10,874.7	182.0	9,676.2	9,677.9	0.00	0.00	0.00
20,400.0	89.65	90.00	10,875.3	182.0	9,776.2	9,777.9	0.00	0.00	0.00
20,500.0	89.65	90.00	10,875.9	182.0	9,876.2	9,877.9	0.00	0.00	0.00
20,582.4	89.65	90.00	10,876.4	182.0	9,958.6	9,960.3	0.00	0.00	0.00

Design Targets

Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (ft)	Easting (ft)	Latitude	Longitude
- hit/miss target									
- Shape									
Chalmers 8T	0.00	0.00	10,876.8	181.7	9,974.9	402,255.11	1,219,964.60	48° 3' 43.067 N	103° 33' 43.255 W
- plan misses target center by 16.3ft at 20582.4ft MD (10876.4 TVD, 182.0 N, 9958.6 E)									
- Point									

Casing Points

Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,150.0	2,150.0	9 5/8"	9.625	13.500
11,106.0	10,818.0	7"	7.000	8.750

Oasis Petroleum

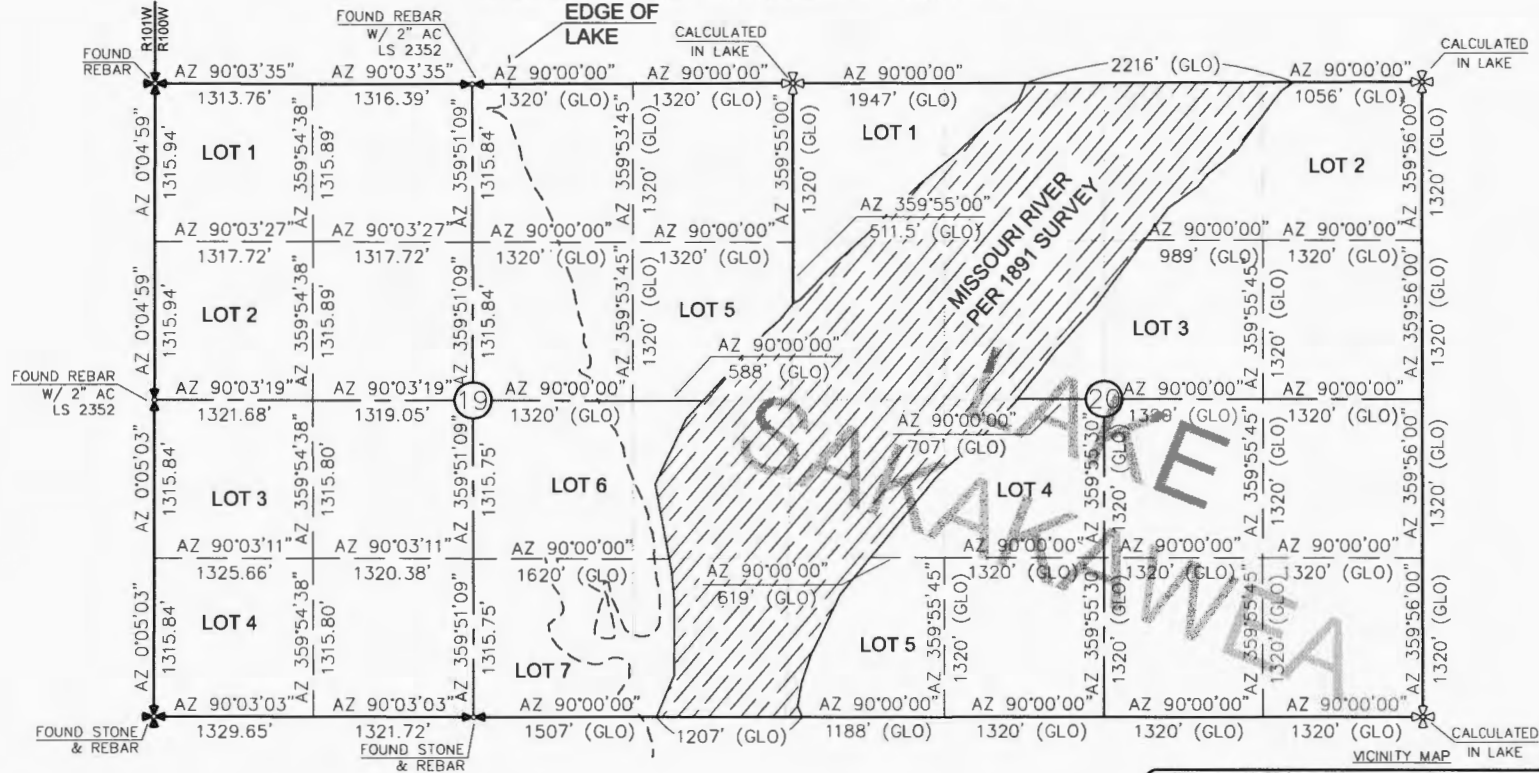
Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Formations						
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)	
2,021.0	2,021.0	Pierre				
4,624.1	4,624.0	Greenhorn				
5,029.1	5,029.0	Mowry				
5,417.1	5,417.0	Dakota				
6,463.2	6,463.0	Rierdon				
6,891.2	6,891.0	Dunham Salt				
6,960.2	6,960.0	Dunham Salt Base				
7,257.2	7,257.0	Pine Salt				
7,290.2	7,290.0	Pine Salt Base				
7,351.2	7,351.0	Opeche Salt				
7,426.2	7,426.0	Opeche Salt Base				
7,662.2	7,662.0	Amsden				
7,828.2	7,828.0	Tyler				
8,032.2	8,032.0	Otter/Base Minnelusa				
8,384.2	8,384.0	Kibbey Lime				
8,534.2	8,534.0	Charles Salt				
9,209.2	9,209.0	Base Last Salt				
9,429.2	9,429.0	Mission Canyon				
9,993.2	9,993.0	Lodgepole				
10,757.0	10,706.0	False Bakken				
10,772.9	10,716.0	Upper Bakken Shale				
10,799.8	10,732.0	Middle Bakken				
10,866.0	10,766.0	Lower Bakken Shale				
10,899.1	10,780.0	Pronghorn				
10,955.9	10,799.0	Threeforks				
11,009.4	10,811.0	Threeforks(Top of Target)				

Plan Annotations					
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates			
		+N/-S (ft)	+E/-W (ft)	Comment	
2,150.0	2,150.0	0.0	0.0	Start Build 5.00	
2,160.0	2,160.0	0.0	0.0	Start 5719.7 hold at 2160.0 MD	
7,879.7	7,879.4	50.0	0.0	Start Drop -5.00	
7,889.7	7,889.4	50.0	0.0	Start 2110.6 hold at 7889.7 MD	
10,000.2	10,000.0	50.0	0.0	Start 340.4 hold at 10000.2 MD	
10,340.7	10,340.4	50.0	0.0	Start Build 12.00 KOP	
11,087.7	10,817.9	133.8	467.1	Start 18.2 hold at 11087.7 MD EOC	
11,105.9	10,818.0	137.0	485.0	Start DLS 2.00 TFO 90.05 Csg Pt	
11,614.4	10,821.1	182.0	990.8	Start 8984.3 hold at 11614.4 MD	
20,598.7	10,876.5	182.0	9,974.9	TD at 20598.7	

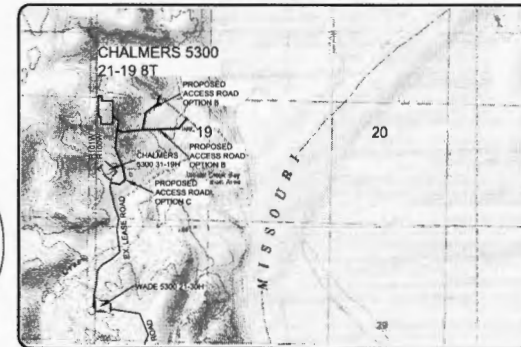
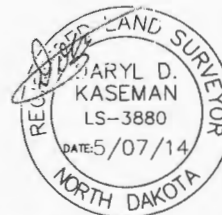
SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5300 21-19 8T"
 2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
 SECTIONS 19 & 20, T153N, R100W, 58th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY ISSUED
 AND SEALED BY DARYL D. KASEMAN,
 PLS, REGISTRATION NUMBER 3880 ON
 5/07/14 AND THE ORIGINAL
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 INC.

ALL AZIMUTHS ARE BASED ON G.P.S.
 OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
 AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
 WAS 1891. THE CORNERS FOUND ARE AS
 INDICATED AND ALL OTHERS ARE COMPUTED FROM
 THOSE CORNERS FOUND AND BASED ON G.L.O.
 DATA. THE MAPPING ANGLE FOR THIS AREA IS
 APPROXIMATELY -0°03'



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Revision	By	Date	Description
REV 1	JLH	3/27/14	ADDED WELLS TO PAD
REV 2	JLH	3/27/14	ADDED WELLS TO PAD/REVISED PAD
REV 3	JLH	3/27/14	ADDED WELLS TO PAD/REVISED PAD

OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTIONS 19 & 20, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: 813-08-282-03 Drawn By: JLH Checked By: JLD
--	--

Interstate Engineering, Inc. P.O. Box 648 423 E. Main Street Sioux Falls, Minnesota 55070 Ph: (605) 433-5617 Fax: (605) 433-5618 www.interstateeng.com	Drawn By: JLH Checked By: JLD
--	----------------------------------



2/8
SHEET NO.

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5300 21-18 BT"

SECTION 19, T33N, R100W, S01P.M., MCKENZIE COUNTY, NORTH DAKOTA

SECTION LINE

33' RIGHT-OF-WAY

SCREENING BERM ELEV. = 2050.0, 20' WIDE TOP, 1:1 SIDE SLOPES

PIT

1% SLOPE

10'x20' PROPPANT PITS

V-DITCH (SEE DETAIL)

1% SLOPE

1:1 SLOPE

1.5:1 FILL SLOPES

CUT SLOPES = 1:1

BERM (SEE DETAIL)

CULVERT

PROPOSED STOCKPILE

TOP OF STOCKPILE

RIPPRAP

DETENTION BASIN

CHALMERS 5300 21-19 5T
GROUND ELEV. = 2057.8'
FINISH PAD ELEV. = 2051.0'

CHALMERS 5300 21-19 6B
GROUND ELEV. = 2056.1'
FINISH PAD ELEV. = 2051.0'

CHALMERS 5300 21-19 7T2
GROUND ELEV. = 2055.3'
FINISH PAD ELEV. = 2051.0'

CHALMERS 5300 21-19 8T
GROUND ELEV. = 2054.5'
FINISH PAD ELEV. = 2051.0'

CHALMERS 5300 21-19 9T2
GROUND ELEV. = 2053.6'
FINISH PAD ELEV. = 2051.0'

CHALMERS 5300 21-19 10B
GROUND ELEV. = 2052.4'
FINISH PAD ELEV. = 2051.0'

CHALMERS 5300 21-19 11T
GROUND ELEV. = 2050.9'
FINISH PAD ELEV. = 2051.0'

C-11.2
FG = 2051.1
F-4.7
FG = 2050.2

C-19.4
FG = 2050.2

C-19.2
FG = 2050.2

C-13.9
FG = 2050.2

C-11.5
FG = 2050.2

C-16.3
FG = 2050.2

C-15.4
FG = 2051.0

C-10.0
FG = 2051.0

C-7.1
FG = 2050.6

C-9.8
FG = 2051.0

C-8.4
FG = 2051.0

C-7.1
FG = 2051.0

C-5.8
FG = 2051.0

C-4.5
FG = 2051.0

C-3.7
FG = 2051.1

C-0.4
FG = 2051.1

C-0.6
FG = 2050.2

F-4.5
FG = 2050.2

F-4.6
FG = 2051.0

F-10.9
FG = 2051.0

F-12.1
FG = 2050.6

F-2.7
FG = 2050.0

F-11.3
FG = 2050.0

F-18.5
FG = 2050.0

F-18.5
FG = 2050.0

RECEIVED LAND SURVEYOR
MARYL D. KASEMAN
LS-3880
DATE: 5/07/14
NORTH DAKOTA



INTERSTATE
ENGINEERING

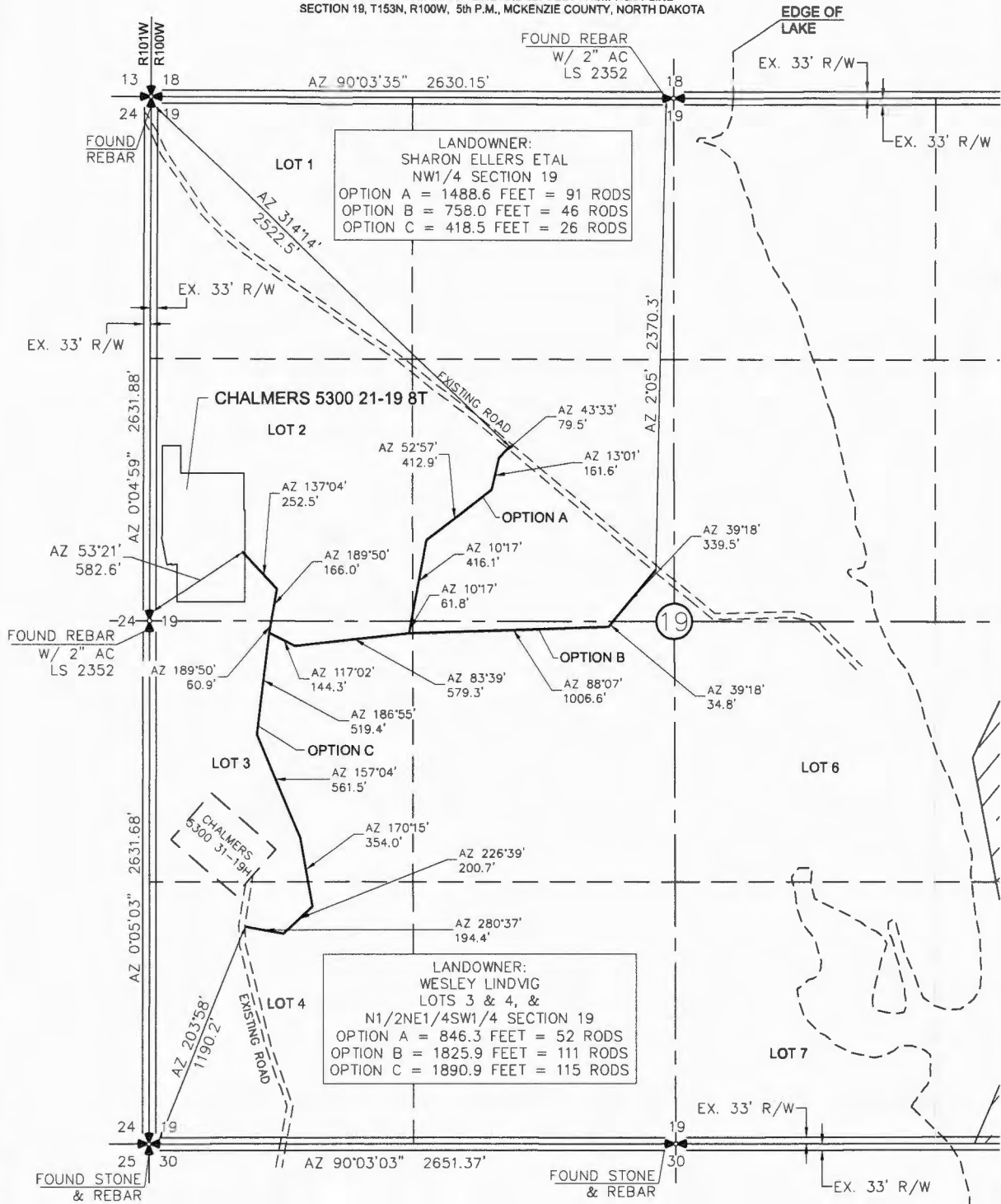
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

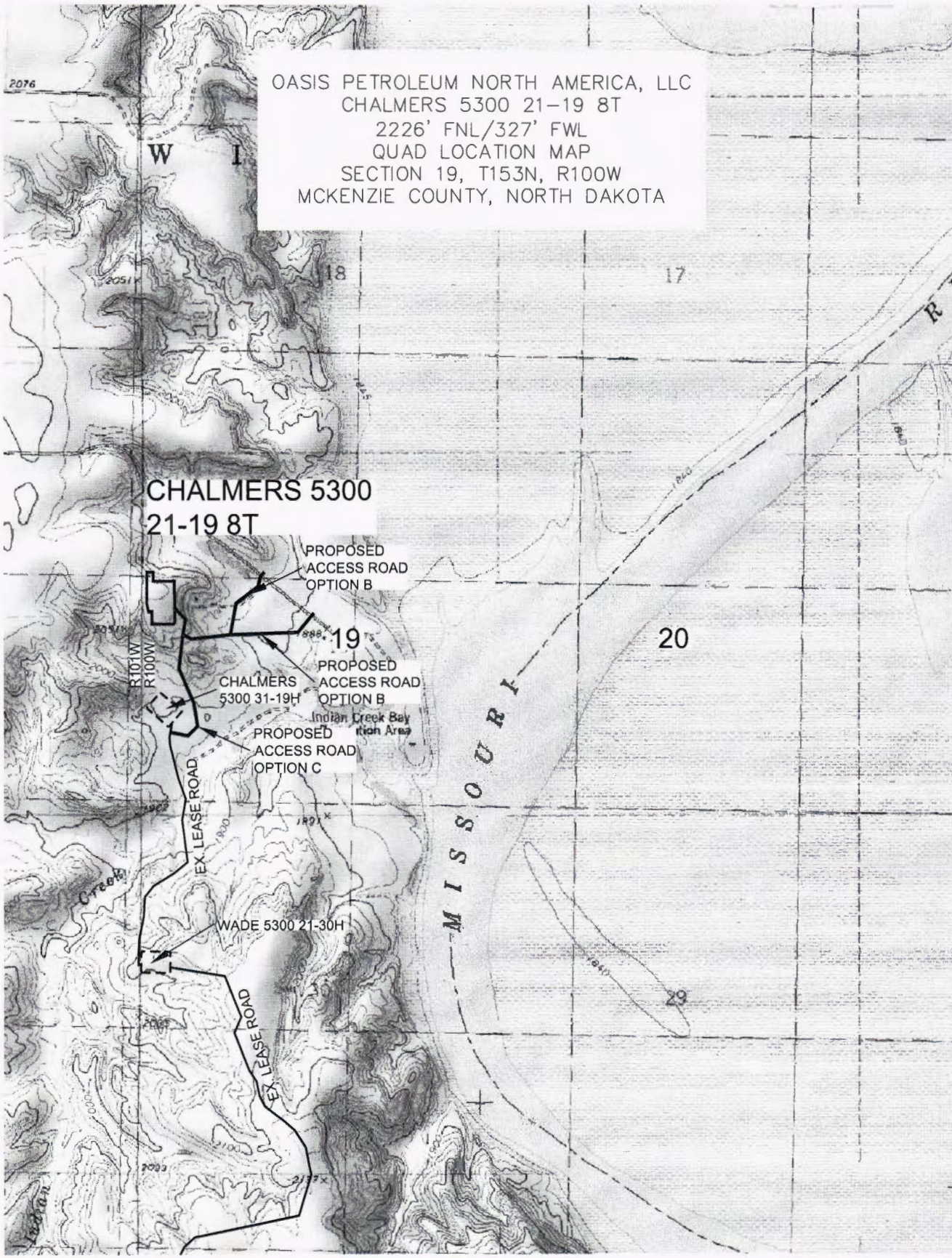
OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 19, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>S.H.H.</u>	Project No.: <u>S13-09-282.03</u>
Checked By: <u>D.D.K.</u>	Date: <u>JAN. 2014</u>

Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD
REV 2	4/22/14	BHH	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/2/14	BHH	MOVED WELLS ON PAD/REVISED PAD

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5300 21-19 8T"
 2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
 SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA





OASIS PETROLEUM NORTH AMERICA, LLC
 CHALMERS 5300 21-19 8T
 2226' FNL/327' FWL
 QUAD LOCATION MAP
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

CHALMERS 5300
 21-19 8T

PROPOSED
 ACCESS ROAD
 OPTION B

PROPOSED
 ACCESS ROAD
 OPTION B

PROPOSED
 ACCESS ROAD
 OPTION C

WADE 5300 21-30H

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 SHEET NO.



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 425 East Main Street
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 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

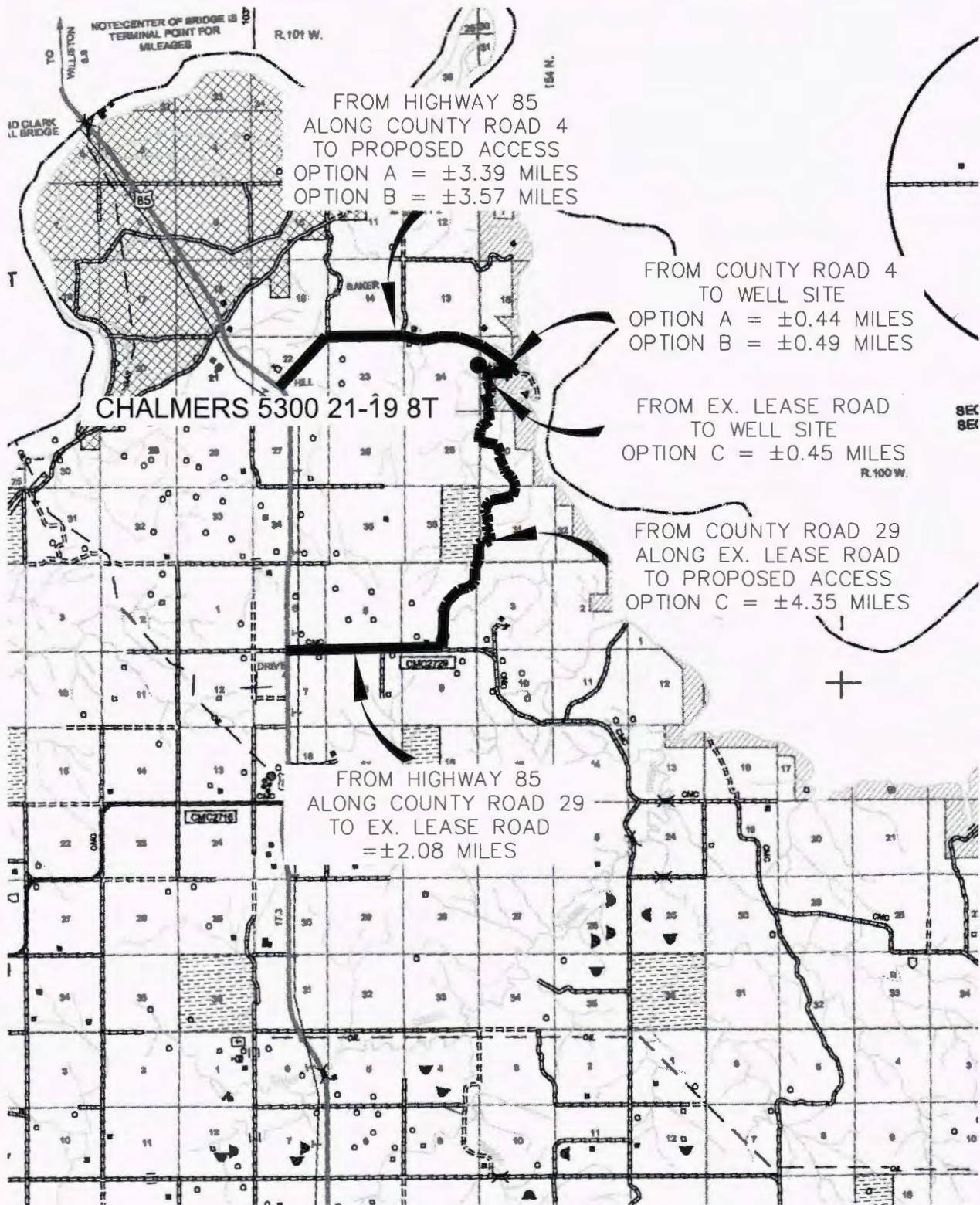
Drawn By: B.H.H. Project No.: S13-09-282.03
 Checked By: D.D.K. Date: JAN. 2014

Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD
REV 2	4/22/14	BHH	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/2/14	BHH	MOVED WELLS ON PAD/REVISED PAD

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5300 21-19 8T"

2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

6/8



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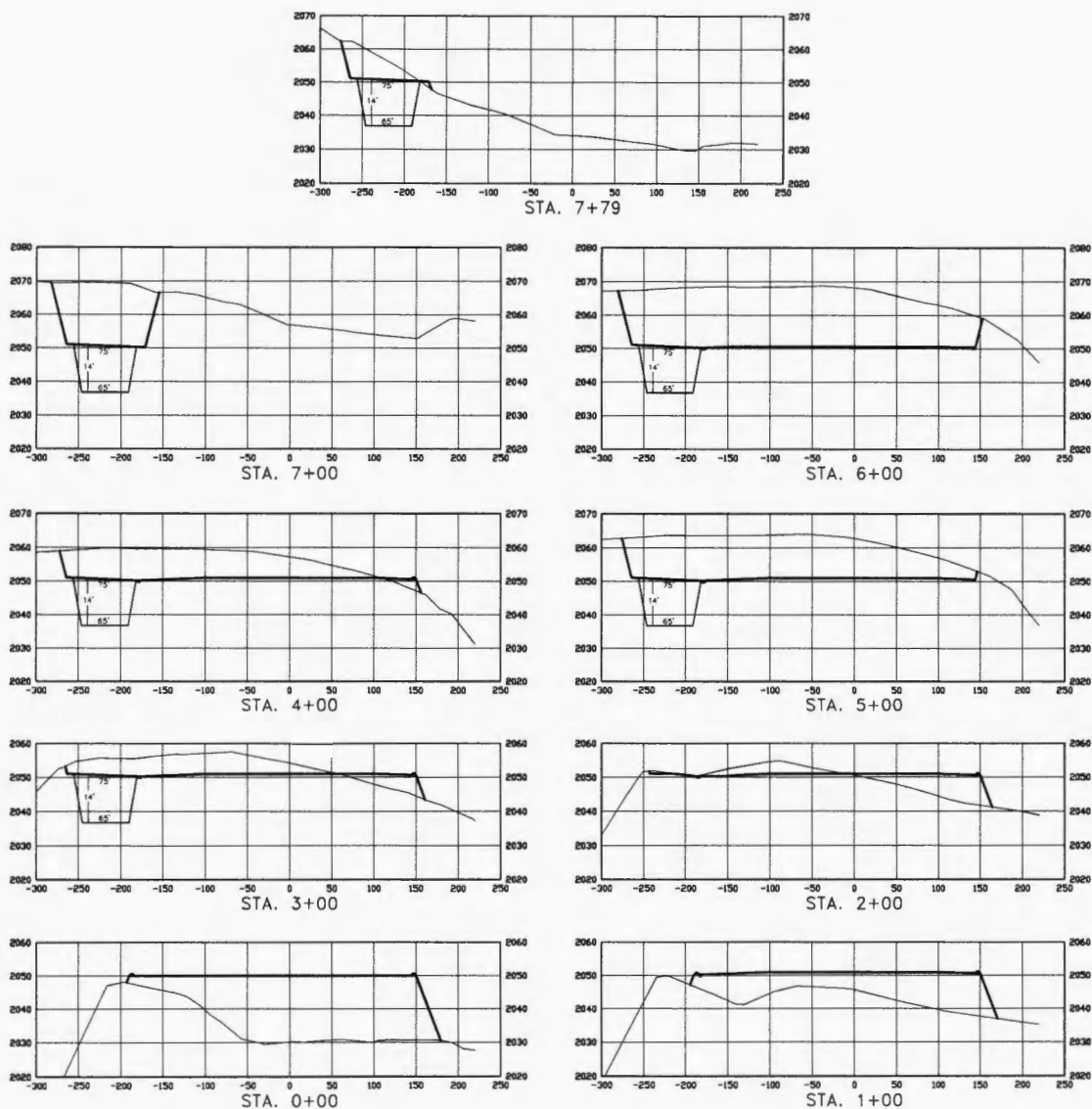
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 19, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No: S13-09-282.03
Checked By: D.D.K. Date: JAN. 2014

Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD
REV 2	4/22/14	BSH	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/3/14	BSH	MOVED WELLS ON PAD/REVISED PAD

CROSS SECTIONS
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5300 21-19 8"
 2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE
 SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED
 AND SEALED BY DARYL D. KASEMAN,
 PLS. REGISTRATION NUMBER 3880 ON
 5/07/14 AND THE ORIGINAL
 DOCUMENTS ARE STORED AT THE
 OFFICES OF INTERSTATE ENGINEERING,
 INC.

SCALE
 HORIZ 1"=140'
 VERT 1"=35'

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7/8
 SHEET NO.



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 425 East Main Street
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 Fax (406) 433-5618
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 CROSS SECTIONS
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.M. Project No.: 613-06-287.03
 Checked By: D.D.K. Date: JUN 2014

Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD
REV 2	4/22/14	BHH	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/2/14	BHH	MOVED WELLS ON PAD/REVISED PAD

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC

1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5300 21-19 8T"

2226 FEET FROM NORTH LINE AND 327 FEET FROM WEST LINE

SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

WELL SITE ELEVATION	2054.5
WELL PAD ELEVATION	2051.0
EXCAVATION	67,041
PLUS PIT	22,050
	89,091
EMBANKMENT	26,714
PLUS SHRINKAGE (25%)	6,679
	33,393
STOCKPILE PIT	22,050
STOCKPILE TOP SOIL (6")	5,434
BERMS	1,007 LF = 326 CY
DITCHES	1,768 LF = 270 CY
SCREENING BERM	27,464 CY
STOCKPILE MATERIAL	694
DISTURBED AREA FROM PAD	6.74 ACRES

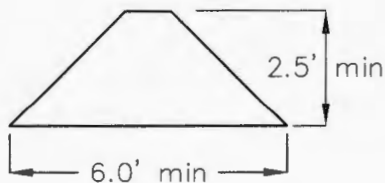
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

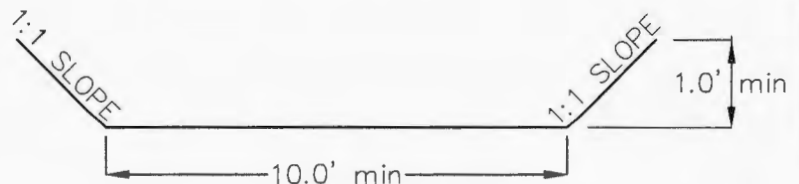
2226' FNL

327' FWL

BERM DETAIL



DITCH DETAIL



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Fax (406) 433-5618
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Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 19, T153N, R100W

MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-282.03
Checked By: D.D.K. Date: JAN. 2014

Revision No.	Date	By	Description
REV 1	3/12/14	JUS	MOVED WELLS ON PAD
REV 2	4/22/14	BNH	MOVED WELLS ON PAD/REVISED PAD
REV 3	5/2/14	BNH	MOVED WELLS ON PAD/REVISED PAD



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28636

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: 6/23/2014

RE: CORES AND SAMPLES

Well Name: **CHALMERS 5300 21-19 8T** Well File No.: **28636**
Location: **LOT2 19-153-100** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **THREE FORKS B1**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

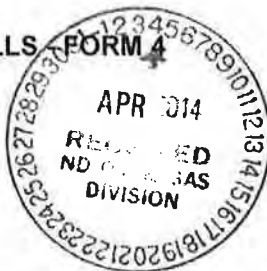
North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely


Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28636

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number Chalmers 5300 21-19 8T					
Footages	Qtr-Qtr	Section	Township	Range	
2226 F N L 326 F W L	LOT2	19	153 N	100 W	
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum/ Chalmers 5300 31-19H (NDIC 2040Z) located within a mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Brandi Terry		
Title Regulatory Specialist	Date April 1, 2014		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 6-16-2014	
By 	
Title Stephen Fried	Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. **28636**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other Suspension of Drilling	

Well Name and Number Chalmers 5300 21-19 8T							
Footages		Qtr-Qtr	Section	Township	Range		
2226 F N L 326 F WL		LOT2	19	153 N	100 W		
Field		Pool		County			
		Bakken		McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services			
Address	City	State	Zip Code

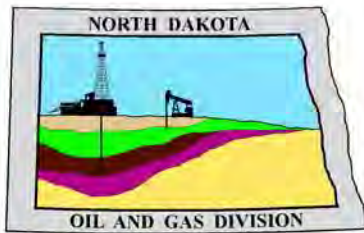
DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC inspector Richard Dunn at 701-770-3554 with spud and TD info.

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 		Printed Name Brandi Terry	
Title Regulatory Specialist		Date April 1, 2014	
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 6/16/14	
By Nathaniel Erbele	
Title Petroleum Resource Specialist	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

June 16, 2014

Brandi Terry
Regulatory Specialist
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
CHALMERS 5300 21-19 8T
LOT2 Section 19-153N-100W
McKenzie County
Well File # 28636**

Dear Brandi:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the 500' setback from the north & south boundaries and 200' setback from the east & west boundaries within the 1280 acre spacing unit consisting of Section 19 & 20 T153N R100W.

PERMIT STIPULATIONS: Due to the proximity of Lake Sakakawea to the well site, a dike is required surrounding the entire location. Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 9972' east.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Brandi Terry
June 16, 2014
Page 2

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Nathaniel Erbele
Petroleum Resource Specialist



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 05 / 1 / 2014	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9491
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name CHALMERS				Well Number 5300 21-19 8T			
Surface Footages 2226 F N L 326 F W L		Qtr-Qtr LOT2	Section 19	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 2089 F N L 811 F W L		Qtr-Qtr LOT2	Section 19	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 137 N From WH 485 E From WH		Azimuth 79.8 °	Longstring Total Depth 11106 Feet MD 10818 Feet TVD				
Bottom Hole Footages From Nearest Section Line 2044 F N L 213 F E L		Qtr-Qtr SENE	Section 20	Township 153 N	Range 100 W	County McKenzie	
Bottom Hole Coordinates From Well Head 182 N From WH 9959 E From WH		KOP Lateral 1 10341 Feet MD	Azimuth Lateral 1 90.0 °		Estimated Total Depth Lateral 1 20582 Feet MD 10876 Feet TVD		
Latitude of Well Head 48 ° 03 ' 41.30 "		Longitude of Well Head -103 ° 36 ' 10.11 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Section 19 & 20 T153N R100W		
Ground Elevation 2054 Feet Above S.L.	Acres in Spacing/Drilling Unit 1280	Spacing/Drilling Unit Setback Requirement 500 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10498 Feet	South Line of Spacing/Drilling Unit 10513 Feet	East Line of Spacing/Drilling Unit 5280 Feet			West Line of Spacing/Drilling Unit 5263 Feet		
Objective Horizons Three Forks B1						Pierre Shale Top 2021	
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2150 Feet	Cement Volume 632 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11106 Feet MD 10818 Feet TVD		Cement Volume 765 Sacks	Cement Top 3917 Feet	Top Dakota Sand 5417 Feet
Base Last Charles Salt (If Applicable) 9209 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

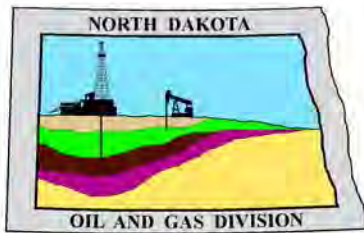
I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.			Date 03 / 31 / 2014
ePermit	Printed Name Brandi Terry	Title Regulatory Specialist	

FOR STATE USE ONLY

Permit and File Number 28636	API Number 33 - 053 - 06021
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 6 / 16 / 2014
By Nathaniel Erbele
Title Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

**RE: Filter Socks and Other Filter Media
Leakproof Container Required
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

Thank you for your cooperation.

Sincerely,

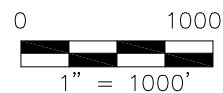
Bruce E. Hicks

Assistant Director

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5300 21-19 8"
2226 FEET FROM NORTH LINE AND 326 FEET FROM WEST LINE
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

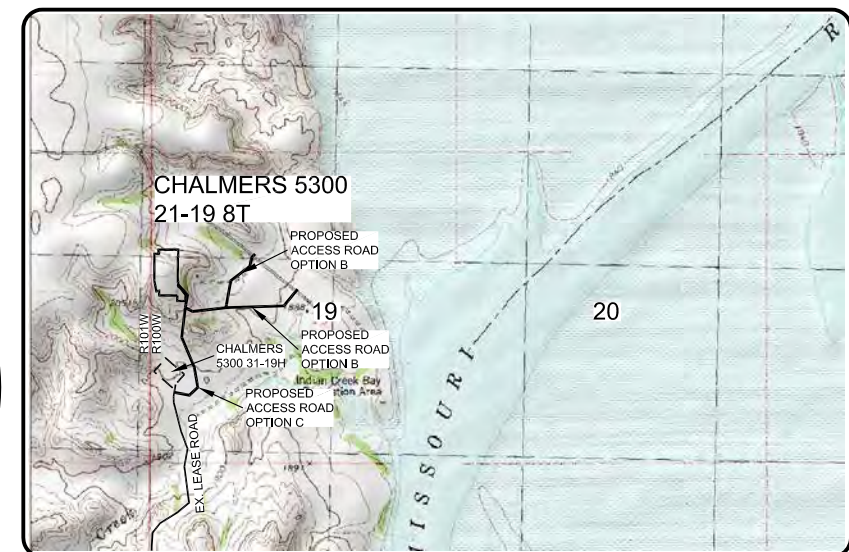


  – MONUMENT – RECOVERED
  – MONUMENT – NOT RECOVERED



THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/13/14
NORTH DAKOTA



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Offices in Minnesota, North Dakota and South Dakota

Other offices in Minnesota, North Dakota and South Dakota



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1/8

1

Refusion No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD

G:\2013\13-06-1982.03 Costs - Petrochem -- 4 of 7 mill wells for
 Customer\GUD\REVISED CHAMBERS 87 deep - 3/7/2014 2:04 PM post schemer

WELL LOCATION PLAT
SECTION 19, T153N, R100W

SECTION 19, 1153N, R100W

B.H.H.	Project No.: S13-09-282.
--------	--------------------------

D.D.K. _____ Date: JAN. 2014

DRILLING PLAN										
OPERATOR	Oasis Petroleum				COUNTY/STATE	McKenzie Co. , ND				
WELL NAME	Chalmers 5300 21-19 8T				RIG	B 25				
WELL TYPE	Horizontal Three Forks									
LOCATION	SW NW 19-153N-100W				Surface Location (survey plat): 2226' FNL		326' FWL			
EST. T.D.	20,582'				GROUND ELEV:		2,046'		Sub Height: 25'	
TOTAL LATERAL:		9,476'		KB ELEV:		2,071'				
MARKER		TVD	Subsea TVD	LOGS:	Type	Interval				
Pierre	NDIC MAP	2,021	50	OH Logs: Request a Sundry for an Open Hole Log Waiver: Oasis Chalmers 5300 31-19H 1,850' to S sec 19 153N 100W						
Greenhorn		4,624	-2,553	CBL/GR: Above top of cement/GR to base of casing						
Mowry		5,029	-2,958	MWD GR: KOP to lateral TD						
Dakota		5,417	-3,346	DEVIATION:	Surf:	3 deg. max., 1 deg / 100'; srvy every 500'				
Rierdon		6,463	-4,392		Prod:	5 deg. max., 1 deg / 100'; srvy every 100'				
Dunham Salt		6,891	-4,820							
Dunham Salt Base		6,960	-4,889							
Pine Salt		7,257	-5,186							
Pine Salt Base		7,290	-5,219							
Opeche Salt		7,351	-5,280							
Opeche Salt Base		7,426	-5,355							
Amsden		7,662	-5,591							
Tyler		7,828	-5,757							
Otter/Base Minnelusa		8,032	-5,961	DST'S:	None planned					
Kibbey Lime		8,384	-6,313	CORES:	None planned					
Charles Salt		8,534	-6,463							
Base Last Salt		9,209	-7,138							
Mission Canyon		9,429	-7,358							
Lodgepole		9,993	-7,922							
False Bakken		10,706	-8,635							
Upper Bakken Shale		10,716	-8,645		MUDLOGGING:	Two-Man:	Begin 200' above Kibbey			
Middle Bakken		10,732	-8,661			30' samples in curve and lateral				
Lower Bakken Shale		10,766	-8,695							
Pronghorn		10,780	-8,709							
Threeforks		10,799	-8,728							
Threeforks(Top of Target)		10,811	-8,740							
Threeforks(Base of Target)		10,822	-8,751							
Claystone		10,822	-8,751	BOP:		11" 5000 psi blind, pipe & annular				
Est. Dip Rate:	-0.35									
Max. Anticipated BHP:	4689				Surface Formation: Glacial till					
MUD:	Interval		Type	WT	Vis	WL	Remarks			
Surface:	0' -	2,150'	FW	8.4-9.0	28-32	NC	Circ Mud Tanks			
Intermediate:	2,150' -	11,106'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks			
Lateral:	11,106' -	20,582'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks			
CASING:	Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks			
Surface:	9-5/8"	36#	13-1/2"	2,150'	To Surface	12	100' into Pierre			
Intermediate:	7"	32#	8-3/4"	11,106'	3917	24	1500' above Dakota			
Production Liner:	4.5"	13.5#	6"	20,582'	TOL @ 10,291'			50' above KOP		
PROBABLE PLUGS, IF REQ'D:										
OTHER:	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI				
Surface:	2,150	2,150	2226 FNL	326 FWL	SEC. 19 T153N R100W		Survey Company:			
KOP:	10,341'	10,341'	2176 FNL	326 FWL	SEC. 19 T153N R100W		Build Rate: 12 deg /100'			
EOC:	11,088'	10,818'	2092 FNL	793 FWL	SEC. 19 T153N R100W	79.8				
Casing Point:	11,106'	10,818'	2089 FNL	811 FWL	SEC. 19 T153N R100W	79.8				
Three Forks Lateral TD:	20,582'	10,876'	2044 FNL	200 FEL	SEC. 20 T153N R100W	90.0				
Comments:										
Request a Sundry for an Open Hole Log Waiver: Oasis Chalmers 5300 31-19H 1,850' to S sec 19 153N 100W										
No frac string planned										
35 packers and 25 sleeves planned 3.6MM lbs 30% ceramic										
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.										
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)										
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)										
										
Geology: N. Gabelman			2/4/2014			Engineering: dad6/11/14				

**Oasis Petroleum
Well Summary
Chalmers 5300 21-19 8T
Section 19 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
9-5/8"	0' - 2150'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 2150'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.00	3520 / 3.49	453 / 2.71

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2150' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2150' setting depth).
- c) Based on string weight in 9 ppg fluid at 2150' TVD plus 100k# overpull. (Buoyed weight equals 67k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2 " hole with 60% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **459 sks** (243 bbls), 11.5 lb/gal, 2.97 cu. Ft./sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: **173 sks** (62 bbls), 13.0 lb/gal, 2.01 cu.ft./sk Varicem with .125 lb/sk Lost Circulation Agent

**Oasis Petroleum
Well Summary
Chalmers 5300 21-19 8T
Section 19 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift**	Minimum	Optimum	Max
7"	0' - 11106'	32	HCP-110	LTC	6.094"	6.000"***	6730	8970	9870

**Special Drift 7" 32# to 6.0"

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' - 11106'	11106'	7", 32#, P-110, LTC, 8rd	11820 / 2.10*	12460 / 1.28	897 / 2.23
6695' - 9433'	2738'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.04**	12460 / 1.29	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10818' TVD.
- c) Based on string weight in 10 ppg fluid, (301k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Mix and pump the following slurry

Pre-flush (Spacer): **100 bbls** Saltwater
 20bbls CW8
 20bbls Fresh Water

Lead Slurry: **183 sks** (84 bbls), 11.8 ppg, 2.55 cu. ft./sk Econocem Cement with .3% Fe-2 and .25 lb/sk Lost Circulation Additive

Tail Slurry: **582 sks** (170 bbls), 14.0 ppg, 1.55 cu. ft./sk Extendcem System with .2% HR-5 Retarder and .25 lb/sk Lost Circulation Additive

**Oasis Petroleum
Well Summary
Chalmers 5300 21-19 8T
Section 19 T153N R100W
McKenzie County, ND**

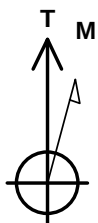
PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10291' - 20582'	13.5	P-110	BTC	3.920"	3.795"	2270	3020	3780

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
10291' - 20582'	10308	4-1/2", 13.5 lb, P-110, BTC	10670 / 1.98	12410 / 1.28	443 / 2.01

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10877' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10877' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 119k lbs.) plus 100k lbs overpull.



Azimuths to True North
Magnetic North: 8.17°

Magnetic Field
Strength: 56490.5snT
Dip Angle: 72.96°
Date: 2/17/2014
Model: IGRF200510



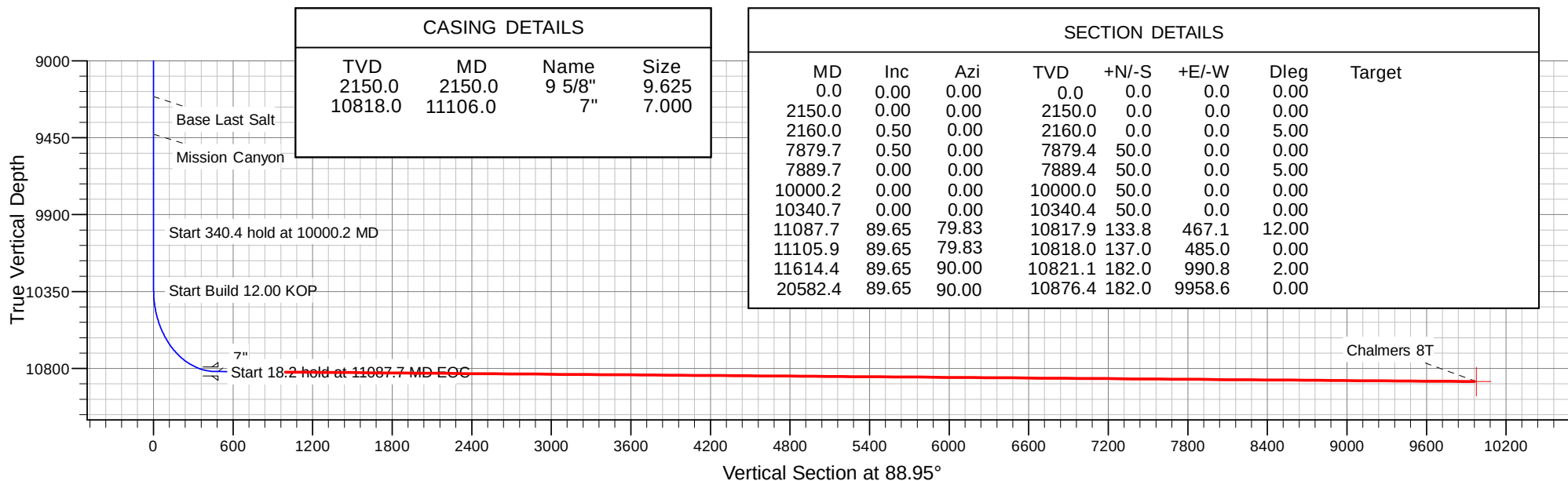
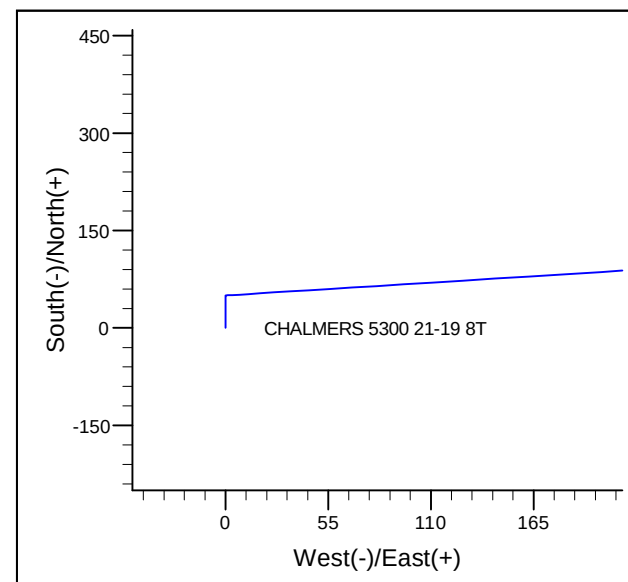
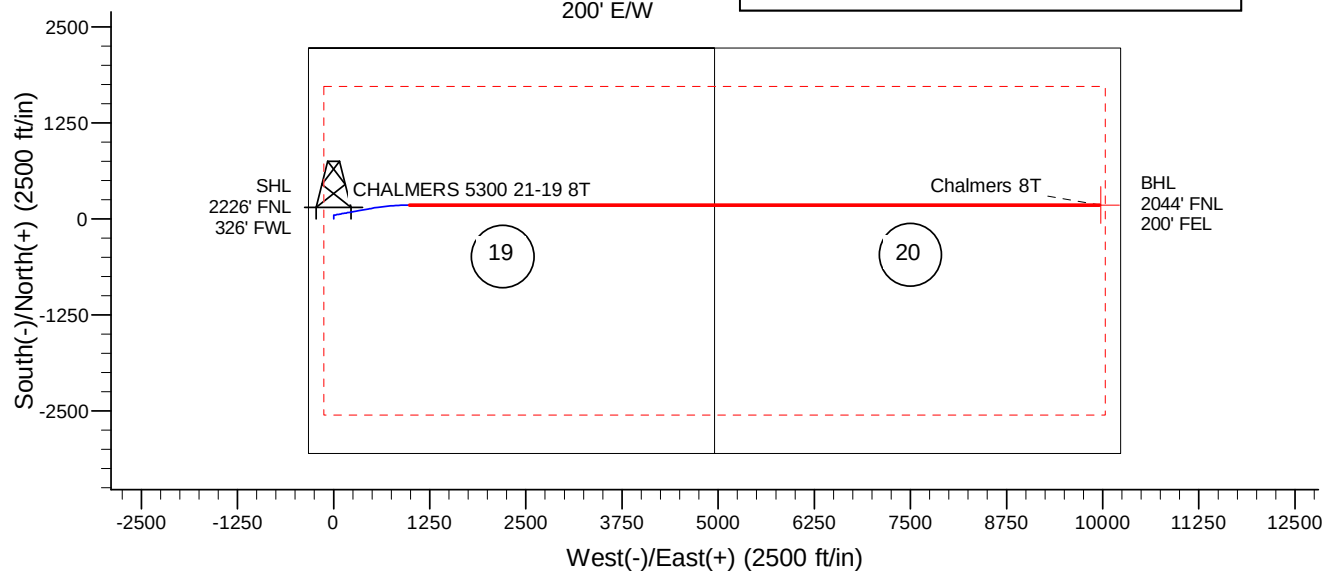
Project: Indian Hills
Site: 153N-100W-19/20
Well: CHALMERS 5300 21-19 8T
Wellbore: CHALMERS 5300 21-19 8T
Design: Plan #1

Setbacks
500' N/S
200' E/W

SITE DETAILS: 153N-100W-19/20

Well Centre Latitude: 48° 3' 41.300 N
Longitude: 103° 36' 10.110 W

Positional Uncertainty: 0.0
Convergence: -2.31
Local North: True



CASING DETAILS

TVD	MD	Name	Size
2150.0	2150.0	9 5/8"	9.625
10818.0	11106.0	7"	7.000

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2150.0	0.00	0.00	2150.0	0.0	0.0	0.00	
2160.0	0.50	0.00	2160.0	0.0	0.0	5.00	
7879.7	0.50	0.00	7879.4	50.0	0.0	0.00	
7889.7	0.00	0.00	7889.4	50.0	0.0	5.00	
10000.2	0.00	0.00	10000.0	50.0	0.0	0.00	
10340.7	0.00	0.00	10340.4	50.0	0.0	0.00	
11087.7	89.65	79.83	10817.9	133.8	467.1	12.00	
11105.9	89.65	79.83	10818.0	137.0	485.0	0.00	
11614.4	89.65	90.00	10821.1	182.0	990.8	2.00	
20582.4	89.65	90.00	10876.4	182.0	9958.6	0.00	

Oasis

Indian Hills

153N-100W-19/20

CHALMERS 5300 21-19 8T

CHALMERS 5300 21-19 8T

Plan: Plan #1

Standard Planning Report

11 June, 2014

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-19/20			
Site Position:		Northing:	402,777.74 ft	Latitude:	48° 3' 44.270 N
From:	Lat/Long	Easting:	1,209,962.51 ft	Longitude:	103° 36' 10.700 W
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence:	-2.31 °

Well	CHALMERS 5300 21-19 8T					
Well Position	+N/-S	-300.9 ft	Northing:	402,475.42 ft	Latitude:	48° 3' 41.300 N
	+E/-W	40.1 ft	Easting:	1,209,990.43 ft	Longitude:	103° 36' 10.110 W
Position Uncertainty		0.0 ft	Wellhead Elevation:		Ground Level:	2,046.0 ft

Wellbore	CHALMERS 5300 21-19 8T				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	2/17/2014	8.17	72.96	56,491

Design	Plan #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	0.0	0.0	0.0	88.95

Plan Sections										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,150.0	0.00	0.00	2,150.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,160.0	0.50	0.00	2,160.0	0.0	0.0	5.00	5.00	0.00	0.00	
7,879.7	0.50	0.00	7,879.4	50.0	0.0	0.00	0.00	0.00	0.00	
7,889.7	0.00	0.00	7,889.4	50.0	0.0	5.00	-5.00	0.00	180.00	
10,000.2	0.00	0.00	10,000.0	50.0	0.0	0.00	0.00	0.00	0.00	
10,340.7	0.00	0.00	10,340.4	50.0	0.0	0.00	0.00	0.00	0.00	
11,087.7	89.65	79.83	10,817.9	133.8	467.1	12.00	12.00	0.00	79.83	
11,105.9	89.65	79.83	10,818.0	137.0	485.0	0.00	0.00	0.00	0.00	
11,614.4	89.65	90.00	10,821.1	182.0	990.8	2.00	0.00	2.00	90.05	
20,582.4	89.65	90.00	10,876.4	182.0	9,958.6	0.00	0.00	0.00	0.00	

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,021.0	0.00	0.00	2,021.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
2,150.0	0.00	0.00	2,150.0	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 5.00 - 9 5/8"									
2,160.0	0.50	0.00	2,160.0	0.0	0.0	0.0	5.00	5.00	0.00
Start 5719.7 hold at 2160.0 MD									
2,200.0	0.50	0.00	2,200.0	0.4	0.0	0.0	0.00	0.00	0.00
2,300.0	0.50	0.00	2,300.0	1.3	0.0	0.0	0.00	0.00	0.00
2,400.0	0.50	0.00	2,400.0	2.1	0.0	0.0	0.00	0.00	0.00
2,500.0	0.50	0.00	2,500.0	3.0	0.0	0.1	0.00	0.00	0.00
2,600.0	0.50	0.00	2,600.0	3.9	0.0	0.1	0.00	0.00	0.00
2,700.0	0.50	0.00	2,700.0	4.8	0.0	0.1	0.00	0.00	0.00
2,800.0	0.50	0.00	2,800.0	5.6	0.0	0.1	0.00	0.00	0.00
2,900.0	0.50	0.00	2,900.0	6.5	0.0	0.1	0.00	0.00	0.00
3,000.0	0.50	0.00	3,000.0	7.4	0.0	0.1	0.00	0.00	0.00
3,100.0	0.50	0.00	3,100.0	8.2	0.0	0.2	0.00	0.00	0.00
3,200.0	0.50	0.00	3,200.0	9.1	0.0	0.2	0.00	0.00	0.00
3,300.0	0.50	0.00	3,300.0	10.0	0.0	0.2	0.00	0.00	0.00
3,400.0	0.50	0.00	3,400.0	10.9	0.0	0.2	0.00	0.00	0.00
3,500.0	0.50	0.00	3,499.9	11.7	0.0	0.2	0.00	0.00	0.00
3,600.0	0.50	0.00	3,599.9	12.6	0.0	0.2	0.00	0.00	0.00
3,700.0	0.50	0.00	3,699.9	13.5	0.0	0.2	0.00	0.00	0.00
3,800.0	0.50	0.00	3,799.9	14.4	0.0	0.3	0.00	0.00	0.00
3,900.0	0.50	0.00	3,899.9	15.2	0.0	0.3	0.00	0.00	0.00
4,000.0	0.50	0.00	3,999.9	16.1	0.0	0.3	0.00	0.00	0.00
4,100.0	0.50	0.00	4,099.9	17.0	0.0	0.3	0.00	0.00	0.00
4,200.0	0.50	0.00	4,199.9	17.8	0.0	0.3	0.00	0.00	0.00
4,300.0	0.50	0.00	4,299.9	18.7	0.0	0.3	0.00	0.00	0.00
4,400.0	0.50	0.00	4,399.9	19.6	0.0	0.4	0.00	0.00	0.00
4,500.0	0.50	0.00	4,499.9	20.5	0.0	0.4	0.00	0.00	0.00
4,600.0	0.50	0.00	4,599.9	21.3	0.0	0.4	0.00	0.00	0.00
4,624.1	0.50	0.00	4,624.0	21.5	0.0	0.4	0.00	0.00	0.00
Greenhorn									

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,700.0	0.50	0.00	4,699.9	22.2	0.0	0.4	0.00	0.00	0.00
4,800.0	0.50	0.00	4,799.9	23.1	0.0	0.4	0.00	0.00	0.00
4,900.0	0.50	0.00	4,899.9	24.0	0.0	0.4	0.00	0.00	0.00
5,000.0	0.50	0.00	4,999.9	24.8	0.0	0.5	0.00	0.00	0.00
5,029.1	0.50	0.00	5,029.0	25.1	0.0	0.5	0.00	0.00	0.00
Mowry									
5,100.0	0.50	0.00	5,099.9	25.7	0.0	0.5	0.00	0.00	0.00
5,200.0	0.50	0.00	5,199.9	26.6	0.0	0.5	0.00	0.00	0.00
5,300.0	0.50	0.00	5,299.9	27.4	0.0	0.5	0.00	0.00	0.00
5,400.0	0.50	0.00	5,399.9	28.3	0.0	0.5	0.00	0.00	0.00
5,417.1	0.50	0.00	5,417.0	28.5	0.0	0.5	0.00	0.00	0.00
Dakota									
5,500.0	0.50	0.00	5,499.9	29.2	0.0	0.5	0.00	0.00	0.00
5,600.0	0.50	0.00	5,599.9	30.1	0.0	0.5	0.00	0.00	0.00
5,700.0	0.50	0.00	5,699.9	30.9	0.0	0.6	0.00	0.00	0.00
5,800.0	0.50	0.00	5,799.9	31.8	0.0	0.6	0.00	0.00	0.00
5,900.0	0.50	0.00	5,899.9	32.7	0.0	0.6	0.00	0.00	0.00
6,000.0	0.50	0.00	5,999.9	33.6	0.0	0.6	0.00	0.00	0.00
6,100.0	0.50	0.00	6,099.8	34.4	0.0	0.6	0.00	0.00	0.00
6,200.0	0.50	0.00	6,199.8	35.3	0.0	0.6	0.00	0.00	0.00
6,300.0	0.50	0.00	6,299.8	36.2	0.0	0.7	0.00	0.00	0.00
6,400.0	0.50	0.00	6,399.8	37.0	0.0	0.7	0.00	0.00	0.00
6,463.2	0.50	0.00	6,463.0	37.6	0.0	0.7	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	0.00	6,499.8	37.9	0.0	0.7	0.00	0.00	0.00
6,600.0	0.50	0.00	6,599.8	38.8	0.0	0.7	0.00	0.00	0.00
6,700.0	0.50	0.00	6,699.8	39.7	0.0	0.7	0.00	0.00	0.00
6,800.0	0.50	0.00	6,799.8	40.5	0.0	0.7	0.00	0.00	0.00
6,891.2	0.50	0.00	6,891.0	41.3	0.0	0.8	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.50	0.00	6,899.8	41.4	0.0	0.8	0.00	0.00	0.00
6,960.2	0.50	0.00	6,960.0	41.9	0.0	0.8	0.00	0.00	0.00
Dunham Salt Base									
7,000.0	0.50	0.00	6,999.8	42.3	0.0	0.8	0.00	0.00	0.00
7,100.0	0.50	0.00	7,099.8	43.2	0.0	0.8	0.00	0.00	0.00
7,200.0	0.50	0.00	7,199.8	44.0	0.0	0.8	0.00	0.00	0.00
7,257.2	0.50	0.00	7,257.0	44.5	0.0	0.8	0.00	0.00	0.00
Pine Salt									
7,290.2	0.50	0.00	7,290.0	44.8	0.0	0.8	0.00	0.00	0.00
Pine Salt Base									
7,300.0	0.50	0.00	7,299.8	44.9	0.0	0.8	0.00	0.00	0.00
7,351.2	0.50	0.00	7,351.0	45.3	0.0	0.8	0.00	0.00	0.00
Opeche Salt									
7,400.0	0.50	0.00	7,399.8	45.8	0.0	0.8	0.00	0.00	0.00
7,426.2	0.50	0.00	7,426.0	46.0	0.0	0.8	0.00	0.00	0.00
Opeche Salt Base									
7,500.0	0.50	0.00	7,499.8	46.6	0.0	0.9	0.00	0.00	0.00
7,600.0	0.50	0.00	7,599.8	47.5	0.0	0.9	0.00	0.00	0.00
7,662.2	0.50	0.00	7,662.0	48.1	0.0	0.9	0.00	0.00	0.00
Amsden									
7,700.0	0.50	0.00	7,699.8	48.4	0.0	0.9	0.00	0.00	0.00
7,800.0	0.50	0.00	7,799.8	49.3	0.0	0.9	0.00	0.00	0.00
7,828.2	0.50	0.00	7,828.0	49.5	0.0	0.9	0.00	0.00	0.00
Tyler									

Oasis Petroleum

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-19/20
Well: CHALMERS 5300 21-19 8T
Wellbore: CHALMERS 5300 21-19 8T
Design: Plan #1

Local Co-ordinate Reference: Well CHALMERS 5300 21-19 8T
TVD Reference: WELL @ 2071.0ft (Original Well Elev)
MD Reference: WELL @ 2071.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,879.7	0.50	0.00	7,879.4	50.0	0.0	0.9	0.00	0.00	0.00
Start Drop -5.00									
7,889.7	0.00	0.00	7,889.4	50.0	0.0	0.9	5.00	-5.00	0.00
Start 2110.6 hold at 7889.7 MD									
7,900.0	0.00	0.00	7,899.8	50.0	0.0	0.9	0.00	0.00	0.00
8,000.0	0.00	0.00	7,999.8	50.0	0.0	0.9	0.00	0.00	0.00
8,032.2	0.00	0.00	8,032.0	50.0	0.0	0.9	0.00	0.00	0.00
Otter/Base Minnelusa									
8,100.0	0.00	0.00	8,099.8	50.0	0.0	0.9	0.00	0.00	0.00
8,200.0	0.00	0.00	8,199.8	50.0	0.0	0.9	0.00	0.00	0.00
8,300.0	0.00	0.00	8,299.8	50.0	0.0	0.9	0.00	0.00	0.00
8,384.2	0.00	0.00	8,384.0	50.0	0.0	0.9	0.00	0.00	0.00
Kibbey Lime									
8,400.0	0.00	0.00	8,399.8	50.0	0.0	0.9	0.00	0.00	0.00
8,500.0	0.00	0.00	8,499.8	50.0	0.0	0.9	0.00	0.00	0.00
8,534.2	0.00	0.00	8,534.0	50.0	0.0	0.9	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,599.8	50.0	0.0	0.9	0.00	0.00	0.00
8,700.0	0.00	0.00	8,699.8	50.0	0.0	0.9	0.00	0.00	0.00
8,800.0	0.00	0.00	8,799.8	50.0	0.0	0.9	0.00	0.00	0.00
8,900.0	0.00	0.00	8,899.8	50.0	0.0	0.9	0.00	0.00	0.00
9,000.0	0.00	0.00	8,999.8	50.0	0.0	0.9	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	50.0	0.0	0.9	0.00	0.00	0.00
9,200.0	0.00	0.00	9,199.8	50.0	0.0	0.9	0.00	0.00	0.00
9,209.2	0.00	0.00	9,209.0	50.0	0.0	0.9	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	50.0	0.0	0.9	0.00	0.00	0.00
9,400.0	0.00	0.00	9,399.8	50.0	0.0	0.9	0.00	0.00	0.00
9,429.2	0.00	0.00	9,429.0	50.0	0.0	0.9	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	50.0	0.0	0.9	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	50.0	0.0	0.9	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	50.0	0.0	0.9	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	50.0	0.0	0.9	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	50.0	0.0	0.9	0.00	0.00	0.00
9,993.2	0.00	0.00	9,993.0	50.0	0.0	0.9	0.00	0.00	0.00
Lodgepole									
10,000.2	0.00	0.00	10,000.0	50.0	0.0	0.9	0.00	0.00	0.00
Start 340.4 hold at 10000.2 MD									
10,100.0	0.00	0.00	10,099.8	50.0	0.0	0.9	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	50.0	0.0	0.9	0.00	0.00	0.00
10,300.0	0.00	0.00	10,299.8	50.0	0.0	0.9	0.00	0.00	0.00
10,340.7	0.00	0.00	10,340.4	50.0	0.0	0.9	0.00	0.00	0.00
Start Build 12.00 KOP									
10,350.0	1.12	79.83	10,349.8	50.0	0.1	1.0	12.00	12.00	0.00
10,375.0	4.12	79.83	10,374.8	50.2	1.2	2.1	12.00	12.00	0.00
10,400.0	7.12	79.83	10,399.6	50.7	3.6	4.6	12.00	12.00	0.00
10,425.0	10.12	79.83	10,424.3	51.3	7.3	8.3	12.00	12.00	0.00
10,450.0	13.12	79.83	10,448.8	52.2	12.3	13.2	12.00	12.00	0.00
10,475.0	16.12	79.83	10,473.0	53.3	18.5	19.5	12.00	12.00	0.00
10,500.0	19.12	79.83	10,496.8	54.7	25.9	26.9	12.00	12.00	0.00
10,525.0	22.12	79.83	10,520.2	56.2	34.6	35.6	12.00	12.00	0.00
10,550.0	25.12	79.83	10,543.1	58.0	44.5	45.5	12.00	12.00	0.00
10,575.0	28.12	79.83	10,565.5	60.0	55.5	56.6	12.00	12.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,600.0	31.12	79.83	10,587.2	62.1	67.6	68.8	12.00	12.00	0.00
10,625.0	34.12	79.83	10,608.3	64.5	80.9	82.1	12.00	12.00	0.00
10,650.0	37.12	79.83	10,628.6	67.1	95.2	96.4	12.00	12.00	0.00
10,675.0	40.12	79.83	10,648.1	69.8	110.6	111.9	12.00	12.00	0.00
10,700.0	43.12	79.83	10,666.8	72.8	126.9	128.2	12.00	12.00	0.00
10,725.0	46.12	79.83	10,684.6	75.9	144.2	145.6	12.00	12.00	0.00
10,750.0	49.12	79.83	10,701.4	79.1	162.4	163.8	12.00	12.00	0.00
10,757.0	49.96	79.83	10,706.0	80.1	167.7	169.1	12.00	12.00	0.00
False Bakken									
10,772.9	51.87	79.83	10,716.0	82.2	179.8	181.2	12.00	12.00	0.00
Upper Bakken Shale									
10,775.0	52.12	79.83	10,717.3	82.5	181.4	182.9	12.00	12.00	0.00
10,799.8	55.09	79.83	10,732.0	86.1	201.0	202.6	12.00	12.00	0.00
Middle Bakken									
10,800.0	55.12	79.83	10,732.1	86.1	201.2	202.8	12.00	12.00	0.00
10,825.0	58.12	79.83	10,745.9	89.8	221.8	223.4	12.00	12.00	0.00
10,850.0	61.12	79.83	10,758.5	93.6	243.0	244.7	12.00	12.00	0.00
10,866.0	63.04	79.83	10,766.0	96.1	256.9	258.6	12.00	12.00	0.00
Lower Bakken Shale									
10,875.0	64.12	79.83	10,770.0	97.5	264.8	266.6	12.00	12.00	0.00
10,899.1	67.02	79.83	10,780.0	101.4	286.5	288.3	12.00	12.00	0.00
Pronghorn									
10,900.0	67.12	79.83	10,780.3	101.5	287.3	289.1	12.00	12.00	0.00
10,925.0	70.12	79.83	10,789.4	105.6	310.2	312.0	12.00	12.00	0.00
10,950.0	73.12	79.83	10,797.3	109.8	333.5	335.5	12.00	12.00	0.00
10,955.9	73.83	79.83	10,799.0	110.8	339.1	341.0	12.00	12.00	0.00
Threeforks									
10,975.0	76.12	79.83	10,804.0	114.1	357.2	359.3	12.00	12.00	0.00
11,000.0	79.12	79.83	10,809.3	118.4	381.3	383.4	12.00	12.00	0.00
11,009.4	80.25	79.83	10,811.0	120.0	390.4	392.5	12.00	12.00	0.00
Threeforks(Top of Target)									
11,025.0	82.12	79.83	10,813.4	122.7	405.5	407.7	12.00	12.00	0.00
11,050.0	85.12	79.83	10,816.2	127.1	430.0	432.3	12.00	12.00	0.00
11,075.0	88.12	79.83	10,817.6	131.5	454.6	456.9	12.00	12.00	0.00
11,087.7	89.65	79.83	10,817.9	133.8	467.1	469.5	12.00	12.00	0.00
Start 18.2 hold at 11087.7 MD EOC									
11,105.9	89.65	79.83	10,818.0	137.0	485.0	487.4	0.00	0.00	0.00
Start DLS 2.00 TFO 90.05 Csg Pt									
11,106.0	89.65	79.83	10,818.0	137.0	485.1	487.5	0.00	0.00	0.00
7"									
11,200.0	89.65	81.71	10,818.6	152.1	577.8	580.5	2.00	0.00	2.00
11,300.0	89.65	83.71	10,819.2	164.8	677.0	679.9	2.00	0.00	2.00
11,400.0	89.65	85.71	10,819.8	174.0	776.6	779.7	2.00	0.00	2.00
11,500.0	89.65	87.71	10,820.4	179.7	876.4	879.6	2.00	0.00	2.00
11,600.0	89.65	89.71	10,821.0	182.0	976.4	979.6	2.00	0.00	2.00
11,614.4	89.65	90.00	10,821.1	182.0	990.8	994.0	2.00	0.00	2.00
Start 8984.3 hold at 11614.4 MD									
11,700.0	89.65	90.00	10,821.7	182.0	1,076.4	1,079.5	0.00	0.00	0.00
11,800.0	89.65	90.00	10,822.3	182.0	1,176.4	1,179.5	0.00	0.00	0.00
11,900.0	89.65	90.00	10,822.9	182.0	1,276.4	1,279.5	0.00	0.00	0.00
12,000.0	89.65	90.00	10,823.5	182.0	1,376.4	1,379.5	0.00	0.00	0.00
12,100.0	89.65	90.00	10,824.1	182.0	1,476.4	1,479.5	0.00	0.00	0.00
12,200.0	89.65	90.00	10,824.7	182.0	1,576.4	1,579.4	0.00	0.00	0.00
12,300.0	89.65	90.00	10,825.4	182.0	1,676.4	1,679.4	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
12,400.0	89.65	90.00	10,826.0	182.0	1,776.4	1,779.4	0.00	0.00	0.00
12,500.0	89.65	90.00	10,826.6	182.0	1,876.4	1,879.4	0.00	0.00	0.00
12,600.0	89.65	90.00	10,827.2	182.0	1,976.4	1,979.4	0.00	0.00	0.00
12,700.0	89.65	90.00	10,827.8	182.0	2,076.4	2,079.4	0.00	0.00	0.00
12,800.0	89.65	90.00	10,828.4	182.0	2,176.4	2,179.3	0.00	0.00	0.00
12,900.0	89.65	90.00	10,829.1	182.0	2,276.4	2,279.3	0.00	0.00	0.00
13,000.0	89.65	90.00	10,829.7	182.0	2,376.4	2,379.3	0.00	0.00	0.00
13,100.0	89.65	90.00	10,830.3	182.0	2,476.4	2,479.3	0.00	0.00	0.00
13,200.0	89.65	90.00	10,830.9	182.0	2,576.4	2,579.3	0.00	0.00	0.00
13,300.0	89.65	90.00	10,831.5	182.0	2,676.4	2,679.2	0.00	0.00	0.00
13,400.0	89.65	90.00	10,832.1	182.0	2,776.4	2,779.2	0.00	0.00	0.00
13,500.0	89.65	90.00	10,832.8	182.0	2,876.4	2,879.2	0.00	0.00	0.00
13,600.0	89.65	90.00	10,833.4	182.0	2,976.4	2,979.2	0.00	0.00	0.00
13,700.0	89.65	90.00	10,834.0	182.0	3,076.4	3,079.2	0.00	0.00	0.00
13,800.0	89.65	90.00	10,834.6	182.0	3,176.4	3,179.2	0.00	0.00	0.00
13,900.0	89.65	90.00	10,835.2	182.0	3,276.4	3,279.1	0.00	0.00	0.00
14,000.0	89.65	90.00	10,835.8	182.0	3,376.4	3,379.1	0.00	0.00	0.00
14,100.0	89.65	90.00	10,836.5	182.0	3,476.3	3,479.1	0.00	0.00	0.00
14,200.0	89.65	90.00	10,837.1	182.0	3,576.3	3,579.1	0.00	0.00	0.00
14,300.0	89.65	90.00	10,837.7	182.0	3,676.3	3,679.1	0.00	0.00	0.00
14,400.0	89.65	90.00	10,838.3	182.0	3,776.3	3,779.0	0.00	0.00	0.00
14,500.0	89.65	90.00	10,838.9	182.0	3,876.3	3,879.0	0.00	0.00	0.00
14,600.0	89.65	90.00	10,839.5	182.0	3,976.3	3,979.0	0.00	0.00	0.00
14,700.0	89.65	90.00	10,840.2	182.0	4,076.3	4,079.0	0.00	0.00	0.00
14,800.0	89.65	90.00	10,840.8	182.0	4,176.3	4,179.0	0.00	0.00	0.00
14,900.0	89.65	90.00	10,841.4	182.0	4,276.3	4,278.9	0.00	0.00	0.00
15,000.0	89.65	90.00	10,842.0	182.0	4,376.3	4,378.9	0.00	0.00	0.00
15,100.0	89.65	90.00	10,842.6	182.0	4,476.3	4,478.9	0.00	0.00	0.00
15,200.0	89.65	90.00	10,843.2	182.0	4,576.3	4,578.9	0.00	0.00	0.00
15,300.0	89.65	90.00	10,843.9	182.0	4,676.3	4,678.9	0.00	0.00	0.00
15,400.0	89.65	90.00	10,844.5	182.0	4,776.3	4,778.9	0.00	0.00	0.00
15,500.0	89.65	90.00	10,845.1	182.0	4,876.3	4,878.8	0.00	0.00	0.00
15,600.0	89.65	90.00	10,845.7	182.0	4,976.3	4,978.8	0.00	0.00	0.00
15,700.0	89.65	90.00	10,846.3	182.0	5,076.3	5,078.8	0.00	0.00	0.00
15,800.0	89.65	90.00	10,846.9	182.0	5,176.3	5,178.8	0.00	0.00	0.00
15,900.0	89.65	90.00	10,847.6	182.0	5,276.3	5,278.8	0.00	0.00	0.00
16,000.0	89.65	90.00	10,848.2	182.0	5,376.3	5,378.7	0.00	0.00	0.00
16,100.0	89.65	90.00	10,848.8	182.0	5,476.3	5,478.7	0.00	0.00	0.00
16,200.0	89.65	90.00	10,849.4	182.0	5,576.3	5,578.7	0.00	0.00	0.00
16,300.0	89.65	90.00	10,850.0	182.0	5,676.3	5,678.7	0.00	0.00	0.00
16,400.0	89.65	90.00	10,850.6	182.0	5,776.3	5,778.7	0.00	0.00	0.00
16,500.0	89.65	90.00	10,851.3	182.0	5,876.3	5,878.6	0.00	0.00	0.00
16,600.0	89.65	90.00	10,851.9	182.0	5,976.3	5,978.6	0.00	0.00	0.00
16,700.0	89.65	90.00	10,852.5	182.0	6,076.3	6,078.6	0.00	0.00	0.00
16,800.0	89.65	90.00	10,853.1	182.0	6,176.3	6,178.6	0.00	0.00	0.00
16,900.0	89.65	90.00	10,853.7	182.0	6,276.3	6,278.6	0.00	0.00	0.00
17,000.0	89.65	90.00	10,854.3	182.0	6,376.3	6,378.6	0.00	0.00	0.00
17,100.0	89.65	90.00	10,855.0	182.0	6,476.3	6,478.5	0.00	0.00	0.00
17,200.0	89.65	90.00	10,855.6	182.0	6,576.3	6,578.5	0.00	0.00	0.00
17,300.0	89.65	90.00	10,856.2	182.0	6,676.3	6,678.5	0.00	0.00	0.00
17,400.0	89.65	90.00	10,856.8	182.0	6,776.3	6,778.5	0.00	0.00	0.00
17,500.0	89.65	90.00	10,857.4	182.0	6,876.3	6,878.5	0.00	0.00	0.00
17,600.0	89.65	90.00	10,858.0	182.0	6,976.3	6,978.4	0.00	0.00	0.00
17,700.0	89.65	90.00	10,858.7	182.0	7,076.3	7,078.4	0.00	0.00	0.00
17,800.0	89.65	90.00	10,859.3	182.0	7,176.3	7,178.4	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
17,900.0	89.65	90.00	10,859.9	182.0	7,276.3	7,278.4	0.00	0.00	0.00
18,000.0	89.65	90.00	10,860.5	182.0	7,376.3	7,378.4	0.00	0.00	0.00
18,100.0	89.65	90.00	10,861.1	182.0	7,476.3	7,478.4	0.00	0.00	0.00
18,200.0	89.65	90.00	10,861.7	182.0	7,576.3	7,578.3	0.00	0.00	0.00
18,300.0	89.65	90.00	10,862.4	182.0	7,676.3	7,678.3	0.00	0.00	0.00
18,400.0	89.65	90.00	10,863.0	182.0	7,776.3	7,778.3	0.00	0.00	0.00
18,500.0	89.65	90.00	10,863.6	182.0	7,876.3	7,878.3	0.00	0.00	0.00
18,600.0	89.65	90.00	10,864.2	182.0	7,976.3	7,978.3	0.00	0.00	0.00
18,700.0	89.65	90.00	10,864.8	182.0	8,076.3	8,078.2	0.00	0.00	0.00
18,800.0	89.65	90.00	10,865.4	182.0	8,176.3	8,178.2	0.00	0.00	0.00
18,900.0	89.65	90.00	10,866.1	182.0	8,276.3	8,278.2	0.00	0.00	0.00
19,000.0	89.65	90.00	10,866.7	182.0	8,376.3	8,378.2	0.00	0.00	0.00
19,100.0	89.65	90.00	10,867.3	182.0	8,476.3	8,478.2	0.00	0.00	0.00
19,200.0	89.65	90.00	10,867.9	182.0	8,576.3	8,578.1	0.00	0.00	0.00
19,300.0	89.65	90.00	10,868.5	182.0	8,676.3	8,678.1	0.00	0.00	0.00
19,400.0	89.65	90.00	10,869.1	182.0	8,776.2	8,778.1	0.00	0.00	0.00
19,500.0	89.65	90.00	10,869.8	182.0	8,876.2	8,878.1	0.00	0.00	0.00
19,600.0	89.65	90.00	10,870.4	182.0	8,976.2	8,978.1	0.00	0.00	0.00
19,700.0	89.65	90.00	10,871.0	182.0	9,076.2	9,078.1	0.00	0.00	0.00
19,800.0	89.65	90.00	10,871.6	182.0	9,176.2	9,178.0	0.00	0.00	0.00
19,900.0	89.65	90.00	10,872.2	182.0	9,276.2	9,278.0	0.00	0.00	0.00
20,000.0	89.65	90.00	10,872.8	182.0	9,376.2	9,378.0	0.00	0.00	0.00
20,100.0	89.65	90.00	10,873.5	182.0	9,476.2	9,478.0	0.00	0.00	0.00
20,200.0	89.65	90.00	10,874.1	182.0	9,576.2	9,578.0	0.00	0.00	0.00
20,300.0	89.65	90.00	10,874.7	182.0	9,676.2	9,677.9	0.00	0.00	0.00
20,400.0	89.65	90.00	10,875.3	182.0	9,776.2	9,777.9	0.00	0.00	0.00
20,500.0	89.65	90.00	10,875.9	182.0	9,876.2	9,877.9	0.00	0.00	0.00
20,582.4	89.65	90.00	10,876.4	182.0	9,958.6	9,960.3	0.00	0.00	0.00

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (ft)	Easting (ft)	Latitude	Longitude
- hit/miss target									
- Shape									
Chalmers 8T	0.00	0.00	10,876.8	181.7	9,974.9	402,255.11	1,219,964.60	48° 3' 43.067 N	103° 33' 43.255 W
- plan misses target center by 16.3ft at 20582.4ft MD (10876.4 TVD, 182.0 N, 9958.6 E)									
- Point									

Casing Points					
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)	
2,150.0	2,150.0	9 5/8"	9.625	13.500	
11,106.0	10,818.0	7"	7.000	8.750	

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well CHALMERS 5300 21-19 8T
Company:	Oasis	TVD Reference:	WELL @ 2071.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2071.0ft (Original Well Elev)
Site:	153N-100W-19/20	North Reference:	True
Well:	CHALMERS 5300 21-19 8T	Survey Calculation Method:	Minimum Curvature
Wellbore:	CHALMERS 5300 21-19 8T		
Design:	Plan #1		

Formations

Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)
2,021.0	2,021.0	Pierre			
4,624.1	4,624.0	Greenhorn			
5,029.1	5,029.0	Mowry			
5,417.1	5,417.0	Dakota			
6,463.2	6,463.0	Rierdon			
6,891.2	6,891.0	Dunham Salt			
6,960.2	6,960.0	Dunham Salt Base			
7,257.2	7,257.0	Pine Salt			
7,290.2	7,290.0	Pine Salt Base			
7,351.2	7,351.0	Opeche Salt			
7,426.2	7,426.0	Opeche Salt Base			
7,662.2	7,662.0	Amsden			
7,828.2	7,828.0	Tyler			
8,032.2	8,032.0	Otter/Base Minnelusa			
8,384.2	8,384.0	Kibbey Lime			
8,534.2	8,534.0	Charles Salt			
9,209.2	9,209.0	Base Last Salt			
9,429.2	9,429.0	Mission Canyon			
9,993.2	9,993.0	Lodgepole			
10,757.0	10,706.0	False Bakken			
10,772.9	10,716.0	Upper Bakken Shale			
10,799.8	10,732.0	Middle Bakken			
10,866.0	10,766.0	Lower Bakken Shale			
10,899.1	10,780.0	Pronghorn			
10,955.9	10,799.0	Threeforks			
11,009.4	10,811.0	Threeforks(Top of Target)			

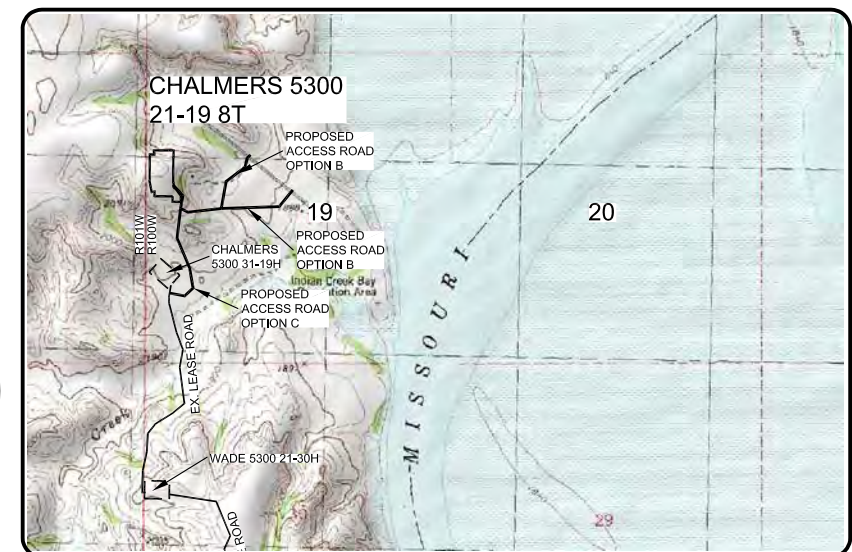
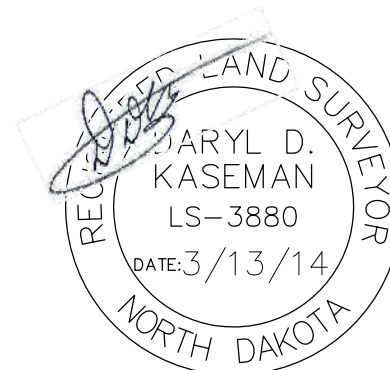
Plan Annotations

Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates		Comment
		+N/-S (ft)	+E/-W (ft)	
2,150.0	2,150.0	0.0	0.0	Start Build 5.00
2,160.0	2,160.0	0.0	0.0	Start 5719.7 hold at 2160.0 MD
7,879.7	7,879.4	50.0	0.0	Start Drop -5.00
7,889.7	7,889.4	50.0	0.0	Start 2110.6 hold at 7889.7 MD
10,000.2	10,000.0	50.0	0.0	Start 340.4 hold at 10000.2 MD
10,340.7	10,340.4	50.0	0.0	Start Build 12.00 KOP
11,087.7	10,817.9	133.8	467.1	Start 18.2 hold at 11087.7 MD EOC
11,105.9	10,818.0	137.0	485.0	Start DLS 2.00 TFO 90.05 Csg Pt
11,614.4	10,821.1	182.0	990.8	Start 8984.3 hold at 11614.4 MD
20,598.7	10,876.5	182.0	9,974.9	TD at 20598.7

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5300 21-19 8T"
2226 FEET FROM NORTH LINE AND 326 FEET FROM WEST LINE
SECTIONS 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY $-0^{\circ}03'$.



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2/8

SHEET NO.

CONCLUSION

Revision No.	Date	By	Description
REV 1	3/12/14	JUS	MOVED WELLS ON PAD

SECTION BREAKDOWN
SECTIONS 19 & 20 T153N R100W

MCKENZIE COUNTY, NORTH DAKOTA

Project No.: S13-09-282.03
Date: JAN. 2014
Drawn By: B.H.H.
Checked By: D.D.K.

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5300 21-19 8T"

2226 FEET FROM NORTH LINE AND 326 FEET FROM WEST LINE
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

WELL SITE ELEVATION 2054.5
WELL PAD ELEVATION 2046.0

EXCAVATION 119,687
PLUS PIT 22,050
141,737

EMBANKMENT 53,703
PLUS SHRINKAGE (30%) 16,111
69,814

STOCKPILE PIT 22,050
STOCKPILE TOP SOIL (6") 7,335

BERMS 1,373 LF = 445 CY

DITCHES 2,044 LF = 313 CY

DETENTION AREA 4,751 CY

SCREENING BERM 27,464 CY

STOCKPILE MATERIAL 19,693

DISTURBED AREA FROM PAD 9.09 ACRES

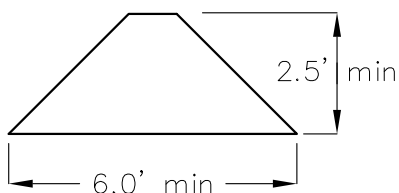
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

2226' FNL

326' FWL

BERM DETAIL



DITCH DETAIL



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Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 19, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S13-09-282.03
Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD

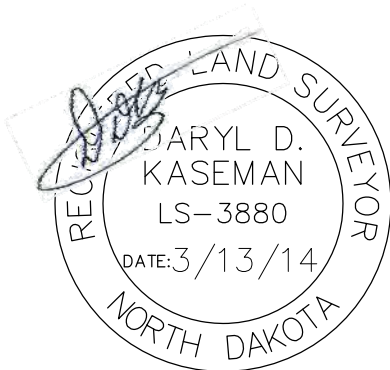
CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

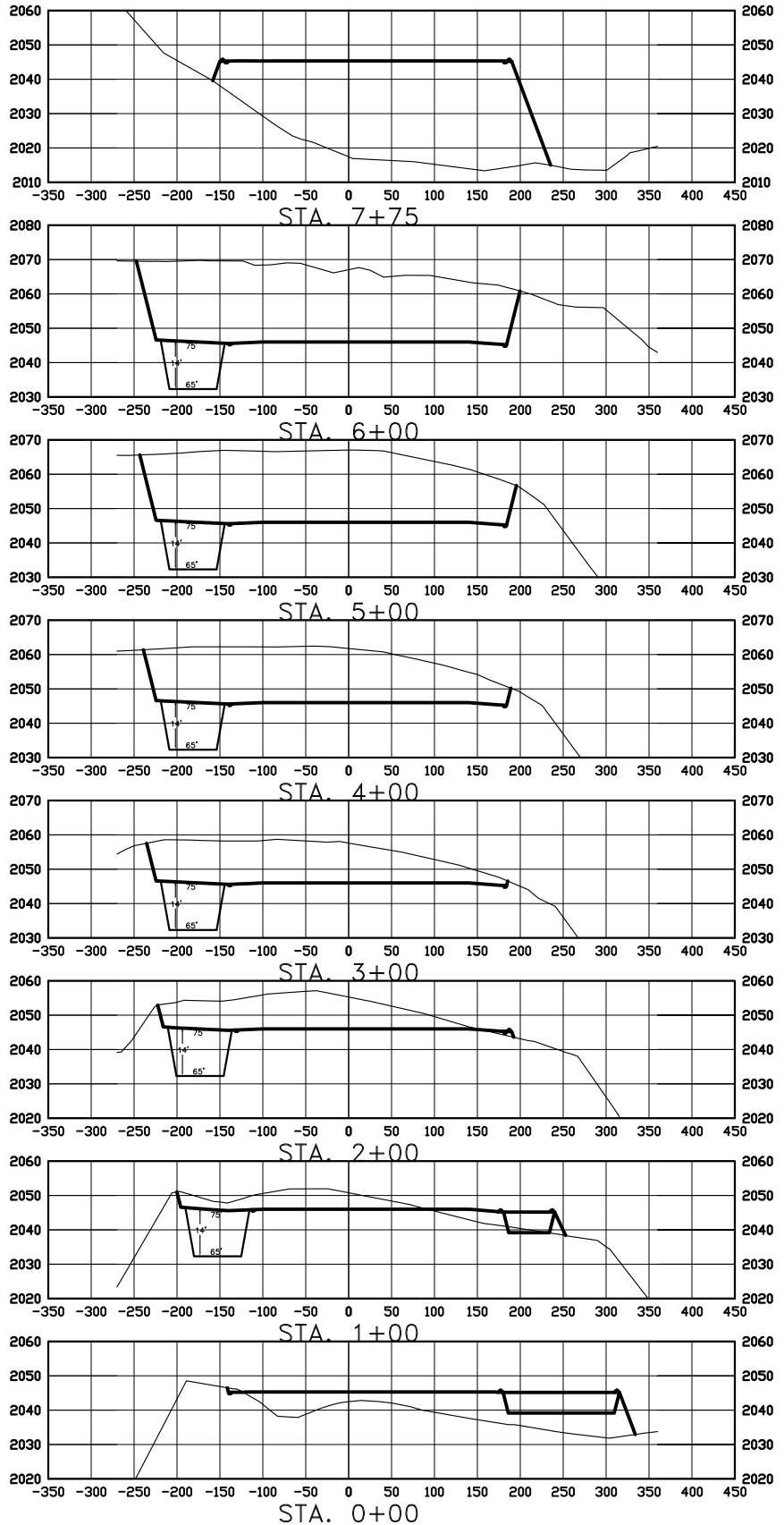
"CHALMERS 5300 21-19 8T"

2226 FEET FROM NORTH LINE AND 326 FEET FROM WEST LINE
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

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SCALE
HORIZ 1"=200'
VERT 1"=50'



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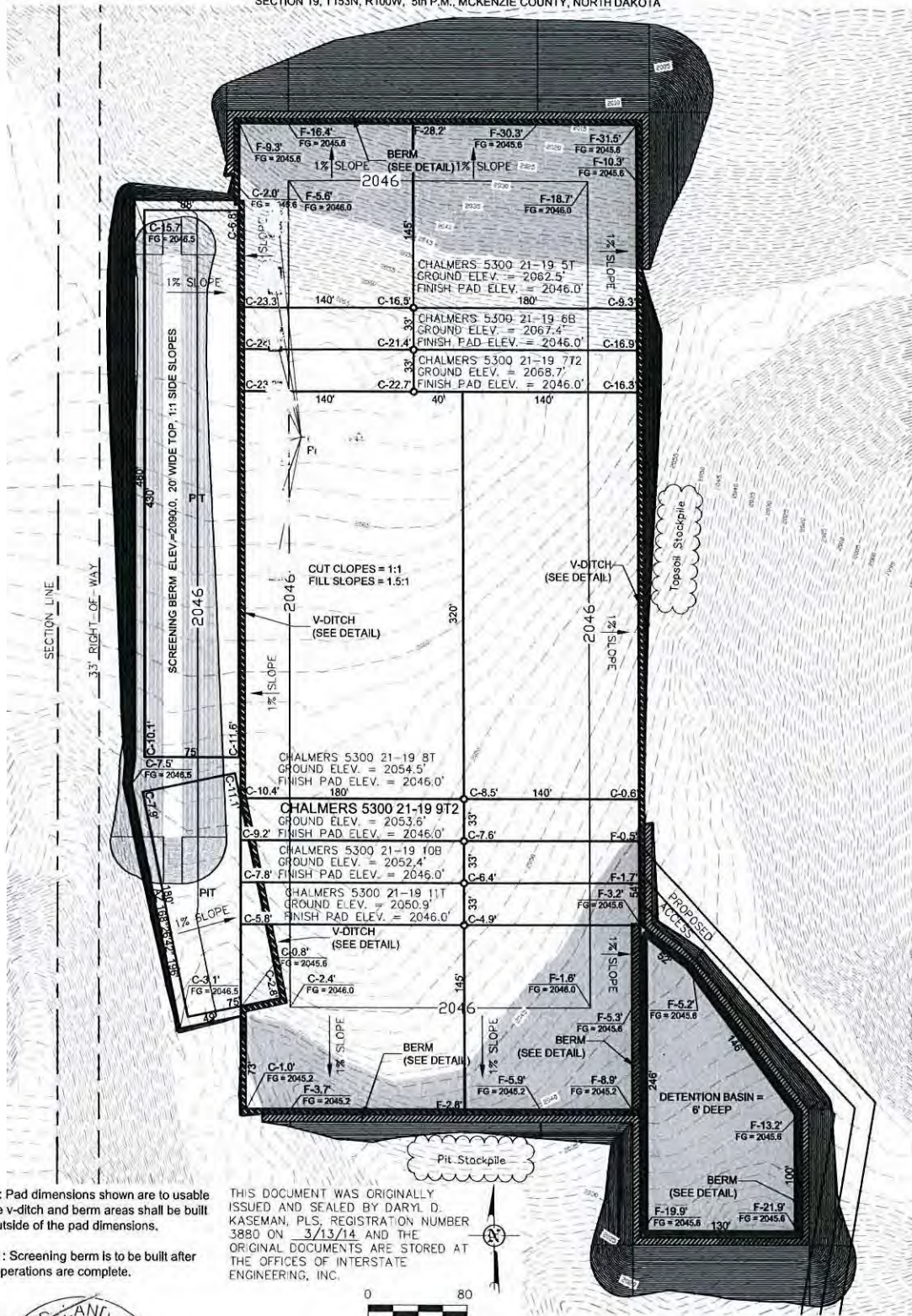
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OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 19, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S13-09-282,03
Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD

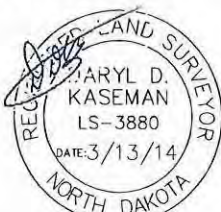
PAD LAYOUT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5300 21-19 9T2"
 2259 FEET FROM NORTH LINE AND 326 FEET FROM WEST LINE
 SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE 1: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE 2: Screening berm is to be built after drilling operations are complete.

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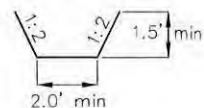


- Proposed Contours
- - - Original Contours
- BERM
- DITCH

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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V-DITCH DETAIL



3/8



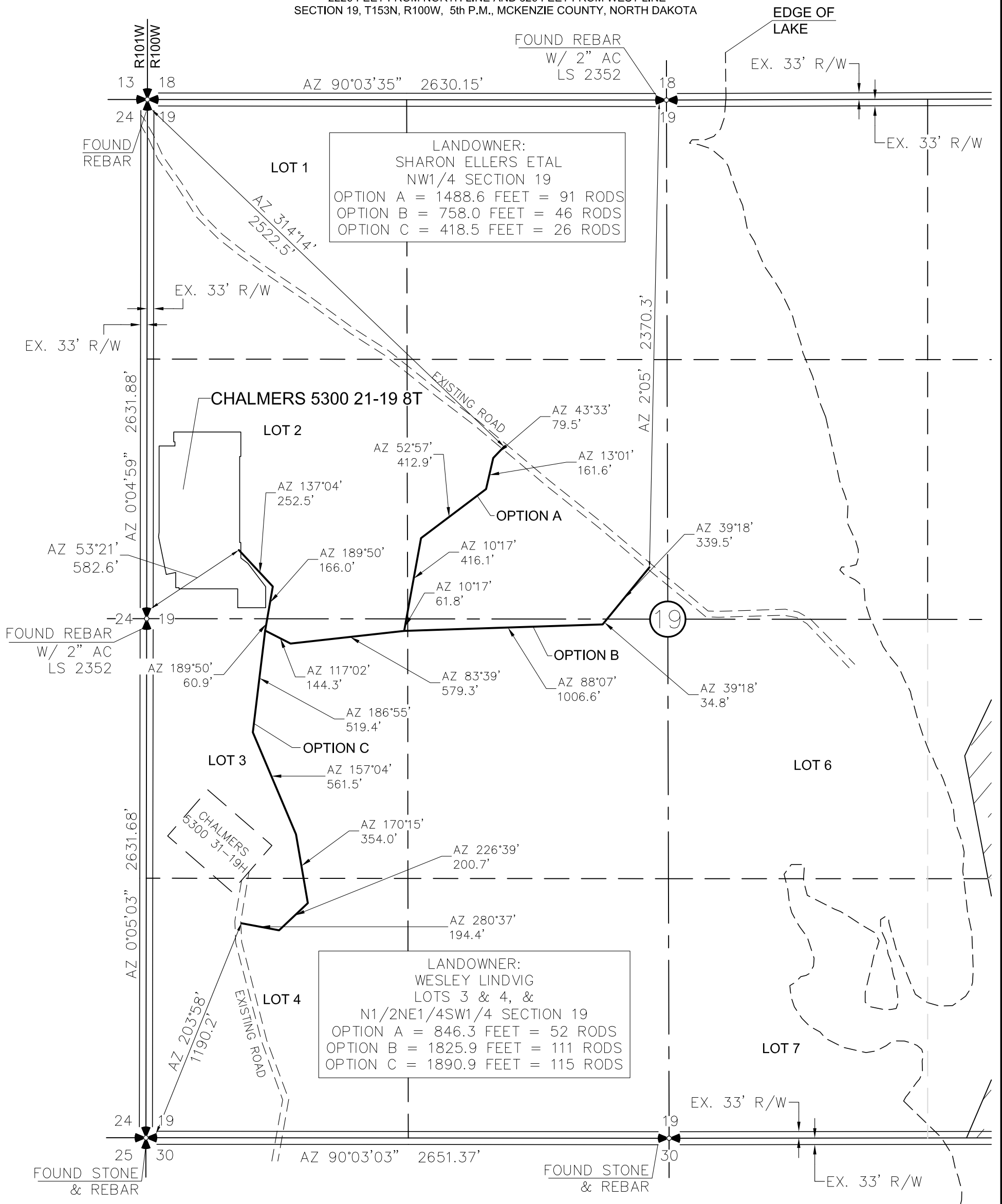
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OASIS PETROLEUM NORTH AMERICA, LLC
 PAD LAYOUT
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: 813-06-282.04
 Checked By: D.D.K. Date: JAN. 2014

Revision No.	Date	By	Description
REV 1	3/12/14	AJS	MOVED WELLS ON PAD

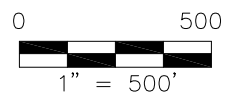
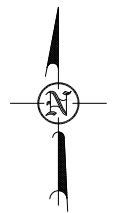
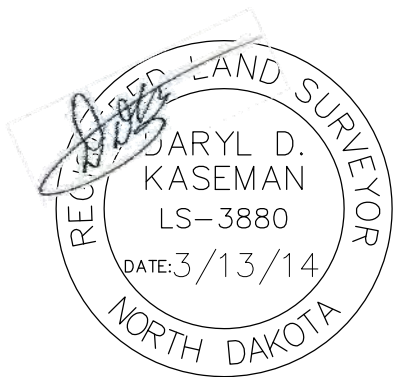
ACCESS APPROACH
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5300 21-19 8T"
2226 FEET FROM NORTH LINE AND 326 FEET FROM WEST LINE
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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NOTE: All utilities shown are preliminary only, a complete
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ACCESS APPROACH
SECTION 19, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-282.03
Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD

Q:\2013\513-09-282.03 Oasis Petroleum - 4 of 7 Infill Wells for Chalmer\CADD\REVISED CHALMERS 8T.dwg - 3/13/2014 2:05 PM josh schmierer

OASIS PETROLEUM NORTH AMERICA, LLC
 CHALMERS 5300 21-19 8T
 2226' FNL/326' FWL
 QUAD LOCATION MAP
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

CHALMERS 5300
 21-19 8T

PROPOSED
 ACCESS ROAD
 OPTION B

PROPOSED
 ACCESS ROAD
 OPTION B

PROPOSED
 ACCESS ROAD
 OPTION C

Indian Creek Bay
 Indian Area

WADE 5300 21-30H

EX LEASE ROAD

EX LEASE ROAD

MISSOURI



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 QUAD LOCATION MAP
 SECTION 19, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

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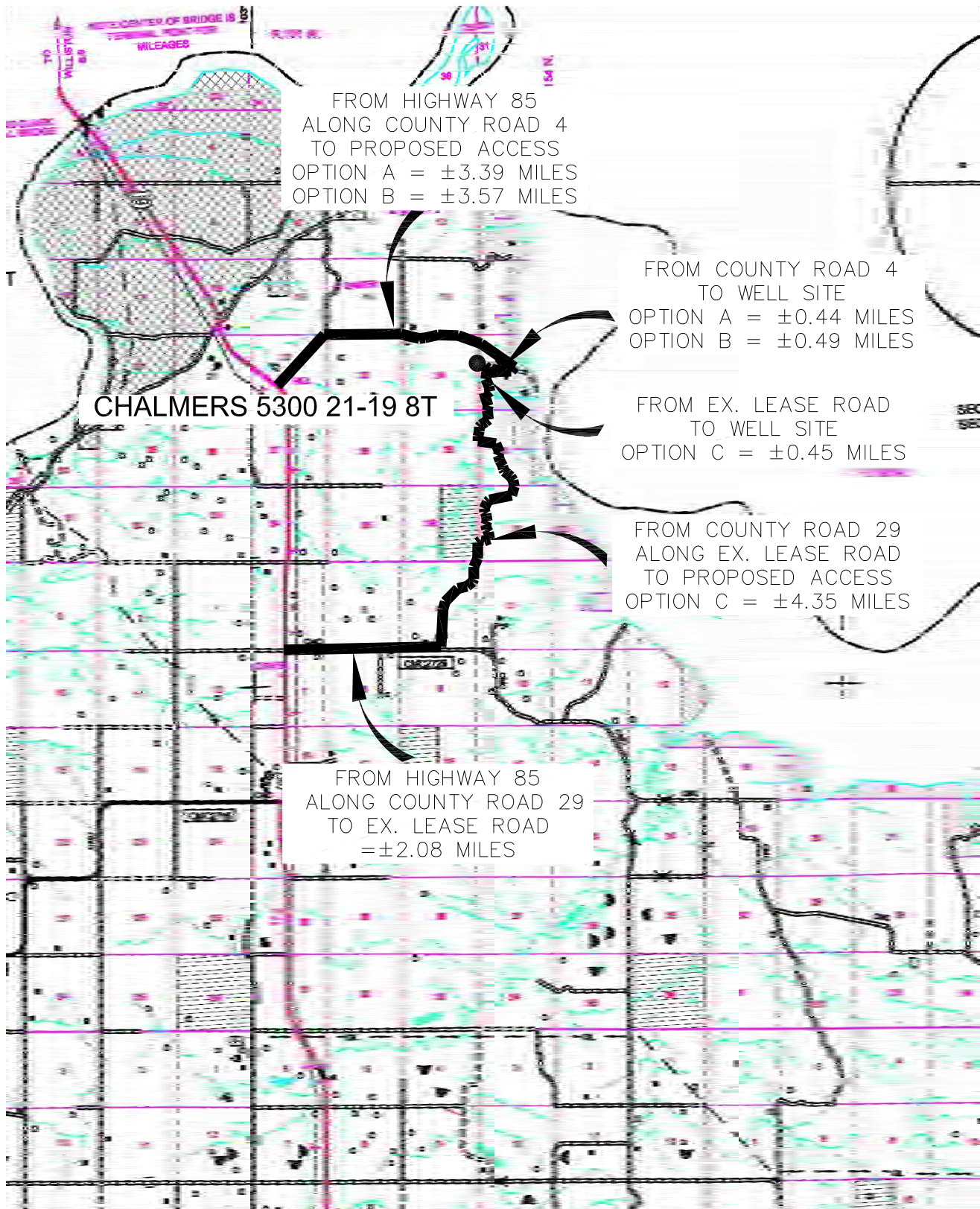
Revision No.	Date	By	Description
REV 1	3/12/14	JJS	MOVED WELLS ON PAD

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5300 21-19 8T"

2226 FEET FROM NORTH LINE AND 326 FEET FROM WEST LINE
SECTION 19, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA





STATEMENT

This statement is being sent in order to comply with NDAC 43-02-03-16 (Application for permit to drill and recomplete) which states (in part that) "confirmation that a legal street address has been requested for the well site, and well facility if separate from the well site, and the proposed road access to the nearest existing public road". On the date noted below a legal street address was requested from the appropriate county office.

April 3, 2014

McKenzie County

Aaron Chisolm – address@co.mckenzie.nd.us

Pad 1

Chalmers 5300 21-19 5T Lot 2 Section 19 T153N R100W

Chalmers 5300 21-19 6B Lot 2 Section 19 T153N R100W

Chalmers 5300 21-19 7T2 Lot 2 Section 19 T153N R100W

Chalmers 5300 21-19 8T Lot 2 Section 19 T153N R100W

Chalmers 5300 21-19 9T2 Lot 2 Section 19 T153N R100W

A handwritten signature in black ink, appearing to read "Chelsea Covington", written over a horizontal line.

Chelsea Covington

Regulatory Assistant

Oasis Petroleum North America, LLC



June 10, 2014

Re: Un-Occupied Trailer House and Seasonal Cabin.

Brandi,

Just to follow up with past conversations about the dwellings east of our proposed Chalmer 5300 21-19 well site. The white trailer is unlivable, it has no water, power or sewer. The cabin is seasonal at best and has not been used for several years. If I can be of further assistance please advise.

Thank you,

A handwritten signature in blue ink, appearing to read "JD DeMorrett", is written over the typed name.

JD DeMorrett

Sr. Staff Landman for Oasis Petroleum North America, LLC

PO Box 1126 Williston ND- Office 701-577-1600 Fax 701-577-1692